

Our Deficit, Your Problem?

Fiscal Sustainability, Exchange Rates, and Inflation^{*}

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Abstract

Fiscal deficits in a financial Hegemon transmit to inflation and exchange rates abroad. Using market-value data on U.K. and U.S. sovereign debt over 1975–2024, we recover model-consistent fiscal deficit innovations and document two empirical regularities: U.S. deficit shocks appreciate the U.K. real exchange rate on impact and raise U.K. consumer prices persistently, with a transmitted magnitude roughly a third that of own-country U.K. deficit shocks. We rationalise these facts in a quantitative continuous-time two-country model in which Hegemon bonds carry a convenience yield. A Hegemon fiscal expansion tightens the foreign intertemporal budget constraint through two channels: a wealth effect on foreign holders of Hegemon debt, and a discount-rate spillover that revalues foreign sovereign liabilities. We call this mechanism *fiscal contagion* and show that it operates across different monetary–fiscal policy mixes. An estimated version of the model attributes roughly two percentage points of cumulative U.K. inflation since 2021 to spillovers from U.S. fiscal policy.

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JEL Classification: E63, F41, E62.

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1 Introduction

Recent debates over fiscal sustainability have focused primarily on the domestic inflationary consequences of debt-financed deficits. Yet fiscal expansions can also move exchange rates, thereby revaluing sovereign debt held across borders. This channel is especially important for the United States, which sits at the centre of the global financial system: its currency anchors international reserves, and its government debt is the benchmark safe asset held on balance sheets worldwide. Shocks to U.S. fiscal conditions therefore travel through channels that do not exist for smaller economies. The dynamics observed in April 2025 offer a recent example. The U.S. dollar depreciated even as Treasury yields rose, a combination at odds with the conventional safe-haven view of U.S. government debt (Jiang et al., 2025), and international holdings of U.S. Treasuries fell by nearly 50 billion USD over the same period (U.S. Treasury, 2026). When investors reallocate out of Hegemon debt on this scale, the price adjustments do not stay within U.S. markets. They propagate to the exchange rates and discount rates of any sovereign whose bonds those investors hold as substitutes. If such episodes recur, as current fiscal pressures suggest they may, then understanding how shocks at the centre of the system transmit to real financing conditions elsewhere is essential.

This paper studies how fiscal deficits in a financial Hegemon affect exchange rates and inflation abroad. Our framework treats representative households as international investors who view Hegemon and Foreign sovereign bonds as substitutes. As expected currency-adjusted returns change, Foreign households reallocate between Hegemon and Foreign government debt. Because global investors price both assets at the margin, equilibrium requires exchange rates and country-specific discount factors to adjust jointly. A Hegemon fiscal shock therefore propagates through two linked valuation margins: exchange-rate movements revalue foreign holdings of Hegemon debt, while adjustments in the common discount-rate system revalue Foreign sovereign liabilities. We call this joint transmission *fiscal contagion*: a Hegemon deficit shock can tighten the intertemporal budget constraint of the Foreign government even when its expected primary surpluses are unchanged.

The empirical challenge in linking deficits, exchange rates, and inflation is that the relevant object is the cumulative present discounted sum of *unfunded* fiscal impulses in any given period. Building on Hall and Sargent (2011), Cochrane (2019), and Jiang et al. (2024c), we use a market-value decomposition of sovereign debt to recover the exact revisions to expected primary surpluses that are consistent with observed debt prices and global discount rates. Using market-value data for U.K. and U.S. sovereign liabilities over the period 1975–2024, this approach quantifies how much of the observed debt dynamics is driven by primary surplus news, by global real-rate shocks, and by exchange rate and risk-premium repricing. The recovered U.K. and U.S. fiscal innovations predict subsequent movements in the U.K. real exchange rate and inflation. In a related note

(Hyland and Kawalec, 2026), we develop the valuation accounting, from which we here use the recovered innovations to study the international transmission of fiscal shocks.

To rationalise these empirical observations, we develop a two-country open-economy model in which fiscal shocks in a financial Hegemon, whose sovereign debt carries a convenience yield, propagate internationally through the uncovered interest parity condition. We show that the threat of fiscal inflation in the Hegemon transmits fiscal pressure to other economies by raising the cost of servicing their debt. In the model, we identify a mechanism through which inflationary pressures at the Hegemon spill over to other economies: a mechanism of “international fiscal contagion” that generates fiscally driven inflation spillovers abroad. A deterioration of the Hegemon’s expected surpluses tightens the Foreign sovereign fiscal-valuation condition under some policy specifications, even when Foreign projected surpluses are unchanged, because currency-denominated discount factors must adjust to clear sovereign debt markets. A deterioration in Hegemon fiscal backing can therefore transmit to Foreign sovereign valuation through the common equilibrium discount-rate system, with the allocation across exchange rates and price levels depending on the policy regime. The mechanism operates through the interest rate charged on Foreign government debt in equilibrium, with the nominal exchange rate driving much of the adjustment.

Mechanisms linking Hegemon deficits to exchange rate movements and to price-level changes in other countries are gaining traction (e.g., Alberola et al., 2021; Ding and Jiang, 2024; Mac Mullen, 2025). The existing empirical evidence supports a close nexus between cross-country deficits and exchange rates but assigns only a limited role to government-debt valuation in equilibrium, and does not rigorously emphasise the possible nominal devaluation of government debt across borders.

In our model, Hegemon government bonds carry convenience yields, which are non-pecuniary benefits from holding quasi-safe assets, documented for sovereign debt by Valchev (2020), Jiang et al. (2024b), and Bianchi and Bigio (2026). Their international allocation is pinned down by factors such as convenience demand, market clearing, and the accumulation of net foreign assets. Under the maintained isoelastic specification of convenience flows, the marginal convenience yield declines with the Hegemon bond position in both countries. Therefore, the size and sign of fiscal contagion is jointly determined by cross-country slope asymmetry, portfolio exposure, and the monetary–fiscal policy mix. In particular, some policy measures attenuate or neutralise the effects on the Foreign economy. The economics of the portfolio-convenience channel is as follows: suppose an unfunded Hegemon transfer expansion raises the outstanding stock of Treasuries, while its central bank leans against the expansion with tight policy. The two countries absorb the additional supply in proportions governed by their convenience-demand curvatures, so their marginal valuations of the asset move apart. The equilibrium can require a higher Foreign real financing rate, delivered jointly through policy rates, inflation, and currency adjustment. Portfolio reallocation and the correlated discount rates move currencies and sovereign yields, revaluing Foreign government lia-

bilities, while the gap in marginal valuations accumulates into a cross-border transfer of purchasing power. Because internationally mobile investors treat Hegemon and Foreign government bonds as substitutes along this convenience margin, discount rate movements at the Hegemon, which anchors global discount rates, transmit across borders.

Literature Review

The title of our contribution echoes the book of [Rogoff \(2025\)](#). Our central focus, however, is the role of sovereign *deficit* dynamics for exchange rates, and how deficits can impact the fiscal position of other countries that are closely financially integrated.

Our primary contribution is to the literature on sovereign deficits and their ties to inflation and exchange rate dynamics. The paper is also adjacent to work on fiscally driven price-level determination, which depends on the broader fiscal-monetary policy mix. This literature was inaugurated by [Sargent and Wallace \(1981\)](#), with [Leeper \(1991\)](#) and [Woodford \(1995\)](#) laying much of the modern groundwork for models that assign at least a share of inflationary dynamics to fiscal aggregate demand management. [Leeper and Leith \(2016\)](#) and [Cochrane \(2023\)](#) provide literature reviews. Support for fiscally driven inflation has been developed both theoretically ([Ascari et al., 2023](#); [Bassetto et al., 2024](#); [Bassetto and Cui, 2018](#); [Bianchi et al., 2023](#); [Bianchi and Melosi, 2017, 2022](#); [Bigio et al., 2024](#); [Campos et al., 2024](#); [Caramp and Silva, 2023](#); [Corsetti and Maćkowiak, 2024](#); [Kaplan et al., 2024](#); [Miao and Su, 2024](#)) and empirically ([Ascari et al., 2024](#); [Banerjee et al., 2022](#); [Cochrane, 2022](#); [Reichlin et al., 2023](#)), with [Barro and Bianchi \(2026\)](#) collecting rich international evidence that the post-Covid inflationary episode can be attributed to a significant part to expansionary fiscal policy.¹ In our framework, we do not explicitly introduce the possibility of regime-switching monetary and fiscal policy, which has been richly explored by [Bianchi \(2013\)](#); [Bianchi and Ilut \(2017\)](#); [Bianchi and Melosi \(2019\)](#) and [Chen et al. \(2022a\)](#). However, we explore in detail the effects induced by changes to the prevalent monetary/fiscal policy mix in the international context, and our estimation exercise is broadly informative over which type of monetary and fiscal policy combination can rationalize the observed macroeconomic time series in the U.S. and the U.K. since 1998 the best.

Most of this work is set in closed-economy environments, but a small and growing literature brings these insights and broader discussions of fiscal-monetary interactions to international models. Early explorations linking fiscal price-level determination and exchange rate dynamics include [Bergin \(2000\)](#), [Dupor \(2000\)](#), [Sims \(1997\)](#), [Sims \(1999\)](#), and [Woodford \(1998\)](#). Initial applications include [Benigno and Missale \(2004\)](#) and [Corsetti and Maćkowiak \(2006\)](#), who focus on the ties between public debt and currency stability, especially during episodes of severe currency turmoil.

¹See [Kaplan \(2025a,b\)](#) for overviews on the relevance of the (Non-)Ricardianness of households within this literature and how these relate to modern heterogeneous-agent models.

International aspects of government debt valuation and their effects on inflation and exchange rates were further explored in the run-up to the Global Financial Crisis by [Giannitsarou and Scott \(2006\)](#) and [Gourinchas and Rey \(2007\)](#).

Alongside the renewed interest in deficit-inflation links, research has revisited the role of deficits in exchange rate determination. [Schmitt-Grohé and Uribe \(2003\)](#) provide an early analysis of exchange rate adjustments relative to inflation in financing fiscal deficits within emerging markets. [Valchev \(2020\)](#) establishes that excess returns on currency holdings can be traced to endogenous movements in bond convenience yields, which in turn depend on the prevailing interactions between monetary and fiscal policy. Particularly related to our contribution are [Alberola et al. \(2021\)](#) and [Jiang \(2022\)](#). [Alberola et al. \(2021\)](#) distinguish between Ricardian and non-Ricardian regimes, tying unexpected exchange rate movements (relative to a complete-market frictionless benchmark) to non-Ricardian fiscal regimes in which government debt lacks sufficient fiscal backing. [Jiang \(2022\)](#) makes explicit some of the implications of applying fiscal price-level determination to exchange rate dynamics, showing that real exchange rate adjustments are linked to government surplus shocks through the cost of issuing government debt, among other factors. More broadly, our approach highlights the possibility that U.S. sovereign debt acts as a reference safe asset through which global financial conditions and business cycles are transmitted, as in [Bodenstein et al. \(2023\)](#). Since we relate most of the variation observed to common discount rate kernels, our intuitions are also related to [Kirsanova et al. \(2026\)](#).

The paper also relates to the large macro-finance literature on nominal and real exchange rate determination, which we cannot survey fully here. A partial review is provided by [Itskhoki \(2021\)](#), including an overview of the many puzzles that standard complete-market models fail to resolve. Of particular relevance are recent contributions on the possibility of an exchange rate disconnect from macroeconomic fundamentals ([Bacchetta and Van Wincoop, 2004](#); [Chari et al., 2002](#); [Corsetti et al., 2008](#); [Duarte, 2003](#); [Engel and West, 2005](#); [Gabaix and Maggiori, 2015](#); [Itskhoki and Mukhin, 2021](#)), as our main argument links the exchange rate back to one specific macroeconomic fundamental, namely aggregate demand managed by the fiscal authority. Our paper therefore also connects to recent work pushing back against the narrative that exchange rates are disconnected from fundamentals. [Bodenstein et al. \(2024\)](#) link exchange rate dynamics to trade balance shocks, which are reduced-form demand disturbances. [Kekre and Lenel \(2024\)](#) attribute the majority of cross-country exchange rate variation to demand movements, viewed through the lens of resulting interest rate differentials. [Mac Mullen \(2025\)](#) shows that these interest rate differentials can be partially attributed to fiscal deficit dynamics, in line with the evidence and models presented here.

By analysing interactions between fiscal Hegemons and countries that lack the sovereign-debt privilege conferred by financial dominance, we also speak to this literature. [Ellison and Scott \(2020\)](#)

provide a long-run perspective on the valuation of U.K. national debt that served as a cornerstone for our analysis. Our contribution adds the international dimension, extends their data sample, and increases the focus on inflationary factors. [Jiang et al. \(2024c\)](#) focus extensively on the U.S. experience in recent decades, finding evidence of a persistent mispricing of U.S. sovereign debt. As [Chen et al. \(2022b\)](#) document, this dominant position of U.S. sovereign debt was inherited from the United Kingdom in the early 20th century. [Ding and Jiang \(2024\)](#) make explicit a number of channels through which monetary and fiscal policy are intertwined in the presence of a dominant fiscal and financial Hegemon.² [Jiang et al. \(2021\)](#) link the valuation of the U.S. Dollar to convenience yields enjoyed by foreign investors holding safe U.S. assets, and [Jiang \(2021\)](#) argues that the U.S. fiscal cycle links currency risk premia of the Dollar to those of other currencies. [Jiang et al. \(2023\)](#) and [Jiang et al. \(2024b\)](#) are closest to our argument, showing how convenience yields on Dollar-denominated debt enter as a wedge in the Euler equations that determine exchange rates.

Finally, through our interpretations of the U.S. economy acting as a financial Hegemon that acts as the anchor for global discount rates, we relate squarely to the literature on the global financial cycle ([Rey \(2015\)](#), [Miranda-Agrippino and Rey \(2020\)](#)), as surveyed by [Miranda-Agrippino and Rey \(2022\)](#). In particular, [Jiang et al. \(2024a\)](#) relates closely to our work, building a model for the global financial cycle with international demand for safe dollar assets, which we broadly approximate through our convenience yield mechanism, while [Farhi and Maggiori \(2018\)](#) study in detail how a financial Hegemon supplying (quasi-)safe assets anchors the international monetary system. Relative to this literature, which traces the global financial cycle to the Hegemon's monetary stance or to the supply of safe dollar assets, we identify a distinctly fiscal origin: the cycle can be driven by the perceived backing of the Hegemon's debt, and propagates not merely to asset prices and exchange rates but to the nominal valuation of *foreign* sovereign liabilities through the common discount kernel the Hegemon anchors. To our knowledge, this cross-border revaluation of foreign-issued debt has not been made explicit in this literature, which has centred on the co-movement of financial prices and quantities rather than on fiscal positions and price levels.

2 Fiscal Impulses and International Spillovers

We first establish the empirical relevance of our international deficit-inflation channel. A natural question concerns the method of identifying fiscal innovations that is not marred by the simultaneity bias inherent to fiscal policy. Motivated by our focus on the international dimension of bond holdings and on joint revaluations of discount rates, we use unexplained changes to the market value of sovereign debt, derived consistently with [Campbell and Shiller \(1988\)](#), [Hall and Sargent \(2011\)](#), and [Cochrane \(2019\)](#), to establish a causal link between deficits, exchange rates, and

²This literature is in turn connected to the dominant currency paradigm, surveyed in [Gopinath and Itskhoki \(2022\)](#).

inflation with an additional international dimension. Most of the derivations are provided in [Hyland and Kawalec \(2026\)](#), where we detail the methodology used to recover surpluses. Here, we only briefly summarise some of the core aspects of the analysis.

2.1 Recovering Deficit Innovations from Market-Value Debt

Our method relies on unexplained changes to the *market* value of government debt. The focus on market values is important because the market value of government debt reflects exactly how bond investors perceive it to be backed by surpluses, growth, interest rate payments, and nominal devaluation.

In what follows, we let $X \in \{H, F\}$ denote the two countries. H denotes the U.S., which we take to be a ‘financial Hegemon’, and F denotes the Foreign economy, which we take to be the U.K. to keep a narrow but well-identified focus. Following [Cochrane \(2019\)](#), the market value of government debt can be decomposed *exactly* (without approximation error) as

$$\rho v_{t+1}^X = v_t^X + r_{t+1}^X - \pi_{t+1}^X - g_{t+1}^X - s_{t+1}^X, \quad (1)$$

where ρ is a discount rate, v_t is the log ratio of the market value of debt to GDP at the end of period t , r_{t+1}^n is the log nominal return on the portfolio of government bonds, π_t is inflation, g_{t+1} is the log of GDP growth, and s_{t+1} is the scaled real primary surplus to GDP ratio. Focus now on country F . Positing country H as the financial Hegemon, we assume that the *discount rate* in country F is tied to the discount rate in country H by a simple UIP condition: $r_{t+1}^{n,F} = r_{t+1}^{n,H} - \Delta e_{t+1} - \varphi_{t+1}$, where Δe_{t+1} is the change in the log exchange rate (defined by the usual convention that an increase in e corresponds to a *depreciation* for country H), and φ_{t+1} is a deviation from the exact UIP condition, with $\mathbb{E}_t[\varphi_{t+1}] = 0$. The debt valuation equation for country F then becomes

$$\rho_F v_{t+1}^F = v_t^F + \left(r_{t+1}^{n,H} - \Delta e_{t+1} - \varphi_{t+1} \right) - \pi_{t+1}^F - g_{t+1}^F - s_{t+1}^F. \quad (2)$$

Rolling this equation backwards, we can express *exactly* what the changes to the cumulative expected value of surpluses/deficits at any given point must have been to rationalise the observed market value of government debt: all other terms save for s_{t+1} are observable, and the sum of these terms gives the deficit innovations we are after. Iterating the Foreign flow identity from a base year and setting $\rho_F \simeq 1$ for notational simplicity gives

$$v_t^F = v_0^F + \sum_{j=1}^t \left(r_j^{n,H} - \Delta e_j - \varphi_j \right) - \sum_{j=1}^t \pi_j^F - \sum_{j=1}^t g_j^F - \sum_{j=1}^t s_j^F. \quad (3)$$

Equation (3) is useful for two reasons. First, we can directly infer $\sum_{j=1}^t s_j^X$ for both countries $X = H, F$, giving the surplus/deficit innovations consistent with the observed market pricing of

government debt. Second, we can infer how much changes in the discount rate kernel in the U.S. (country H) matter for the value of government debt in the U.K. (country F), giving a first pass that rationalises the role of correlated discount rates in the revaluation of government debt in the face of fiscal expansions. We conduct this decomposition using quarterly data (which constitutes part of our contribution, as it has not been done before) from the end of the Bretton Woods system around 1975 through to the end of 2025.

As an illustration, Figure 1 decomposes the value of government debt. Two channels matter most for our purposes: the orange line, which captures the importance of deficits (the negative of surpluses), and the purple and brown lines, which capture the exchange-rate and UIP violation channels, respectively.

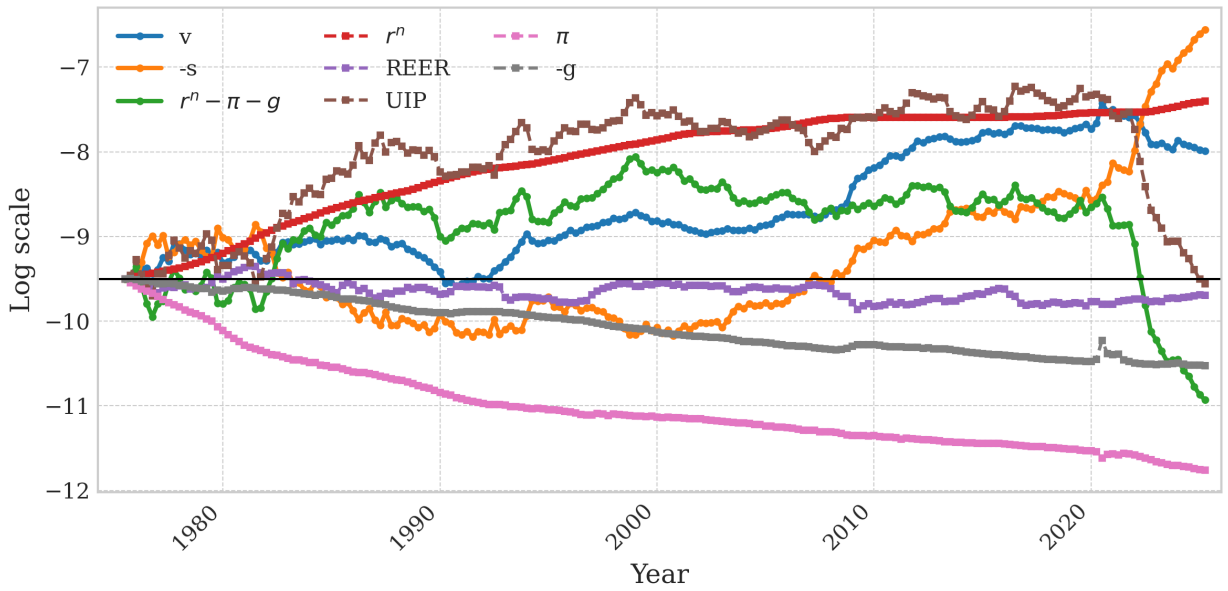


Figure 1: Backwards decomposition of the value of government debt, following equation (3).

The period-by-period difference of the orange line gives our deficit innovations. As expected, the post-GFC and post-Covid periods are characterised by very large deficit innovations, pushing $-s$ up significantly. Of the exchange-rate and UIP channels, UIP violations matter most: from the end of the Bretton Woods system until Covid, UIP violations mostly pushed the required return on government debt down, but the large post-Covid fiscal expansion almost fully eroded these gains from the perspective of the U.K.'s debt valuation equation. Rival theories of UIP violations exist; in what follows we focus in our modelling framework on the role of *convenience yields* and how they contribute to revaluations of international bond holdings. In sum, changes to the discount kernel driven by Hegemon reference rates can matter significantly for the discount kernel of other-country government debt, and hence for its value.

We now use the deficit series from Figure 1 (alongside a similarly derived series for the U.S.,

see [Hyland and Kawalec, 2026](#)) to uncover how these debt-valuation-consistent deficit innovations from *both* the U.K. and the U.S. matter for U.K. exchange rates and inflation.

2.2 Exchange Rates, Inflation, and Deficit Shocks

We now use the recovered model-consistent *deficit* processes for the U.K. and the U.S. to quantify how much each contributes to short- and medium-run movements in U.K. real exchange rates and inflation, in a standard but robust local projections exercise. Economic theory is not a priori conclusive about the influence of U.S. deficits (which we treat as Hegemon deficits) on U.K. exchange rates and inflation: U.S. deficits could contribute to an appreciation of the U.K. exchange rate if the lack of fiscal backing makes U.S. savings assets less attractive, but the opposite could also hold if weaker fiscal backing is accompanied by an increase in the promised return on U.S. savings assets.

Our exercise provides a clear method for identifying the effect of deficits on U.K. exchange rates and inflation. We first consider the effect of U.S. and U.K. deficits on *real* exchange rates.³ The deficits are the negative of the surplus series recovered above; we set $\widehat{\text{Def}}_t^{UK} \equiv -\hat{s}_t^F$ and $\widehat{\text{Def}}_t^{US} \equiv -\hat{s}_t^H$. We estimate the following long-differenced local projection:

$$\Delta e_{t+h} = \alpha_h + \beta_h^{UK} \widehat{\text{Def}}_t^{UK} + \beta_h^{US} \widehat{\text{Def}}_t^{US} + \gamma_h e_{t-1} + \Gamma_h X_{t-1} + u_{h,t}, \quad h = 0, 1, \dots, H, \quad (4)$$

where Δe_{t+h} is the change in the log real exchange rate index and the right-hand side variables are the U.K. and U.S. deficit measures.⁴ We additionally control for the past level of the real exchange rate, the past level and first difference of U.K. inflation, lagged U.K. output growth, the lagged U.K. unemployment rate, and the lagged U.K. bank rate. Figure 2 plots the local projection coefficients. The left panel shows the effects of U.K. deficits, and the right panel shows the effects of U.S. deficits, in both cases for a normalised deficit shock of 1% of U.K. GDP.

The results are striking. Focus first on the right panel: an innovation in U.S. deficits leads on impact to an *appreciation* of the U.K. exchange rate by 0.1 percentage points. The effect is highly persistent and both economically and statistically significant for up to three years. The role of U.K. deficits is somewhat less clear-cut: the impact is small at first, but over time the U.K. real exchange rate depreciates in response to the identified deficit shock, by almost 0.2 percentage points four years out. Taken together, these patterns are consistent with the standard logic of international capital mobility in response to otherwise unfunded fiscal deficits: an increase in U.S.

³Our main measure of real exchange rates is the BIS Real Effective Exchange Rate Index, but our results are fully robust to using the measure developed by [Darvas \(2021\)](#).

⁴We weight the absolute size of U.S. deficits by the ratio of U.K. to U.S. real GDP multiplied by the Pound Sterling-U.S. Dollar exchange rate, to ensure that size differences across the two economies do not drive our results.

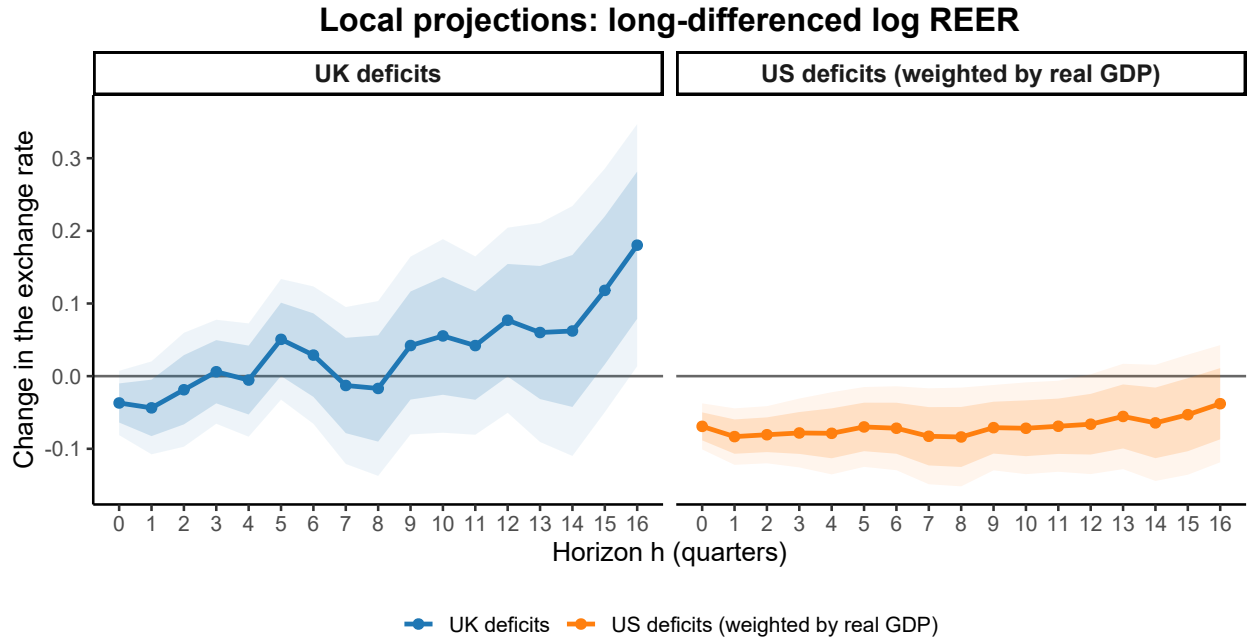


Figure 2: The effect of deficit surprises in the U.K. and the U.S. on the U.K. REER. Shaded areas indicate 68% and 90% confidence bands, respectively, calculated using HAC-robust standard errors.

deficits makes the U.S. savings asset relatively less attractive to foreign investors, and the resulting capital flight contributes to an appreciation of sterling. Likewise, larger U.K. deficits make the domestic savings asset less attractive, contributing to a real depreciation of sterling.

The link between exchange rates and inflation is significant and interesting in its own right, but it also motivates considering the direct effect of deficit innovations on U.K. inflation. We therefore estimate the following local projection to shed light on the link between deficits, international spillovers, and inflation:

$$\Delta P_{t+h}^{UK} = \alpha_h + \beta_h^{UK} \widehat{Def}^{UK} + \beta_h^{US} \widehat{Def}^{US} + \gamma_h P_{t-1}^{UK} + \Gamma_h X_{t-1} + u_{h,t}, \quad h = 0, 1, \dots, H, \quad (5)$$

where ΔP_{t+h}^{UK} is the change in the chosen log price index (i.e., cumulative inflation) and the right-hand side variables are the recovered U.K. and U.S. deficit measures. γ_h controls for the past level of the chosen price index, and X_{t-1} contains the same controls as in equation (4). For the price index, we report results for both headline CPI and headline PPI, though our results are quantitatively robust to using the U.K. GDP deflator. Figure 3 plots the local projection coefficients with CPI as the outcome variable, and Figure 4 plots the implied response of the PPI to U.K. and U.S. deficit innovations. The message is clear regardless of the outcome variable: both U.K. and U.S. deficit surprises matter, quantitatively and economically, for price movements, and both push inflation up significantly and persistently.

Two to four years out, a 1% deficit innovation in the U.K. raises the CPI price level by 0.08

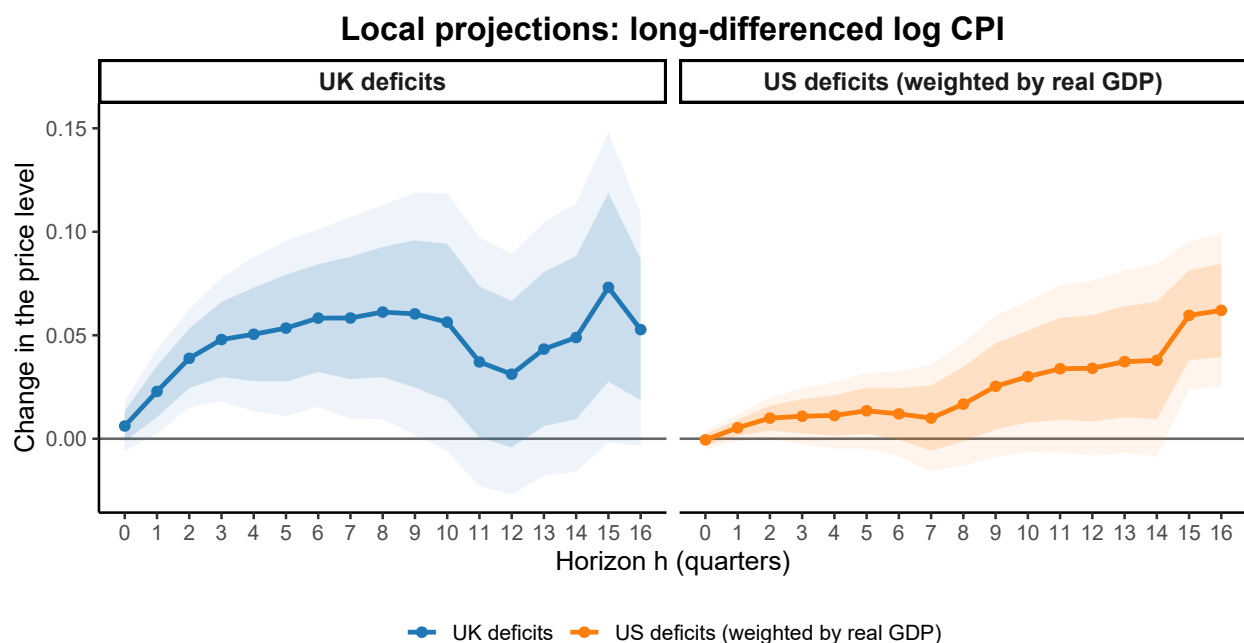


Figure 3: The effect of deficit surprises in the U.K. and the U.S. on the U.K. CPI. Shaded areas indicate 68% and 90% confidence bands, respectively, calculated using HAC-robust standard errors.

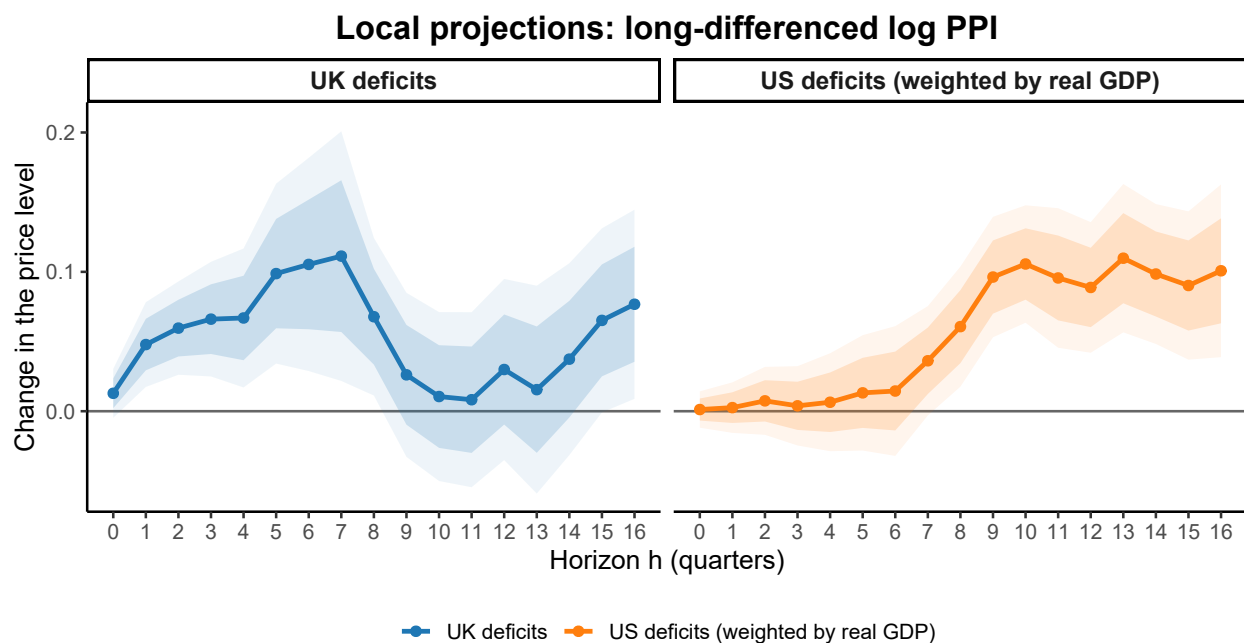


Figure 4: The effect of deficit surprises in the U.K. and the U.S. on the U.K. PPI. Shaded areas indicate 68% and 90% confidence bands, respectively, calculated using HAC-robust standard errors.

percentage points. The spillover from U.S. deficit innovations is only marginally smaller, peaking at about 0.07. The response of PPI inflation is larger, peaking at about 0.11 for U.K. deficits two years after the initial impulse, and at a similar magnitude for U.S. deficits after three to four years. Regardless of the precise measure of inflation, we observe economically and statistically significant changes in the price level. While the effect of U.K. deficits is not surprising given the existing evidence ([Hazell and Hobler, 2024](#)), the international transmission of inflationary pressure is novel: the fact that U.S. deficits meaningfully push U.K. inflation up is novel evidence of international contagion in inflationary pressure in response to expansionary fiscal policy.⁵

The empirical section established three regularities. First, UIP violations matter significantly for the relative attractiveness of U.K. and U.S. government bonds in debt-valuation equations capturing the relative attractiveness of either asset to investors with internationally mobile capital. Second, deficits consistent with market pricing of government debt imply significant exchange rate movements in response to deficit shocks: own-country deficits contribute to depreciations, other-country deficits to appreciations of the local currency. Third, own-country *and* other-country deficits both push up own-country inflation rates significantly and persistently. While we have focused on the bilateral relationship between the U.K. and the U.S., [Hyland and Kawalec \(2026\)](#) rationalises a similar mechanism for a wider subsample of G7 countries using annual data.

The model we develop next aims to rationalise these three facts jointly through two distinct channels: the role of monetary-fiscal interactions for the perceived backing of government debt, and the presence of convenience yields on U.S. government debt.

3 Continuous-Time Model

We develop a continuous-time two-country model to study how fiscal shocks in a financial Hegemon transmit abroad. The countries are labelled H for (financial) *Hegemon*, and F for *Foreign*. At its core, the model highlights how fiscal shocks propagate internationally through (a) exchange rate adjustments that revalue bond holdings, (b) the associated discount rate adjustments, and (c) time-varying convenience yields that affect the relative attractiveness of Hegemon bonds. The model is an enriched two-country New Keynesian model, in which both countries produce and trade differentiated goods, so that fiscal policy affects the Foreign economy not only through asset-pricing channels but also through shifts in aggregate demand. Nominal rigidities in both countries generate a supply-side response to these demand disturbances, giving rise to fiscally driven inflation spillovers. Households and firms in each country transact freely in goods markets, and each economy features its own intermediate and final-good producers operating under Rotemberg price adjustment costs.

⁵In Appendix B we provide additional results based on the [Engel and West \(2005\)](#) reverse-regression approach, showing that exchange rates predict materialised deficits in the U.K. The link between exchange rates and deficits follows an “overshooting” pattern.

Finally, the monetary and fiscal authorities in each country operate under distinct policy regimes. The combination of wealth effects from bond revaluations and expenditure switching, together with the fiscal-monetary policy mix, determines the international transmission of fiscal shocks.

The model introduces two asymmetries between the Hegemon and Foreign economies. First, households in both countries derive *convenience yields* from Hegemon bonds, whereas Foreign government debt provides no such benefit. The second concerns financial market access. Foreign households can hold both Hegemon and Foreign sovereign debt, while Hegemon households hold only Hegemon debt. This second asymmetry is what allows the model to speak to net worth transfers between the two countries rather than only to relative prices, and we return to it in Section 3.1. These two asymmetries capture the Hegemon's privileged position as the issuer of an internationally valued sovereign asset. We therefore interpret country H as the United States.

After log-linearising the model around its stationary steady state and substituting the auxiliary relationships, the competitive equilibrium is reduced to eight first-order equations in eight endogenous variables, driven by seven exogenous state variables. Unless stated otherwise, we study the deterministic perfect-foresight transition path following an unanticipated MIT shock at $t = 0$. After the impact innovation, the subsequent transition path is perfectly foreseen. This formulation keeps the analysis at first order without having to track the variance correction terms, which appears under aggregate risk that is not of measure-zero. Bars denote steady-state values, hats on positive level variables denote log deviations unless explicitly defined otherwise, hats on inflation, interest rates, and $\hat{\pi}_t$ denote level deviations, and starred variables denote Foreign-country quantities. One unit of model time is one quarter.

As is convention, an increase in the nominal exchange rate E_t is a depreciation of the Hegemon currency. The law of one price and the two relative prices used throughout are

$$P_{Ht} = E_t P_{Ht}^*, \quad P_{Ft} = E_t P_{Ft}^*, \quad \text{tot}_t \equiv \frac{P_{Ft}}{P_{Ht}}, \quad Q_t \equiv \frac{E_t P_t^*}{P_t}. \quad (6)$$

An increase in $\text{tot}_t = P_{Ft}/P_{Ht}$ raises the price of Foreign goods relative to Hegemon goods, while an increase in Q_t is a real depreciation of the Hegemon currency. Relative prices are normalised to one in steady state, so lowercase logarithms such as $q_t = \log Q_t$ and $\ell_t = \log \text{tot}_t$ also equal their log deviations from steady state.

3.1 Household problem

Hegemon household. The representative Hegemon household chooses consumption C_t , labour supply N_t , and holdings of (real) Hegemon government debt (b_{Ht}) to maximise

$$\int_0^\infty e^{-\rho t} \left\{ \log C_t - \frac{N_t^{1+\phi}}{1+\phi} + \nu(b_{Ht}, C_t; \theta_t) \right\} dt. \quad (7)$$

The function $\nu(b_{Ht}, C_t; \theta_t)$ captures the convenience services generated by these holdings, with θ_t being an exogenous shifter that governs their time-varying value. The household faces the nominal flow budget constraint

$$P_t C_t + dB_{Ht}/dt = i_t B_{Ht} + W_t N_t + \Pi_t + \Pi_t^l + E_t P_t^* \bar{\mathfrak{Z}} - T_t, \quad \Pi_t^l \equiv \kappa_t B_{Ht}^*. \quad (8)$$

Here, $i_t B_{Ht}$ is interest income, $W_t N_t$ is labour income, Π_t denotes the profits of Hegemon intermediate producers, and T_t denotes nominal net taxes. A lump-sum financial-sector rebate Π_t^l consists of fees collected on Hegemon debt held abroad and is separated from productive profits because it is a financial redistribution that uses no goods. The term $E_t P_t^* \bar{\mathfrak{Z}}$ is the Hegemon-currency value of a constant international transfer received from the Foreign household. Section 3.5 sets $\bar{\mathfrak{Z}}$ to support a symmetric steady state with a non-zero net foreign asset position.

Consumption C_t is a CES aggregate of Hegemon and Foreign goods, with expenditure weight γ on Foreign goods and intratemporal elasticity of substitution η . Cost minimisation over this bundle yields the corresponding demand system and the consumer price index (CPI)

$$P_t = [(1 - \gamma)P_{Ht}^{1-\eta} + \gamma P_{Ft}^{1-\eta}]^{1/(1-\eta)}. \quad (9)$$

We define CPI inflation as $\pi_t \equiv \dot{P}_t/P_t$ and Hegemon producer-price inflation as $\pi_{Ht} \equiv \dot{P}_{Ht}/P_{Ht}$. Appendix C.3 provides the complete intratemporal allocation problem and the associated demand functions.

Market segmentation. We impose asymmetric participation in sovereign-bond markets: Hegemon households hold only Hegemon government debt, whereas Foreign households hold both sovereign bonds. We interpret this restriction as a limiting case in which the Foreign households are the marginal (international) investors in both sovereign-bond markets, while Hegemon investors are marginal only in their domestic market. This asymmetry creates the model's relative-wealth margin. If both households could trade both bonds freely, each would satisfy two bond Euler equations. Differencing these conditions would produce two modified uncovered interest parity relations, tying the effective convenience yields together at every date. Relative consumption would then be determined only up to a constant, and net foreign assets would drop out of the allocation. Removing the Hegemon household's Euler equation for Foreign debt allows the two effective convenience yields to diverge outside the steady state, with their cumulative difference becoming an endogenous state.

Convenience yield. Following Jiang et al. (2023, 2024b), we allow Hegemon government debt to provide liquidity services. These services generates a departure from perfect risk-sharing (Backus and Smith, 1993). We adopt an isoelastic specification in real bond holdings relative to steady

state,

$$\nu(b_{Ht}, C_t; \theta_t) = \frac{\bar{\Psi} \theta_t \bar{b}_H}{(1 - \zeta_H) C_t} \left[\left(\frac{b_{Ht}}{\bar{b}_H} \right)^{1 - \zeta_H} - 1 \right], \quad \zeta_H > 0. \quad (10)$$

Here, $\bar{b}_H$ normalises bond holdings, $\bar{\Psi}$ determines their steady-state marginal convenience value, ζ_H governs the curvature of convenience demand, and θ_t shifts the perceived liquidity services provided by Hegemon debt. The specification is normalised so that the convenience flow is zero in the stationary steady state. Scaling the flow by $1/C_t$ makes convenience services non-separable from consumption and implies $\nu_C = -\nu_t/C_t$, where $\nu_t \equiv \nu(b_{Ht}, C_t; \theta_t)$.

The marginal convenience value, or *convenience yield*, of real Hegemon bond holdings, expressed in consumption units, is

$$\Psi_t \equiv \nu_b C_t = \bar{\Psi} \theta_t \left(\frac{b_{Ht}}{\bar{b}_H} \right)^{-\zeta_H}, \quad (11)$$

where ν_b denotes the derivative of the convenience flow ν with respect to the *real* quantity of debt held. Its effective contribution to the household's pricing kernel (on the Euler equation) is

$$\Phi_t \equiv \frac{\Psi_t}{1 - \nu_t}. \quad (12)$$

The effective convenience yield Φ_t is the non-pecuniary return that lowers the nominal yield required on Hegemon debt. Time variation in the underlying convenience flow also shifts the pricing kernel directly through $\dot{\nu}_t/(1 - \nu_t)$.

Optimality conditions. The first-order condition for Hegemon government debt yields the consumption Euler equation

$$\frac{\dot{C}_t}{C_t} = i_t - \pi_t - \rho + \Phi_t - \frac{\dot{\nu}_t}{1 - \nu_t}. \quad (13)$$

The household's total return on Hegemon debt consists of the real pecuniary return, $i_t - \pi_t$, and the effective convenience yield, Φ_t . A higher convenience yield therefore lowers the nominal interest rate required to support a given consumption path. The final term captures the direct effect of changes in the underlying convenience flow on the marginal utility of consumption. The standard intratemporal conditions, including labour supply, are collected in [Appendix C.3](#).

Foreign household. The Foreign household chooses consumption, labour supply, and holdings of both sovereign bonds. The Foreign household problem is similar to the Hegemon household's, with two important asymmetries. First, foreign households also derive a convenience yield from holding *Hegemon-currency* bonds, not from their own government's debt. Let

$$v_{Ht}^* \equiv \frac{B_{Ht}^*}{E_t P_t^*} \quad (14)$$

denote the Foreign household's Hegemon debt holdings measured in Foreign consumption units. Because Foreign government debt is held domestically, while the Hegemon households do not hold Foreign debt, v_{Ht}^* is also the Foreign economy's net foreign asset position.

The second asymmetry becomes clear in the budget constraint: we allow the Foreign household to hold *both* Hegemon sovereign debt as well as its own. The Foreign budget constraint is therefore given by

$$P_t^* C_t^* + dB_{Ft}^* / dt + d\left(\frac{B_{Ht}^*}{E_t}\right) / dt = \left[i_t^* B_{Ft}^* + \left(i_t - \frac{\dot{E}_t}{E_t} - \kappa_t\right) \frac{B_{Ht}^*}{E_t} + W_t^* N_t^* + \Pi_t^* - T_t^* - P_t^* \bar{\mathcal{Z}}\right]. \quad (15)$$

The pecuniary return on Hegemon debt, measured in Foreign currency, is therefore $i_t - \dot{E}_t / E_t - \kappa_t$. An appreciation of the Hegemon currency raises the Foreign-currency value of a given nominal Treasury position, whereas a depreciation lowers it. The return wedge κ_t is defined below.

The Foreign convenience flow has the same isoelastic form as (10), with v_{Ht}^* replacing domestic Hegemon-bond holdings. The marginal and effective convenience yields are thus given by

$$\psi_t^* = \bar{\Psi}^* \theta_t^* \left(\frac{v_{Ht}^*}{\bar{v}_H^*}\right)^{-\zeta_F}, \quad \Phi_t^* \equiv \frac{\psi_t^*}{1 - \nu_t^*}.$$

Exchange-rate movements consequently affect the Foreign household through two linked margins: they revalue its Treasury wealth and, by changing v_{Ht}^* , alter the marginal convenience value of those holdings. The complete household problem and its optimality conditions are given in Appendix C.4.

The international holding wedge. Foreign holdings of Hegemon debt are subject to an exogenous proportional intermediation wedge κ_t , measured as a rate per unit of the position per unit of time. The Foreign holder receives the net pecuniary return $i_t - \dot{E}_t / E_t - \kappa_t$. The wedge is collected by a consolidated Hegemon financial sector, whose nominal fee-revenue flow is $\Pi_t^I = \kappa_t B_{Ht}^*$, and is rebated lump-sum to the Hegemon household. The financial sector is an accounting device: it does not optimise or use real resources. Thus, κ_t changes the Foreign investor's marginal return without creating an unaccounted loss of world income.

We interpret κ_t as a reduced-form wedge between the gross currency-adjusted return on Hegemon debt and the return received by Foreign investors.⁶ The wedge plays two roles. First, its steady-state value permits domestic and Foreign investors to assign different marginal convenience values to the same Hegemon bond while satisfying their respective Euler equations. Second, following the international financial-shock interpretation in [Itskhoki and Mukhin \(2021\)](#), deviations $\hat{\kappa}_t \equiv \kappa_t - \bar{\kappa}$ represent cross-border financial disturbances that move international portfolios and the exchange rate independently of fiscal and convenience-yield innovations.

⁶The wedge may capture cross-border intermediation spreads, balance-sheet charges, regulatory fees, and hedging-related payments accruing to the Hegemon financial sector.

The Foreign household's Euler equations for the two sovereign bonds are derived in Appendix C.4. Differencing them gives the modified uncovered interest parity condition

$$i_t - i_t^* = \frac{\dot{E}_t}{E_t} + \varkappa_t - \Phi_t^*. \quad (16)$$

For a given path of the exchange rate, a higher Foreign convenience yield Φ_t^* lowers the pecuniary return on Hegemon debt required by the Foreign investor, whereas a higher holding wedge $\varkappa_t$ raises it. The composite wedge $\varkappa_t - \Phi_t^*$ is therefore the model counterpart of the UIP residual φ_t used in Section 2. Equation (16) makes the Foreign household's marginal valuation of Hegemon debt part of exchange-rate determination. This is the channel through which Foreign demand for Hegemon bonds feeds into exchange rate determination and, ultimately, into the cross-country transmission of fiscal shocks.

Steady-state convenience levels. Because both households hold Hegemon debt at an interior optimum, their steady-state pecuniary return must satisfy both households' Euler equations. In a stationary steady state with zero trend inflation, a constant nominal exchange rate, and $\bar{\nu} = \bar{\nu}^* = 0$, the conditions on the steady-state pecuniary return are $\bar{r} = \rho - \bar{\Phi}$ for the Hegemon household and $\bar{r} = \rho^* + \bar{\varkappa} - \bar{\Phi}^*$ for the Foreign household. The Foreign household's Euler equation for its own-bond additionally implies $\bar{r}^* = \rho^*$. Under the maintained restriction that both countries' households are equally patient ($\rho = \rho^*$), it follows that

$$\bar{\varkappa} = \bar{\Phi}^* - \bar{\Phi} = \bar{\Psi}^* - \bar{\Psi}, \quad \bar{r} = \rho - \bar{\Psi}, \quad \bar{r}^* = \rho. \quad (17)$$

The steady-state holding wedge therefore absorbs the difference between the two households' marginal convenience valuations of Hegemon debt. Without this wedge, the two households would price the same instrument at the same effective rate and the two steady-state convenience levels would be forced to coincide.

Combining the household Euler equations with modified UIP yields the model's international risk-sharing condition. Define

$$\Omega_t \equiv \log C_t - \log C_t^* - q_t, \quad \varpi_t \equiv \Omega_t - \log \left(\frac{1 - \nu_t}{1 - \nu_t^*} \right). \quad (18)$$

The first object measures relative consumption taking into account movements in the real exchange rate. The second adjusts this measure for the effect of convenience services on the two households' marginal utilities of consumption. We refer to ϖ_t as the relative-wealth state. Under the maintained restriction $\rho = \rho^*$, this state evolves according to

$$\dot{\varpi}_t = \Phi_t - \Phi_t^* + \varkappa_t. \quad (19)$$

The right-hand side is the difference between the effective valuation assigned by the Hegemon household and the net valuation assigned by the Foreign household, $\Phi_t^* - \varkappa_t$. Differences in these valuations therefore accumulate into persistent movements in relative consumption. Importantly, ϖ_t is not itself a stock of international assets. Its relationship with the Foreign net foreign asset position is determined jointly by bond-market clearing and the international budget constraint developed below. Appendix C.5 derives the underlying risk-sharing condition.

3.2 Production and price setting

In each country, competitive final-good firms aggregate a continuum of differentiated intermediate goods with elasticity of substitution $\varepsilon > 1$. Intermediate producer j operates the linear technology $Y_{Ht}(j) = A_t N_t(j)$ in the Hegemon and $Y_{Ft}^*(j) = A_t^* N_t^*(j)$ in Foreign. These producers are monopolistically competitive and adjust prices subject to Rotemberg costs (Rotemberg, 1982). Let ϑ_H and ϑ_F denote the adjustment-cost parameters and define the corresponding Phillips-curve slopes as $\kappa_H \equiv (\varepsilon - 1)/\vartheta_H$ and $\kappa_F \equiv (\varepsilon - 1)/\vartheta_F$. Appendices C.6 and C.7 provide the resource constraints, firm problems, and pricing derivations.

The distinctive feature of the production block is the discounting of Hegemon firms' nominal profits. Hegemon households own these firms, but equity provides no convenience services. Their stochastic discount factor therefore prices a nominal equity claim using the pecuniary return plus the effective convenience yield, $i_t + \Phi_t$. Foreign firms are owned by Foreign households and discount nominal profits at i_t^* . This asymmetry does not change the steady-state discount coefficient in the linearised Hegemon Phillips curve: $\bar{\gamma} + \bar{\Phi} = \rho$. Under the maintained restriction $\rho = \rho^*$, the coefficient is therefore ρ in both countries. Outside the steady state, convenience yields continue to affect inflation through household discounting and the portfolio-induced demand effects derived below.

3.3 Portfolio revaluation channel

Hegemon government debt is held by both households, whereas Foreign government debt is held only domestically, therefore the bond market clearing conditions are:

$$B_t = B_{Ht} + B_{Ht}^*, \quad B_t^* = B_{Ft}^*.$$

Writing $b_t \equiv B_t/P_t$ and using the definition of v_{Ht}^* in (14), the Hegemon bond market clearing becomes

$$b_t = b_{Ht} + Q_t v_{Ht}^*. \quad (20)$$

The real exchange rate Q_t converts the Foreign household's Treasury position into Hegemon consumption units. Because Treasuries are the Foreign economy's only external asset and Hegemon households hold no Foreign claims, v_{Ht}^* is also the Foreign economy's net foreign asset position.

Throughout, a superscript n denotes the flexible price allocation evaluated at the same states as the sticky price economy. In particular, the flexible price allocation is evaluated at the same productivity levels A_t, A_t^* , liquidity shifters θ_t, θ_t^* , holding wedge $\varkappa_t$, aggregate real debt stocks b_t, b_t^* , net foreign asset position v_{Ht}^* , and the relative-wealth state ϖ_t . It therefore removes nominal rigidity without eliminating the portfolio and convenience-service wedges. We measure activity and relative prices relative to the flexible-price allocation. Define

$$\begin{aligned} x_t &\equiv \log\left(\frac{Y_{Ht}}{Y_{Ht}^n}\right), & x_t^* &\equiv \log\left(\frac{Y_{Ft}^*}{Y_{Ft}^{*n}}\right), \\ \ell_t &\equiv \log \text{tot}_t, & \ell_t^n &\equiv \log \text{tot}_t^n, & \tau_t &\equiv \ell_t - \ell_t^n. \end{aligned} \quad (21)$$

Here, Y_{Ht} and Y_{Ft}^* denote actual output in the Hegemon and Foreign economies, while Y_{Ht}^n and Y_{Ft}^{*n} denote the corresponding outputs in the conditional flexible-price allocation. Thus, x_t and x_t^* are the Hegemon and Foreign output gaps.

The variable ℓ_t is the log level of the terms of trade, $\text{tot}_t = P_{Ft}/P_{Ht}$, while ℓ_t^n is its value in the conditional flexible-price allocation. Consequently, τ_t is the terms-of-trade gap. A positive τ_t means that Foreign goods are more expensive relative to Hegemon goods than in the reference allocation. Unlike the steady-state deviation ℓ_t , the gap τ_t removes the time-varying reference level generated by productivity, fiscal positions, international portfolios, and relative wealth. We derive ℓ_t^n in Section 3.6.

We now show how terms-of-trade movements revalue Foreign-held Hegemon debt and thereby affect convenience yields and international risk sharing. Recall that θ_t and θ_t^* shift the two households' convenience demand for Hegemon debt, b_{Ht} and v_{Ht}^* denote their respective real holdings of that debt (with the latter measured in Foreign consumption units) and Φ_t and Φ_t^* are the corresponding effective convenience yields. Using the hat notation introduced above, write

$$\begin{aligned} \hat{\theta}_t &\equiv \log(\theta_t/\bar{\theta}), & \hat{\theta}_t^* &\equiv \log(\theta_t^*/\bar{\theta}^*), \\ \hat{b}_{Ht} &\equiv \log(b_{Ht}/\bar{b}_H), & \hat{v}_{Ht}^* &\equiv \log(v_{Ht}^*/\bar{v}_H^*), \\ \hat{\Phi}_t &\equiv \log(\Phi_t/\bar{\Phi}), & \hat{\Phi}_t^* &\equiv \log(\Phi_t^*/\bar{\Phi}^*). \end{aligned}$$

To first order, the two effective convenience yields satisfy

$$\hat{\Phi}_t = \hat{\theta}_t - \mathcal{A}_H \hat{b}_{Ht}, \quad \hat{\Phi}_t^* = \hat{\theta}_t^* - \mathcal{A}_F \hat{v}_{Ht}^*, \quad (22)$$

where

$$\mathcal{A}_H \equiv \zeta_H - s_H, \quad \mathcal{A}_F \equiv \zeta_F - s_F, \quad s_H \equiv \frac{\bar{\Psi} \bar{b}_H}{\bar{C}}, \quad s_F \equiv \frac{\bar{\Psi}^* \bar{v}_H^*}{\bar{C}^*}. \quad (23)$$

The coefficients $\mathcal{A}_H$ and $\mathcal{A}_F$ measure the net sensitivity of effective convenience yields to bond holdings. The curvature parameters ζ_H and ζ_F capture diminishing marginal convenience services, while ς_H and ς_F capture the associated first-order effect on the marginal utility of consumption. Thus, when $\mathcal{A}_j > 0$, an increase in a household's Treasury position lowers the effective convenience yield it assigns to an additional unit of Hegemon debt.

To derive the corresponding gap relations, let

$$\omega_H \equiv \frac{\bar{b}_H}{\bar{b}}, \quad \omega_F \equiv \frac{\bar{v}_H^*}{\bar{b}} = 1 - \omega_H$$

denote the shares of Hegemon debt held in steady state by Hegemon and Foreign households. Bond-market clearing therefore requires the Hegemon household's position to absorb the valuation effect generated by the terms-of-trade gap:

$$\hat{b}_{Ht} - \hat{b}_{Ht}^n = -\frac{(1 - 2\gamma)\omega_F}{\omega_H} \tau_t. \quad (24)$$

Combining (24) with the convenience-yield equations with the flexible price allocation gives

$$\Phi_t - \Phi_t^n = \Gamma_\phi \tau_t, \quad \Phi_t^* - \Phi_t^{*n} = 0, \quad \Omega_t - \Omega_t^n = \Gamma_\xi \tau_t, \quad (25)$$

where

$$\Gamma_\phi \equiv \bar{\Phi} \mathcal{A}_H \frac{(1 - 2\gamma)\omega_F}{\omega_H}, \quad \Gamma_\xi \equiv \varsigma_H \frac{(1 - 2\gamma)\omega_F}{\omega_H}. \quad (26)$$

The coefficient Γ_ϕ maps the revaluation-induced portfolio adjustment into the Hegemon effective convenience-yield gap, whereas Γ_ξ maps it into the international risk-sharing gap. Moreover, the change in the convenience component of Hegemon marginal utility appears as a change in Ω_t .

These two coefficients summarise the portfolio revaluation channel in the equilibrium system. The convenience-yield effect Γ_ϕ enters the Hegemon IS curve, while the risk-sharing effect Γ_ξ shifts relative demand and generates the asymmetric terms-of-trade coefficients in the two Phillips curves derived below.

3.4 Aggregate supply

Log-linearising firms' pricing conditions around the conditional flexible-price allocation yields a Phillips curve for each country. The terms-of-trade gap affects marginal costs through two forces. The first is standard expenditure switching: changes in relative goods prices redirect demand across countries. The second is the portfolio revaluation channel derived above: because only the Hegemon household's Treasury position adjusts across the sticky- and flexible-price allocations, the resulting change in relative consumption affects marginal costs asymmetrically. The two Phillips

curves are

$$\dot{\hat{\pi}}_{Ht} = \rho \hat{\pi}_{Ht} - \kappa_H \left\{ (1 + \phi) x_t - [\mathcal{S} + (1 - \gamma) \Gamma_\xi] \tau_t \right\}, \quad (27)$$

$$\dot{\hat{\pi}}_{Ft}^* = \rho \hat{\pi}_{Ft}^* - \kappa_F \left\{ (1 + \phi^*) x_t^* + [\mathcal{S} - \gamma \Gamma_\xi] \tau_t \right\}, \quad (28)$$

where $\mathcal{S} \equiv 2\gamma(1 - \gamma)(\eta - 1)$ is what we label the expenditure-switching coefficient. It summarises how home bias, γ , and the elasticity of substitution between Hegemon and Foreign goods, η , translate a terms-of-trade gap into demand and marginal cost. The coefficient Γ_ξ instead captures the change in relative consumption induced by the revaluation of Foreign-held Hegemon debt. It enters the two Phillips curves with different weights because the associated portfolio adjustment occurs on the Hegemon household's balance sheet. Thus, $\mathcal{S}$ governs the standard trade channel, whereas Γ_ξ governs the additional portfolio channel.

3.5 Policy

We close the model by specifying fiscal and monetary policy in each country. Fiscal authorities adjust primary surpluses in response to outstanding government debt, while central banks set nominal interest rates in response to inflation and output gaps. The interaction between these rules determines how fiscal and convenience-demand disturbances are absorbed through debt valuations, inflation, and international portfolios.

Fiscal rules. Let $s_t \equiv T_t/P_t$ and $s_t^* \equiv T_t^*/P_t^*$ denote real primary surpluses at the Hegemon and Foreign, respectively. Hegemon and Foreign fiscal authorities each face a consolidated nominal flow budget constraint, $\dot{B}_t = i_t B_t - T_t$ and $\dot{B}_t^* = i_t^* B_t^* - T_t^*$. Iterating them forward along the perfect-foresight path and imposing the transversality conditions gives the intertemporal debt valuation relations

$$b_t = \int_t^\infty \exp\left\{-\int_t^u (i_z - \pi_z) dz\right\} s_u du, \quad b_t^* = \int_t^\infty \exp\left\{-\int_t^u (i_z^* - \pi_z^*) dz\right\} s_u^* du. \quad (29)$$

These relations equate the current real value of government debt to the present value of future primary surpluses, discounted at the respective equilibrium real interest rates. Note that convenience yields affect valuation through equilibrium interest rates, and that they do not enter the government budget constraint directly. The effect is nonetheless first order for the Hegemon, since the steady-state real discount rate applied to Hegemon surpluses is $\bar{r} = \rho - \bar{\Phi}$ rather than ρ , so that a given surplus stream supports a larger debt stock. This convenience rent, and not the Phillips curve discount rate, is where the level of $\bar{\Phi}$ matters.

Each fiscal authority chooses a feedback rule for the primary surplus (Leeper, 1991),

$$\hat{s}_t = \psi_b \hat{b}_t - d_t, \quad \hat{s}_t^* = \psi_b^* \hat{b}_t^* - d_t^*, \quad \hat{s}_t \equiv \frac{s_t - \bar{s}}{\bar{b}}, \quad \hat{s}_t^* \equiv \frac{s_t^* - \bar{s}^*}{\bar{b}^*}, \quad (30)$$

where a positive d_t lowers the primary surplus and $\bar{s} = \bar{r}\bar{b}$, $\bar{s}^* = \bar{r}^*\bar{b}^*$ with $\bar{r} = \rho - \bar{\Phi}$ and $\bar{r}^* = \rho^*$. The parameter ψ_b governs fiscal *activeness*. When $\psi_b \geq \bar{r}$ the surplus adjusts sufficiently to stabilise debt and the fiscal authority is *passive*, and when $\psi_b < \bar{r}$ the government commits to deficit finance and the fiscal authority is *active*. The Foreign block is symmetric.

Monetary rules. Both central banks follow standard continuous-time Taylor rules with constant intercepts,

$$i_t = \bar{i} + \phi_\pi \hat{\pi}_t + \phi_x x_t, \quad i_t^* = \bar{i}^* + \phi_\pi^* \hat{\pi}_t^* + \phi_x^* x_t^*, \quad (31)$$

where $\hat{\pi}_t$ and $\hat{\pi}_t^*$ denote CPI inflation relative to the zero inflation target, and x_t and x_t^* are the respective output gaps. The intercepts are the steady-state nominal rates, $\bar{i} = \rho - \bar{\Phi}$ and $\bar{i}^* = \rho^*$.

We hold these intercepts fixed rather than indexing them to the time-varying natural rates. A constant intercept rule is closer to the specification the empirical Taylor rule literature estimates, and it avoids requiring the central bank to observe and respond to the drift of the terms of trade gap. The monetary authority is considered *active* when $\phi_\pi > 1$ under $\phi_x = 0$ (the Taylor principle holds) and *passive* when $\phi_\pi < 1$ under $\phi_x = 0$.

Fiscal and monetary policy jointly determine how the government valuation conditions in (29) are satisfied. Under an active-monetary, passive-fiscal regime, monetary policy stabilises inflation while primary surpluses adjust to stabilise debt. Under an active-fiscal, passive-monetary regime, future surpluses do not adjust sufficiently, so the price level moves to reconcile the real value of nominal debt with the present value of surpluses. The quantitative analysis considers both symmetric and asymmetric combinations of these policy regimes across the two countries.

The international budget constraint. Because Hegemon government debt is the Foreign economy's only external asset, its international budget constraint determines the evolution of v_{Ht}^* . Consolidating the Foreign household and government budget constraints yields the Foreign balance of payments. This is the independent household accounting condition not implied by Walras' law, which, together with the international risk-sharing condition, closes the paths of net foreign assets and relative consumption. Appendix C.15 provides the nonlinear constraint and its linearisation.

A constant international transfer allows the model to sustain a positive steady-state Foreign asset position while preserving a symmetric real allocation. Specifically, we set

$$\bar{\mathfrak{T}} = \bar{r}_v \bar{v}_H^*, \quad \bar{r}_v \equiv \rho^* - \bar{\Phi}^*,$$

where $\bar{r}_v$ is the steady-state net real return received by Foreign holders of Hegemon debt. The

transfer offsets steady-state net investment income, permitting $\bar{v}_H^* > 0$ together with

$$\bar{\text{tot}} = \bar{Q} = 1, \quad \bar{C} = \bar{C}^* = \bar{Y}_H = \bar{Y}_F^* = \bar{Y}, \quad \bar{\Omega} = 0.$$

Because the transfer is constant, it affects the steady-state accounting but introduces no additional first-order disturbance.

The linearised balance of payments is then given by

$$\hat{v}_{Ht}^* = \bar{r}_v \hat{v}_{Ht}^* + (\phi_\pi - 1) \hat{\pi}_{Ht} + \phi_x x_t + [\gamma \phi_\pi - (1 - \gamma)] \dot{\ell}_t - \hat{\kappa}_t + \left(\frac{\bar{v}_H^*}{\bar{Y}} \right)^{-1} [-\mathcal{S} \ell_t + \gamma \Omega_t].$$

$\hat{\kappa}_t$ lowers the net return received by Foreign investors. The final term is the Foreign trade balance, in which $-\mathcal{S} \ell_t$ captures expenditure switching induced by relative goods prices, while $\gamma \Omega_t$ captures the effect of relative consumption demand.

Equation (32) completes the model's international balance-sheet feedback. An increase in Hegemon debt changes the Treasury positions that clear the bond market, and the resulting changes in convenience valuations and relative wealth affect the trade balance and hence the subsequent path of Foreign net assets. A disturbance to κ_t affects this adjustment both by changing the return on Foreign-held Treasuries and, through (19), by shifting the relative-wealth state directly.

3.6 Aggregate dynamics

We now collect the mappings needed to express the model's aggregate dynamics. The model combines standard open-economy adjustment with the portfolio revaluation channel derived above. The coefficients Γ_ϕ and Γ_ξ map the revaluation of Foreign-held Hegemon debt into, respectively, the Hegemon convenience-yield gap and the international risk-sharing gap. Because the conditional flexible-price allocation inherits the fiscal and international-wealth states, these states also affect the natural terms of trade and the natural real rate.

Define productivity deviations by $a_t \equiv \log(A_t/\bar{A})$ and $a_t^* \equiv \log(A_t^*/\bar{A}^*)$, and aggregate debt deviations by $\hat{b}_t \equiv \log(b_t/\bar{b})$ and $\hat{b}_t^* \equiv \log(b_t^*/\bar{b}^*)$. The following auxiliary relations determine the natural terms of trade, recover their actual level and the Hegemon household's Treasury position, and link the two countries' output gaps:

$$\ell_t^n = \Lambda_n^{-1} [(a_t - a_t^*) + L_b \hat{b}_t + L_v \hat{v}_{Ht}^* - \Delta \varpi_t], \quad (32)$$

$$\begin{aligned} \ell_t &= \ell_t^n + \tau_t, & \hat{b}_{Ht} &= \omega_H^{-1} [\hat{b}_t - \omega_F \hat{v}_{Ht}^* - \omega_F (1 - 2\gamma) \ell_t], \\ x_t^* &= x_t - [1 + 2\mathcal{S} + (1 - 2\gamma) \Gamma_\xi] \tau_t. \end{aligned} \quad (33)$$

The supply-side weights are $\lambda_H \equiv 1/(1 + \phi)$ and $\lambda_F \equiv 1/(1 + \phi^*)$. The remaining composite

coefficients are

$$\Delta \equiv (1 - 2\gamma) + \gamma(\lambda_H + \lambda_F), \quad \Lambda_n \equiv 1 + 2\mathcal{S} - \mathcal{S}(\lambda_H + \lambda_F) + (\Delta - \lambda_H)\Gamma_\xi,$$

and

$$L_b \equiv \frac{(\Delta - \lambda_H)\varsigma_H}{\omega_H}, \quad L_v \equiv (\lambda_F - \Delta)\varsigma_F - L_b\omega_F.$$

The coefficient Δ governs the effect of relative wealth on the natural terms of trade, while Λ_n scales its response to the underlying fundamentals. The coefficients L_b and L_v measure the effects of aggregate Hegemon debt and Foreign Treasury holdings, respectively.

Let $\hat{r}_t^n \equiv r_t^n - \bar{r}$ denote the deviation of the Hegemon natural real rate associated with its own government bond. Convenience yields enter this natural rate directly. They do not enter the corresponding Foreign natural rate because Foreign government debt provides no convenience services.

Finally, define the log nominal exchange rate by $\hat{e}_t \equiv \log(E_t/\bar{E})$. To first order, the price indices relate CPI inflation and nominal exchange-rate depreciation to producer-price inflation and terms-of-trade growth according to

$$\hat{\pi}_t = \hat{\pi}_{Ht} + \gamma\dot{\ell}_t, \quad \hat{\pi}_t^* = \hat{\pi}_{Ft}^* - \gamma\dot{\ell}_t, \quad \hat{e}_t = \dot{\ell}_t - \hat{\pi}_{Ft}^* + \hat{\pi}_{Ht}. \quad (34)$$

3.7 Equilibrium

The model is driven by seven exogenous states: productivity in each country, the two convenience-demand shifters, the two fiscal disturbances, and the cross-border holding wedge. Between innovations, these states revert deterministically toward their steady-state values:

$$\begin{aligned} \dot{a}_t &= -\varphi_a a_t, & \dot{a}_t^* &= -\varphi_a^* a_t^*, & \dot{\hat{\theta}}_t &= -\varphi_\theta \hat{\theta}_t, & \dot{\hat{\theta}}_t^* &= -\varphi_\theta^* \hat{\theta}_t^*, \\ \dot{d}_t &= -\varphi_d d_t, & \dot{d}_t^* &= -\varphi_d^* d_t^*, & \dot{\hat{\kappa}}_t &= -\varphi_\kappa \hat{\kappa}_t. \end{aligned} \quad (35)$$

The coefficients $\varphi_j > 0$ govern the corresponding speeds of mean reversion.

We collect these states in the vector

$$z_t \equiv (a_t, a_t^*, \hat{\theta}_t, \hat{\theta}_t^*, d_t, d_t^*, \hat{\kappa}_t)',$$

and compute impulse responses to an unanticipated one-time (MIT) innovation to component j , imposed at $t = 0$ as

$$z_{0+} = z_{0-} + \varepsilon e_j, \quad (36)$$

where e_j is the j th unit vector and ε is the size of the innovation. Conditional on this impact, no further innovations occur. If the economy is initially in its stationary steady state, so that $z_{0-} = 0$,

the shocked component subsequently follows

$$[z_t]_j = \varepsilon e^{-\varphi_j t}, \quad t \geq 0.$$

Formally, we define a *competitive equilibrium* as a collection of paths for $\{x_t, \hat{\pi}_{Ht}, \hat{\pi}_{Ft}^*, \tau_t, \hat{b}_t, \hat{b}_t^*, \hat{v}_{Ht}^*, \varpi_t\}$, conditional on the exogenous processes in (35) and the impact condition (36). These paths are piecewise continuously differentiable away from the impact date. Nominal debt positions are predetermined at impact, whereas the nominal exchange rate and forward-looking variables may jump.

For $t > 0$, the equilibrium is characterised by eight differential equations: the Hegemon IS curve, the two Phillips curves, the terms-of-trade equation, the two government debt laws, the Foreign balance of payments, and the relative-wealth law of motion:

$$\dot{x}_t = (\phi_\pi - 1)\hat{\pi}_{Ht} + \phi_x x_t + \Gamma_\phi \tau_t - \hat{r}_t^n + [\mathcal{S} + (1 - \gamma)\Gamma_\xi + \gamma\phi_\pi]\dot{\tau}_t + \gamma(\phi_\pi - 1)\dot{\ell}_t^n, \quad (37)$$

$$\dot{\hat{\pi}}_{Ht} = \rho\hat{\pi}_{Ht} - \kappa_H \left\{ (1 + \phi)x_t - [\mathcal{S} + (1 - \gamma)\Gamma_\xi]\tau_t \right\}, \quad (38)$$

$$\dot{\hat{\pi}}_{Ft}^* = \rho\hat{\pi}_{Ft}^* - \kappa_F \left\{ (1 + \phi^*)x_t^* + [\mathcal{S} - \gamma\Gamma_\xi]\tau_t \right\}, \quad (39)$$

$$[1 - \gamma(\phi_\pi + \phi_\pi^*)]\dot{\ell}_t = (\phi_\pi - 1)\hat{\pi}_{Ht} - (\phi_\pi^* - 1)\hat{\pi}_{Ft}^* + \phi_x x_t - \phi_x^* x_t^* + \bar{\Phi}^*(\hat{\theta}_t^* - \mathcal{A}_F \hat{v}_{Ht}^*) - \hat{\chi}_t, \quad (40)$$

$$\dot{\hat{b}}_t = (\bar{r} - \psi_b)\hat{b}_t + (\phi_\pi - 1)\hat{\pi}_{Ht} + \gamma(\phi_\pi - 1)\dot{\ell}_t + \phi_x x_t + d_t, \quad (41)$$

$$\dot{\hat{b}}_t^* = (\bar{r}^* - \psi_b^*)\hat{b}_t^* + (\phi_\pi^* - 1)\hat{\pi}_{Ft}^* - \gamma(\phi_\pi^* - 1)\dot{\ell}_t + \phi_x^* x_t^* + d_t^*, \quad (42)$$

$$\dot{\hat{v}}_{Ht}^* = \bar{r}_v \hat{v}_{Ht}^* + (\phi_\pi - 1)\hat{\pi}_{Ht} + \phi_x x_t + [\gamma\phi_\pi - (1 - \gamma)]\dot{\ell}_t - \hat{\chi}_t + \left(\frac{\bar{v}_H^*}{\bar{Y}} \right)^{-1} [-\mathcal{S}\ell_t + \gamma\Omega_t], \quad (43)$$

$$\dot{\varpi}_t = \bar{\Phi}[\hat{\theta}_t - \mathcal{A}_H \hat{b}_{Ht}] - \bar{\Phi}^*[\hat{\theta}_t^* - \mathcal{A}_F \hat{v}_{Ht}^*] + \hat{\chi}_t. \quad (44)$$

We impose the parametric restriction $1 - \gamma(\phi_\pi + \phi_\pi^*) \neq 0$. The mappings in (22), (32), (33) and (34) then close the system. Equilibrium paths also satisfy the household and government transversality conditions, as is standard. Because the exogenous disturbances are transitory, admissible paths remain bounded and converge to the stationary steady state as $t \rightarrow \infty$.

3.8 Parametrization

Before turning to the data-driven estimation in Section 4, we first parametrise the model for an illustrative analysis of the core mechanisms, in particular, the role of convenience yields and the

dependence of fiscal contagion on the policy mix. Table 1 summarises the baseline parametrisation. The left panel specifies four policy configurations that we use throughout the next section, while the right panel lists the remaining structural parameters.

					Param.	Value	Targets/Sources
					κ	0.019	Hazell et al. (2022)
					$\phi = \phi^*$	2.00	Standard value
					$\delta = \delta^*$	0.005	Standard quarterly discount factor
					$\rho = \rho^*$	0.005	Standard quarterly discount factor
					γ	0.25	(Inverse of) home bias
					η	6	Bajzik et al. (2020)
					$\bar{\Psi}$	0.0002	Conv. yield steady-state at Hegemon
					$\bar{\Psi}^*$	0.0001	Conv. yield steady-state at Foreign
					ζ_H	2	Convenience yield sensitivity Hegemon
					ζ_F	8	Convenience yield sensitivity Foreign
					ω_H	0.68	U.S. Treasury (2025)
					B/Y	1.07	Debt-to-GDP ratio in the U.S. in 2025
					B^*/Y^*	1.2	Debt-to-GDP ratio in the U.K. in 2025
					$\bar{E}$	0.8	Nominal exchange rate, assumed

Param.	AM/PF	AF/PM	HD	H-AF/ F-AM
ϕ_π	1.5	0.0	1.5	0.0
ϕ_π^*	1.5	0.7	0.7	1.5
ϕ_x	0.0	0.0	1.0	0.0
ϕ_x^*	0.0	0.0	0.0	0.0
ψ_b	1.0	-0.2	-0.2	-0.2
ψ_b^*	1.0	-0.2	1.0	1.0

(a) Policy scenarios

(b) Baseline calibration

Table 1: Calibration: policy scenarios and baseline parameters.

The four policy configurations capture distinct points in the space of fiscal-monetary regimes: a monetary-led mix at both countries (AM/PF), a fiscally-led mix at both (AF/PM), Hegemon-dominant policy (HD), and a mixed case in which the Hegemon runs deficits while the Foreign central bank fights inflation aggressively (H-AF/F-AM). The first column describes the *monetary-led policy mix* (AM/PF), in which both monetary authorities satisfy the Taylor principle so that inflationary pressure is met with a tightening of the real interest rate. This behaviour induces an explosive root in the evolution of inflation, requiring each fiscal authority to cover deviations of debt from steady state through corresponding adjustments to lump-sum taxes or transfers, and hence to the primary surplus. We therefore set $\psi_b \geq \rho$ and $\psi_b^* \geq \rho^*$. The second column describes the *fiscally-led policy mix* (AF/PM), in which fiscal policy is expansionary in response to deviations of the debt stock from steady state and taxes are not raised to cover deficits. To ensure non-explosive behaviour of nominal variables, monetary policy accommodates fiscal policy by not raising the real interest rate in response to inflationary pressure, indicated by $\phi_\pi < 1$ and $\phi_\pi^* < 1$. The third column describes *Hegemon Dominance* (HD), in which the Hegemon monetary *and* fiscal authorities are both active while both Foreign authorities are passive, and the Hegemon monetary authority additionally also takes into account the output gap in its policy rule as part of its aggressive stance. The fourth column considers the mixed case (H-AF/F-AM), in which the Hegemon fiscal authority and the Foreign monetary authority are active while the Hegemon monetary authority and the Foreign fiscal authority are passive. This configuration reflects a scenario in which the Hegemon commits to deficit finance while the Foreign monetary authority responds aggressively to

inflationary threats.

The right panel summarises the non-policy parameters. Most deep parameters, such as the slope of the Phillips curve and the Frisch elasticity of labour supply, are set to standard values from [Hazell et al. \(2022\)](#) and [Bajzik et al. \(2020\)](#). Empirically disciplined parameters, such as the debt-to-GDP ratio and the share of bonds held abroad, are set to their post-2010 means or their 2024 levels in the U.S. (the financial Hegemon H) and the U.K. (the Foreign country F). The convenience yield level and shape parameters are chosen to capture two empirical regularities documented in the literature on safe-asset demand ([Du et al., 2018](#); [Jiang et al., 2024b](#)): convenience yields are small as a share of overall utility but higher at the Hegemon, reflecting the deeper demand for the liquidity of U.S. Treasury markets in the U.S., and the convenience-yield function is relatively sensitive to changes in bond supply. These calibrated values are fully consistent with our headline estimates reported in Section 4.

With this calibration at hand, we examine the model-implied responses to changes in Hegemon convenience yields and Hegemon fiscal policy, conditioning on the prevailing fiscal-monetary policy mix. The conditioning matters because, as we will see, the policy mix determines the strength of relative wealth effects in each country in response to demand-side shocks.

4 Quantitative Results: Fiscal Spillovers

In Section 4.1, we simulate the response of the model to persistent shocks to Hegemon convenience yields and Hegemon fiscal transfers, conditioning on each of the four policy mixes. Section 4.2 decomposes the transmission of a Hegemon fiscal shock into the convenience-yield channel, the policy-mix channel, and the residual cross-country wealth effects. Section 4.3 then estimates the model on the last thirty years of U.K. and U.S. data and delivers our headline result: the post-2021 U.S. fiscal expansion raised the U.K. price level by roughly two percentage points. We generally set dt equal to 1 quarter in the following analysis.

4.1 Simulation

4.1.1 Convenience Yield Shocks

We first consider shocks to the convenience yield on Hegemon bonds, captured by an exogenous change to θ . Figure 5 shows the response of domestic and Foreign output gaps, inflation, terms of trade, and the nominal exchange rate to a temporary but persistent 1 basis point *decrease* in the convenience yield on Hegemon bonds enjoyed by Hegemon households.

Under the monetary-led policy mix, the shock has no effect on the terms of trade gap, but it directly works as a negative demand disturbance both at the Hegemon and at the Foreign

Impulse-responses: Shock to Hegemon Convenience Yield

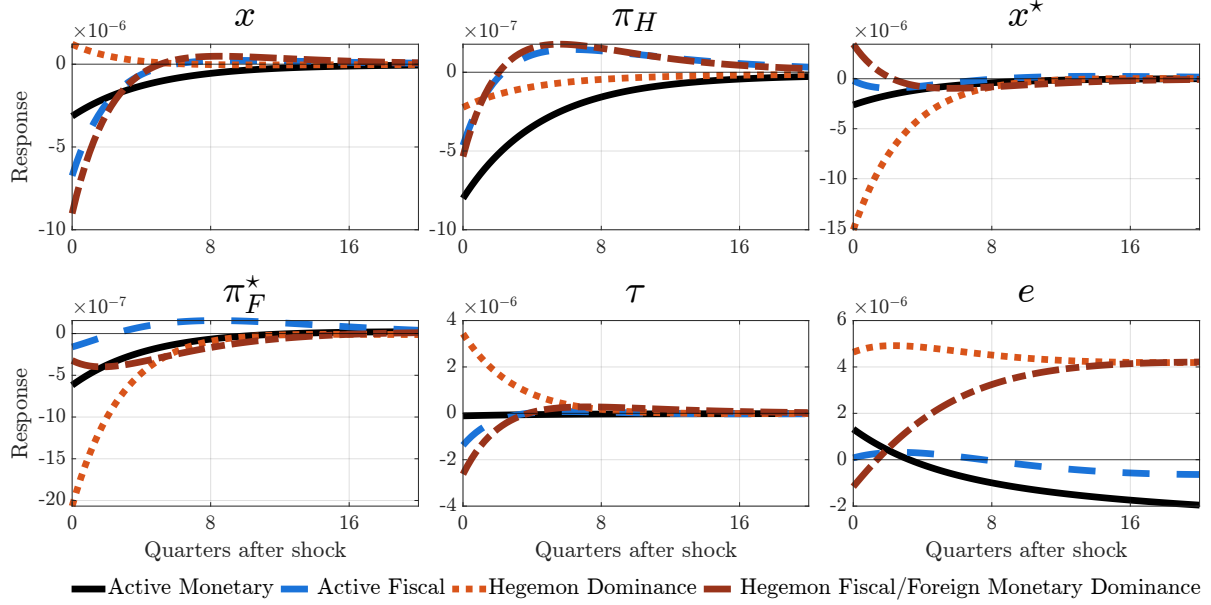


Figure 5: Impulse response functions following a Hegemon convenience yield shock.

economy, paired with a slow appreciation of the Hegemon currency. The loss of convenience value translates into a wedge in the Euler equations of both households, which generally reduce fiscal space that must be recouped through taxation, reducing the present budget of the household at the Hegemon. The Foreign household only experiences the Euler equation wedge, which also dis-incentivizes consumption today.

Under the fiscally-led policy mix, the decrease in convenience yields operates as a negative demand shock at the Hegemon: the output gap and inflation fall domestically, however, their reaction is comparatively muted in Foreign (with the output gap barely moving and inflation falling only slightly below steady-state before increasing in a hump-shape), consistent with a relative shift of wealth. That relative wealth shift comes about since Hegemon households are the majority holders of Hegemon government debt, which has declined in relative value. The terms of trade and the exchange rate appreciate, because the high elasticity of substitution leads Hegemon households to substitute toward Hegemon goods, preventing a favourable terms-of-trade shift for Foreign.

Under Hegemon Dominance, the dominant country attempts to export the entire shock through a depreciation of its own currency, nullifying most of the effects on output and inflation. This depreciation is accommodated by a central bank that preserves the attractiveness of Hegemon government bonds by adjusting the interest rate accordingly, retaining Hegemon bonds as sufficiently attractive for Foreign households. Under the mixed-policy configuration (H-AF/F-AM), the Foreign monetary authority's restrictive stance insulates the Foreign economy from the detrimental real effects of the shock, so that outcomes approximate the fiscally-led case but with the allocative effects on Foreign avoiding recessionary dynamics.

4.1.2 Fiscal Transfer Shocks

Figure 6 presents the consequences of an expansionary Hegemon fiscal transfer shock. As is to be expected, the shock generates no real effects under the monetary-led policy mix, since Ricardian Equivalence holds. Under the fiscally-led mix, the fiscal expansion is not expected to be met by a future increase in Hegemon taxes, which boosts Hegemon household wealth: the government's obligations are not expected to be covered by future surpluses, and Hegemon households feel richer. Foreign households, by contrast, face additional pressure on their holdings of Hegemon government bonds, since those bonds are not backed by a commensurate stream of future surpluses, depressing their currency-adjusted value as the Hegemon currency depreciates. The fiscal expansion at the Hegemon thus translates into a positive demand shock domestically and a negative demand shock abroad, although - and this is crucial - the negative demand impulse *still* induces inflationary pressure, since the backing of Foreign debt is also reduced as the Hegemon bond holdings of Foreign households are worth less, depressing Foreign household total net value and thereby worsening the degree of taxes that can be raised. Simultaneously, the Foreign monetary authority must engage in slight monetary tightening to ensure market-clearing, further raising the net outlays on Foreign debt and worsening the Foreign fiscal stance. All this is paired with the terms-of-trade gap appreciation from the perspective of Foreign, which ensures market clearing and generates further price pressure on Foreign PPI. These are the mechanisms by which “Our” deficit at the Hegemon becomes “Your” problem in Foreign.

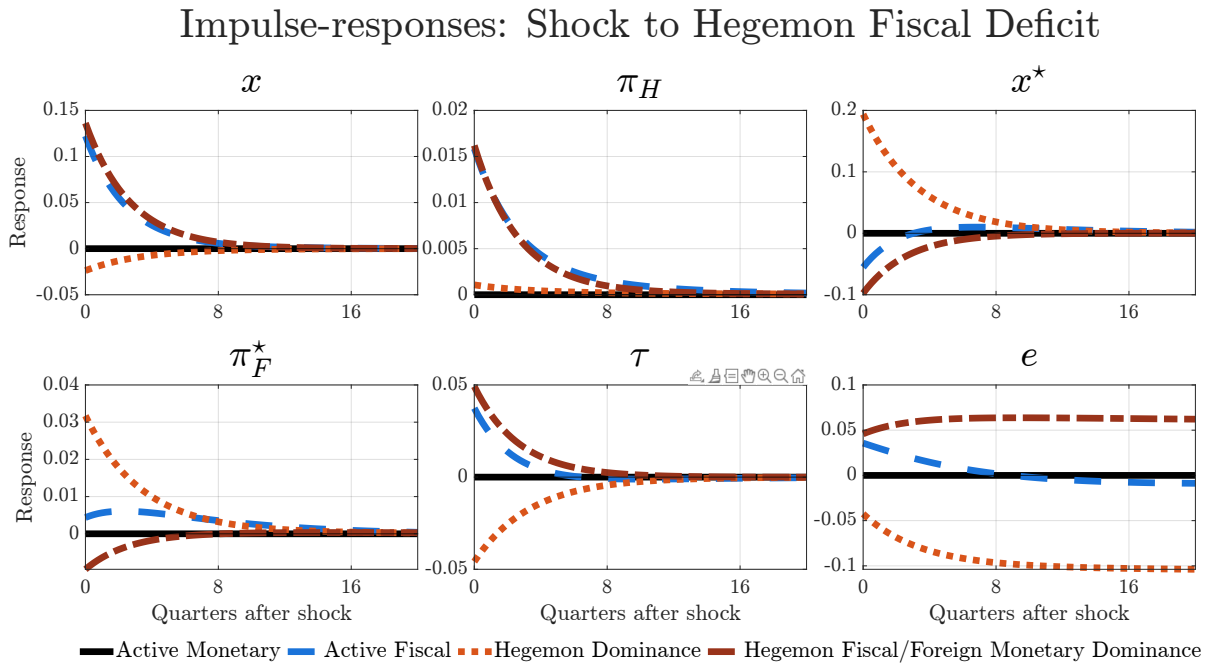


Figure 6: Impulse response functions following an expansionary Hegemon fiscal transfer shock.

Under Hegemon Dominance, the aggressive monetary stance at the Hegemon ensures that the

fiscal expansion does barely generate an output gap or producer-side price pressure domestically. As interest rates rise significantly, the Hegemon currency appreciates, but Hegemon households do not translate this into a real or nominal expansion because the interest rate tightening offsets the fiscal stimulus. Foreign households, however, benefit from the appreciation of their Hegemon bond holdings in terms of their own currency as the Hegemon currency appreciates, contributing to a wealth-driven expansion abroad, which translates into effectively a positive demand shock at Foreign. Under the mixed-policy configuration, the Hegemon fiscal expansion continues to generate a positive domestic demand at the Hegemon shock alongside a more persistent depreciation. This too can be a problem for Foreign when the inflationary pressure leads to welfare losses through a revaluation of their international asset holdings. We consider this to be a possibly relevant case in our analysis of the historical data below.

4.1.3 The Role of Convenience Yield Levels

Both the level and the shape of the convenience yield function matter significantly for model dynamics, operating through changes to the non-pecuniary value of government bonds as well as through the budget path itself. In the baseline parametrisation, an increase in bond supply decreases the convenience yield earned by households in both countries symmetrically.

We now investigate what happens when the convenience yield is several orders of magnitude larger, setting $\bar{\Psi} = \bar{\Psi}^* = 0.1$. We keep the rest of the model unchanged, including the elasticity of the convenience yield functions, placing us in a regime in which both households derive *substantial* utility from Hegemon bonds.⁷ We view this specification as an illustrative exercise of the power of higher convenience yield levels overall, in a bid to showcase their ability to overturn canonical wisdom on the dynamics of international macroeconomic models all on their own.

While the model dynamics in the baseline parametrisation are fairly standard, high levels of convenience yields overturn some of the previous results. We again first analyse the response of the economy to a decrease in convenience yields. Figure 7 presents the responses to a decrease in Hegemon convenience yields.

With higher steady-state convenience flows, policy generally cannot fully nullify the effects of the unexpected change to the perceived convenience benefits inherent to Hegemon bonds, at least within the Hegemon. Because these benefits are quite significant, households will necessarily engage in substitution across both types of bonds and consumption over time. The decrease in convenience yields amounts to a tightening of the Hegemon fiscal position, since budget balance requires more stringent fiscal policy when households in both countries are temporarily less willing to pay a premium to hold Hegemon government debt, reducing Hegemon government windfall gains from issuing bonds.

⁷We also set the rate of time preference, ρ , to 0.11 such that the steady-state nominal rate remains positive.

Impulse-responses: Shock to Hegemon Convenience Yield

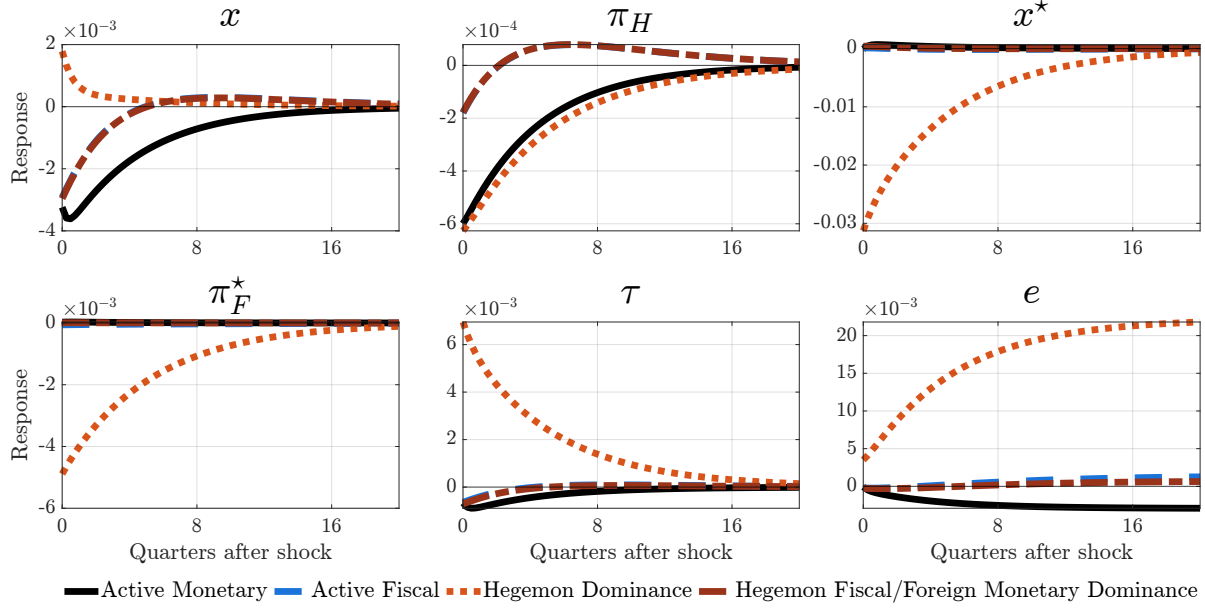


Figure 7: Impulse response functions following a Hegemon convenience yield shock under a higher level of convenience yields.

Under the monetary-led mix, the central bank accommodates by easing, reducing the future fiscal burden. This reduces the income earned by Hegemon households on the debt it holds, leading to a negative overall change in its wealth levels. The Foreign economy, however, does overall stay neutral: the output gap and inflation barely move. Intuitively, since *only* the Hegemon bond is traded internationally, the substitution between Hegemon and Foreign debt is only materializing for Foreign households. Here, joint fiscal-monetary coordination, amounting to a tightening of the rate of return earned on Foreign bonds paired with slight tax easing can nullify the real effects from the swap of Foreign household demand towards Foreign assets. Despite the initial loss of confidence in the Hegemon fiscal position and the subsequent monetary reaction, the terms of trade and the exchange rate appreciate from the Hegemon's perspective, since the policy reactions from the two policy authorities still facilitate an overall increase in weighted return flows Φ_t in general equilibrium, improving the net asset balance of the Hegemon.

Under Hegemon dominance, the quantitative effects on output and inflation at Foreign are significantly larger than under any of the other policy specifications: here, the shock to the Hegemon-bond convenience yield acts clearly akin to a negative demand disturbance, with the output gap and inflation falling below their steady-state levels. This happens despite the appreciation of the Hegemon currency. The decrease in Hegemon inflation, coming from the negative wealth effect on Hegemon households, is paired with a monetary loosening at the Hegemon, while the policy rate stays fairly flat at Foreign. Foreign households see this as a good opportunity to invest in Foreign government bonds, depressing current output through a substitution effect and also curbing

inflationary pressure through the NKPC.

The Hegemon Fiscal/Foreign Monetary dominance configuration coincides almost exactly with the fiscally-led mix, highlighting how the Hegemon fiscal authority, who issues the global quasi-safe asset, is the major determinant of the scope of the responses in the aggregate economy when convenience yields are sufficiently high. In these cases, Hegemon government bonds remain attractive due to the higher relative real interest paid, increasing the Hegemon households' propensity to save in these bonds, acting therefore as a temporary negative demand disturbance.

We next turn to the effects of an expansionary Hegemon government transfer shock under the high-convenience-yield parametrisation. Figure 8 presents the results.

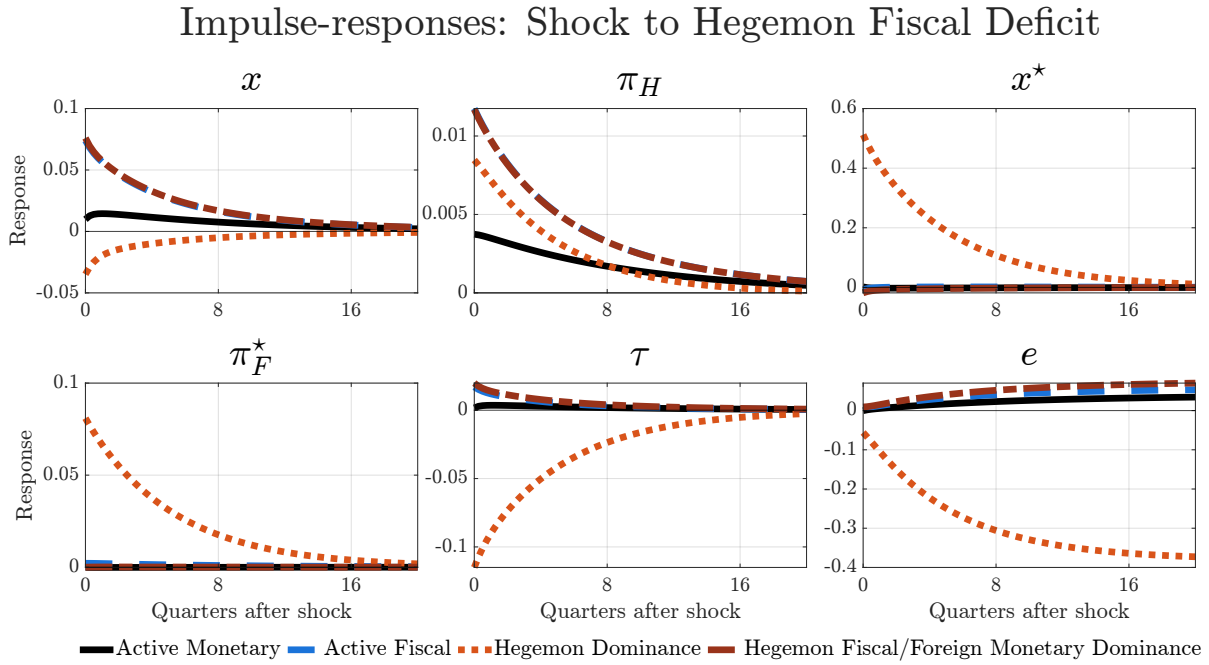


Figure 8: Impulse response functions following an expansionary Hegemon government transfer shock under a higher level of convenience yields.

Consider first the monetary-led case. The monetary authorities cannot fully neutralise the effect of the Hegemon fiscal expansion: because the initial convenience yield level is so high, Hegemon households in particular must be incentivized to take up the additional bond supply without losing utility, which under the monetary-led policy mix occurs naturally as monetary policy strives to fight off the fiscal expansion through a real tightening. However, the Hegemon currency nonetheless depreciates due to the pressure of Foreign households substituting relatively away from Hegemon debt that delivers lower convenience flows. Foreign households are comparatively insulated from the shock, since their own fiscal stance does not change because Hegemon households end up not engaging in substitution to the Foreign savings asset, which they cannot access.

Under the fiscally-led mix, the fiscal transfer measure is expansionary at the Hegemon and leads to no change in the nominal and real outcomes at Foreign. Relative to the case with low steady-

state convenience yields, the major change is the insulation of the Foreign economy: because Hegemon government bonds are perceived as so useful in steady-state for both countries, the Foreign economy is effectively insulated against revaluation pressure through the appreciation of its own currency, which it can afford by being able to substitute away more in relative terms from Hegemon debt. Interestingly, such high convenience yields limit the degree of fiscal contagion the model is designed to isolate under most parametrizations of fiscal and monetary policy. Only under Hegemon Dominance, we observe some contagion as a type of *exported fiscal inflation*, induced through a positive wealth effect on Foreign households as they perceive additional wealth due to the concurrent appreciation of their Hegemon-currency denoted bond holdings, alongside an increase in interest outlays for Foreign bonds because of a no-arbitrage argument needed to clear both bond markets, which rationalizes the inflationary pressure simultaneously on the fiscal side through discount rate contagion.

Finally, under the mixed-policy configuration, the Foreign monetary authority is again able to fully neutralise the output and inflation effects of the Hegemon fiscal expansion on the Foreign economy. The aggressive monetary stance at Foreign acts primarily as a credible threat against an inflationary fiscal stance at Foreign itself, neutralising the wealth effects of Foreign bonds, which constitute the majority of Foreign household savings. This exercise highlights the importance of Foreign authorities coordinating to keep their own bonds attractive in response to a Hegemon fiscal expansion, though the optimal coordination depends on the shape and slope of the convenience flows associated with sovereign debt in both countries.

4.2 Decomposing the Spillover Channels

Our exercises so far have shown that the model gives rise to international fiscal contagion: fiscal expansions and deteriorations in the perceived convenience benefits of Hegemon bonds both contribute to exchange rate movements and to inflation or deflation in Foreign, depending on the policy configuration. We now want to disentangle these channels in greater detail.

We are interested in the extent to which the various international margins of the model contribute to the international spillover of Hegemon inflation. Since we work with a log-linear model, we can characterise this neatly. For each of the four calibrations of interest,⁸ we decompose the total response of three endogenous variables of interest (Hegemon inflation π_H , Foreign inflation π_F^* , and the nominal exchange rate e) into four channels: the convenience-yield channel, an international portfolio adjustment channel and its associated wealth effects, the role played by Home bias in shaping goods baskets, and a remaining term that captures *discount rate adjustments attributable to the monetary and fiscal policy choice*. We perform this decomposition by shutting

⁸Active Monetary in both - H-M, F-F; Active Fiscal in both - AF; Hegemon Dominance - HD; Hegemon-Active Fiscal, Foreign-Active Monetary - H-F, F-M.

off all channels except the one of interest, whose contribution is then given by the depicted bar. Figure 9 shows the responses for the baseline case with low convenience-yield steady states.⁹

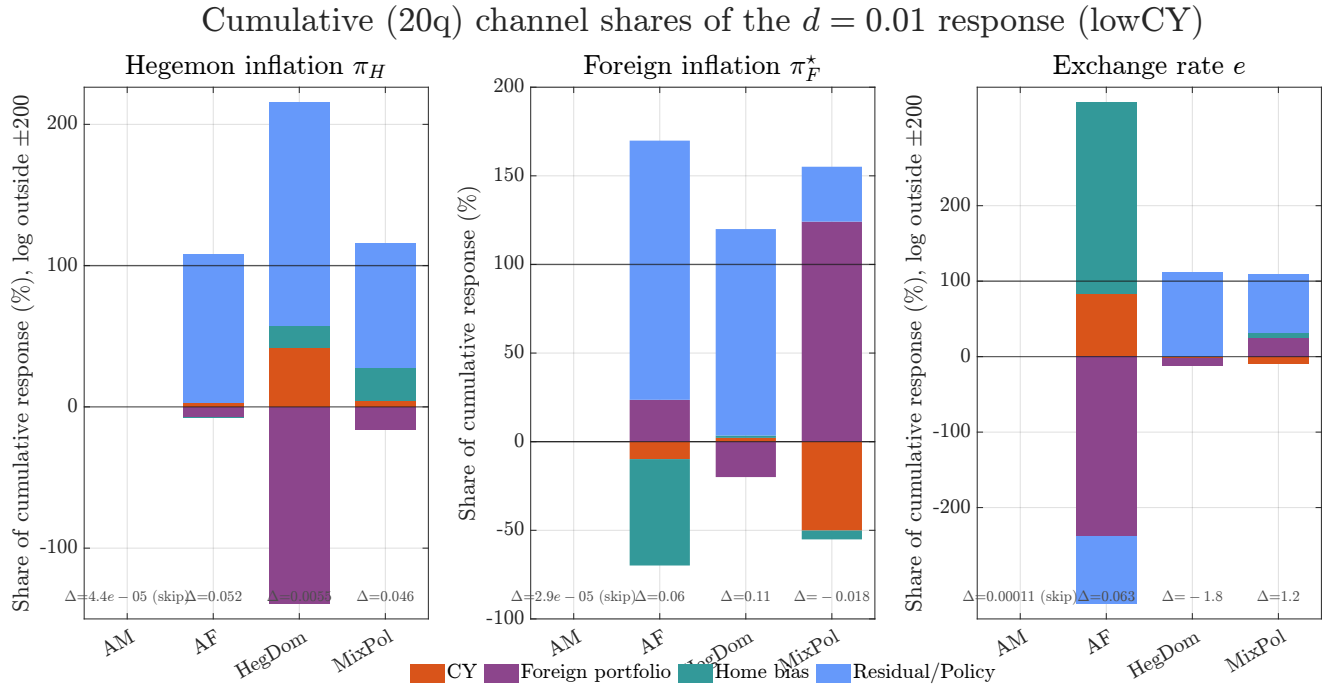


Figure 9: Decomposition of the total response of π_H , π_F^* , and e in response to a Hegemon government transfer shock under the *low convenience-yield* calibration. The numbers at the bottom of each column indicate the observed total response of each variable.

The three right-hand bars in each panel depict the policy combinations that yield to any effects (recall that under the "Active Monetary" specification, the fiscal expansion has no effects). The total change in the variables echoes what we observed in the IRFs: the active-fiscal case leads to Hegemon and Foreign inflation, and Hegemon depreciation; Hegemon Dominance induces barely any movement of the price level at the Hegemon, paired with Foreign inflation and Hegemon appreciation; and the mixed case shows Hegemon inflation from the non-Ricardian fiscal policy at the Hegemon together with Hegemon depreciation.

Across all three policy specifications, the discount rate adjustment coming from the policy choice matters the most for the adjustment of Hegemon inflation. The discount rate adjustment and the associated spillovers, which revalue the government bonds held by the representative household in each country, drive the perceived wealth of each household and, thereby, the distortionary adjustments in nominal variables operating through the New Keynesian Phillips curve and the UIP condition in general equilibrium. This holds regardless of the direction of the total change in the variable: Hegemon inflation increases under AF but decreases under HD, yet in both cases most of the variation is attributable to discount rate movements driven by policy. Under AF, discount rates

⁹We perform this exercise for both the low-CY and the high-CY steady states, given the differences in propagation observed above.

adjust to yield a windfall gain to Hegemon households. Under HD, the same adjustment implies a slight loss of perceived real wealth.

For Foreign inflation, discount rate channels matter the most again, except in the mixed-policy specification. Here, the Foreign portfolio adjustment delivers the major margin of adjustment, consistent with the idea that the Foreign monetary policy authority here reacts aggressively with its policy rate in response to any inflationary or dis-inflationary pressure, invoking significant portfolio rebalancing on behalf of the Foreign household, which sits on the UIP condition.

Convenience yields mostly offset the variation coming from the discount rate-policy component for Foreign, again regardless of the directional change of the variable considered. Intuitively, to the extent that discount rates are correlated through the substitutability of Hegemon and Foreign assets, convenience yields break that substitutability through the non-pecuniary utility flow on Hegemon bonds, which matters predominantly for the Foreign household that can engage in asset trading. Regardless of whether the fiscal expansion makes the Hegemon asset more or less attractive, convenience yields break the otherwise swift substitutability between Hegemon and Foreign assets used to smooth economic disturbances.

What changes when convenience yields are substantially larger? Figure 10 provides an answer.

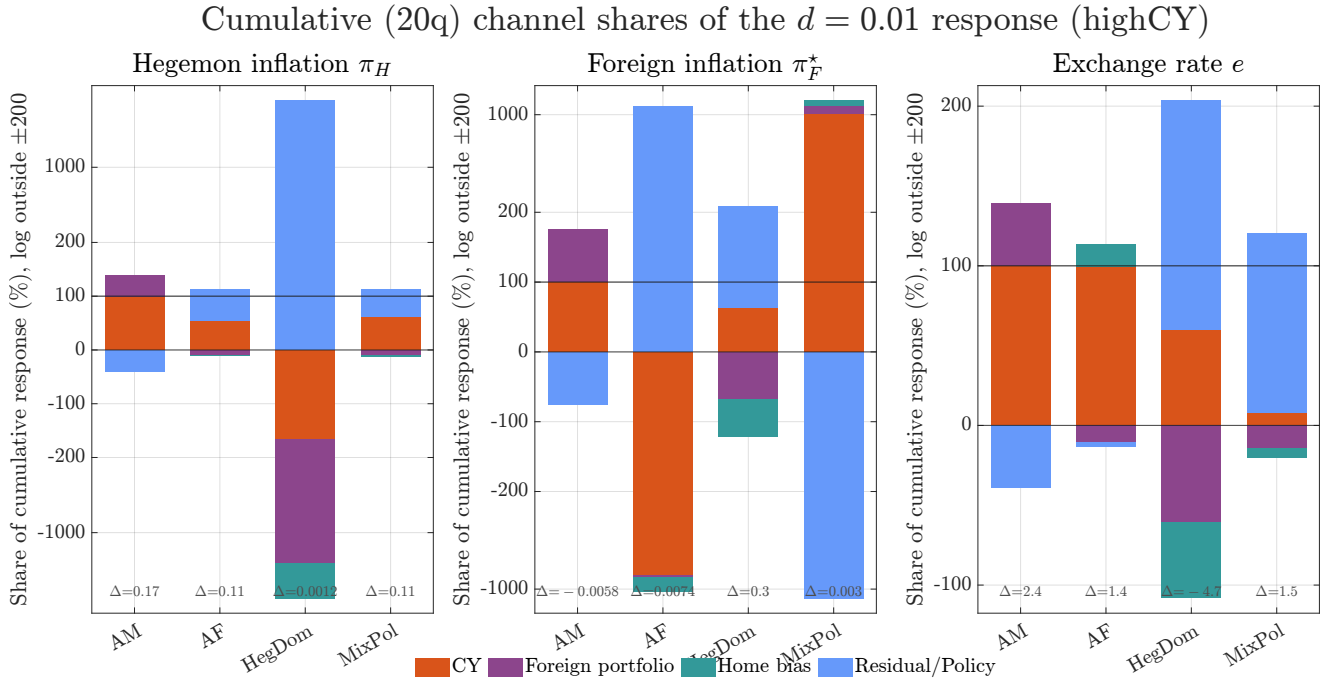


Figure 10: Decomposition of the total response of π_H, π_F^* , and e in response to a Hegemon government transfer shock under the *high convenience-yield* calibration. The numbers at the bottom of each column indicate the observed total response of each variable.

We again observe that the spillover in discount rates induced by the specific behaviour of the fiscal and monetary authorities (summarised by the blue bars) drives a significant portion of the observed adjustment across all specifications. A major general exception is the active-monetary

specification, where convenience yields matter the most. This is not a surprise: recall that under a steady-state with small convenience yields, there was no response of the economy whatsoever following a deficit shock, as Ricardian equivalence was holding. Naturally, to the extent that we now observe effects even under the active-monetary specification, it is attributed to movements induced through convenience yields.

But even under the rival policy specifications, convenience yield adjustments now mechanically matter more for the equilibrium adjustment of nominal variables: for Hegemon inflation, they make up more than half of the variation under the active-fiscal and mixed-policy cases, and under Hegemon Dominance they act as a significant counter-force against the price level pressure coming from the policy specification. It acts as a similar cushion for Foreign inflation under the Active-Fiscal specification, as well as a non-trivial driver of the adjustment of π_F^* in the other policy configurations. Heuristically, convenience yields buffer the external price level spillover when the policy specification is such that the Hegemon fiscal stance is globally worsened (as the larger steady-state convenience flow for the Hegemon and the ability of Foreign to substitute to their own bonds effectively forces a majority of the adjustment to the Hegemon economy in these cases), while the opposite is the case otherwise.

We also want to clarify how much our assumption of international bond holdings matters for the observed inflation spillovers. This is important because there are two distinct channels at play from the perspective of the Foreign household: a direct revaluation of their holdings of Hegemon bonds (which may be mitigated by exchange rate movements), and a revaluation of *Foreign-government-issued* bonds through the correlation of international discount rates. Figure 11 disentangles these channels by showing the cumulative response of the Foreign price level to a Hegemon fiscal expansion under both the Active-Fiscal and the Hegemon-Dominance policy mix, varying the steady-state level of convenience yields and the share of internationally held bonds.

This exercise clarifies the link between convenience yields and the importance of international bond holdings. A substantial baseline level of inflation (accounting for most of the Foreign price level variation) is exported in any case through the policy specification and the corresponding movement of foreign discount rates, even when Foreign holdings of Hegemon bonds are zero ($\omega = 1$): the cumulative change in the price level (measured in percentage points for a 1% deficit/GDP innovation at the Hegemon) lies between 0.05 and 0.08 for the active-fiscal policy specification, and between 0.08 and 0.12 for the Hegemon dominance specification. The second of the two channels therefore accounts for most of the variation: discount rate correlations are the primary driver, depending intricately on the fiscal-monetary policy mix. Figure 11 thus confirms our main observations from Figures 9 and 10.

Interestingly, the link between exported inflation and the share of Hegemon debt held in the Foreign economy is non-linear under the active-fiscal specification. Exported inflationary pressure is

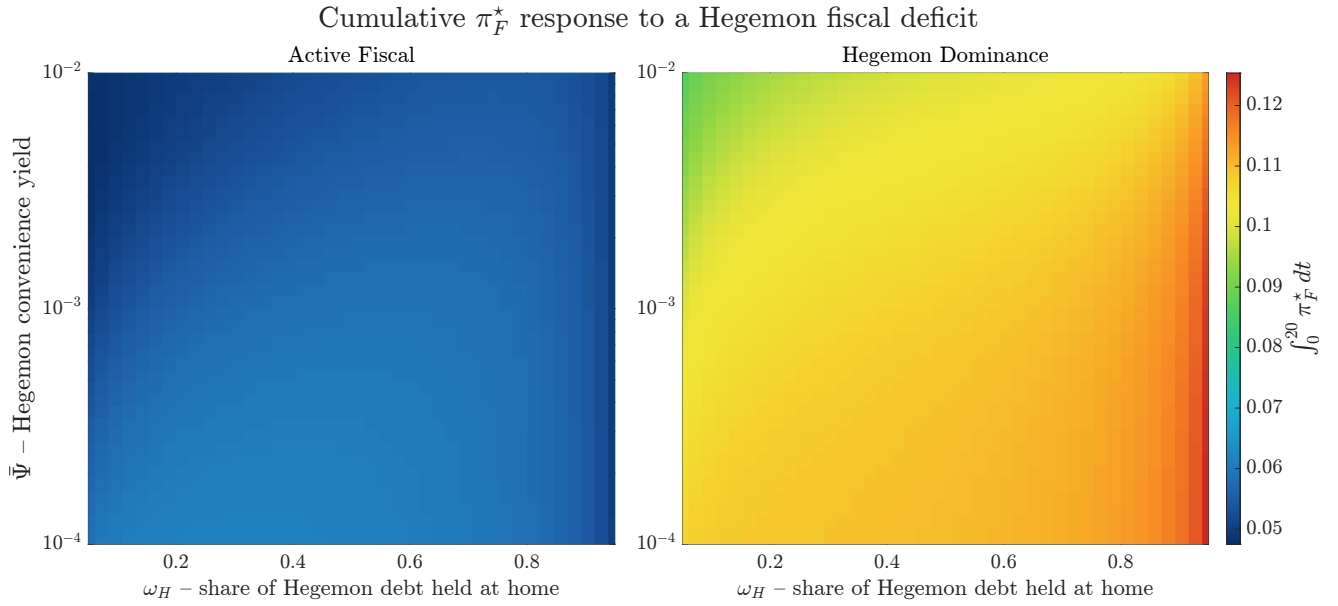


Figure 11: Total inflation in Foreign following a Hegemon fiscal impulse, depending on the steady-state level of convenience yields (y-axis) and the share of Hegemon bonds held at the Hegemon itself (x-axis). The assumed policy mix in the left panel is *Active Fiscal Policy*, while the right-hand panel corresponds to the *Hegemon Dominance* specification.

highest for intermediary splits of Hegemon debt holdings, and also slightly larger for lower levels of convenience yields. Convenience yields generally act as a slight mitigator of exported inflation under the active-fiscal specification, reflecting the previous decomposition. Intuitively, since convenience yields increase the required return on Foreign debt to ensure no arbitrage between the two types of bonds for the Foreign household, a higher level thereof means that the required return compensation paid through Foreign debt increases in the level of convenience yields, which in turn worsens the Foreign fiscal balance, requiring inflation for debt stabilization.

By contrast, under the Hegemon-dominance policy specification, the gradient between debt holdings at the Hegemon and Foreign inflationary pressure is clearer, albeit it also takes a somewhat surprising direction: the more Hegemon debt is held at Home¹⁰, the *larger* is the degree of exported inflation. To understand why, we must begin by analysing the 'dominant' fiscal and especially monetary policy authorities of the Hegemon. Fiscal backing is insufficient within the Hegemon, but, at the same time, the Hegemon monetary authority tightens the policy rate in response to inflationary pressure arising from the fiscal expansion. This would possibly increase demand for Hegemon debt from Foreign, as currency-adjusted rates of return on Hegemon debt are still increasing due to the appreciation of the Hegemon currency. To the extent that this reduces the attractiveness of Foreign bonds, the foregone fiscal backing coming from lower demand for Foreign debt must be counterbalanced with larger nominal devaluation pressures, i.e., inflation.

¹⁰With the exception of $\omega = 1$, which is not modelled here.

4.3 Estimation

We now estimate the model at quarterly frequency, covering the period from 1998-Q1 (the first year following the Bank of England's operational independence) through 2025-Q2. Estimation requires a stochastic state-space with period-by-period innovations, which the above model cannot achieve as it stands now, since the shock space is of probability measure zero. Here, we use Ornstein-Uhlenbeck processes for all exogenous innovations, retaining however the previously derived equilibrium conditions. Concretely,

$$\begin{aligned}
 da_t &= -\varphi_a a_t dt + \sigma_a dW_{a,t}, & da_t^* &= -\varphi_a^* a_t^* dt + \sigma_a^* dW_{a^*,t}, \\
 d\hat{\theta}_t &= -\varphi_\theta \hat{\theta}_t dt + \sigma_\theta dW_{\theta,t}, & d\hat{\theta}_t^* &= -\varphi_\theta^* \hat{\theta}_t^* dt + \sigma_\theta^* dW_{\theta^*,t}, \\
 dd_t &= -\varphi_d d_t dt + \sigma_d dW_{d,t}, & dd_t^* &= -\varphi_d^* d_t^* dt + \sigma_d^* dW_{d^*,t}, \\
 d\hat{\kappa}_t &= -\varphi_\kappa \hat{\kappa}_t dt + \sigma_\kappa dW_{\kappa,t}.
 \end{aligned} \tag{45}$$

Because of Ito's Lemma, there is a small discrepancy in the model now, since the presence of stochasticity requires second-order correction terms to be exact, which we here do not do.¹¹ Heuristically, this can be thought of as effectively using a discrete approximation of the model for this subsection, allowing us to postulate AR(1) shock processes without the need for re-deriving the model with a huge number of tedious variance-correction terms.

Data. We use data obtained from FRED. The observables are conventional output gap estimates for the U.S. and the U.K., demeaned quarterly PPI inflation for the U.S. and the U.K: and the terms of trade derived from the USD-GBP spot exchange rate, adjusted by relative import/export price levels computed from nominal and real trade quantities. Total and local bond issuance are defined as demeaned quarter-on-quarter growth rates in the observed market value of total bond stocks after removing a first-order trend.

In total, we match six data series: the Hegemon output gap, Hegemon and Foreign local inflation, the terms of trade, and Hegemon and Foreign total bond issuance.¹² The estimation varies the policy parameters ϕ_π , ϕ_χ , ϕ_π^* , ϕ_χ^* , ψ_b , and ψ_b^* , the convenience yield shape parameters ζ_H and ζ_F , and the steady-state convenience yield levels $\bar{\Psi}$ and $\bar{\Psi}^*$. We also vary the persistence and standard deviations of all six shock processes.¹³

¹¹This linearity of the model is also helpful in the estimation exercise that we are conducting, which follows [Christensen et al. \(2024\)](#).

¹²This corresponds to the MSV system. We do not use data on the Foreign output gap, for instance, as it is not part of that MSV system.

¹³We impose common persistence and standard deviation for the convenience yield shock at the Hegemon and at Foreign, which ensures that the model matrices are of full rank in the estimation.

Estimation method. Principally, we use the tools developed by Christensen et al. (2024) for estimating continuous-time models using discrete data. Christensen et al. (2024) develop a method to estimate continuous-time models using a discrete state-space representation with a likelihood function based on Kalman filtering, which we also use to provide model estimates. Our headline exercise here specifies a Bayesian estimation (following Herbst and Schorfheide (2016)), primarily over the space of the shock and policy parameters, in which the likelihood for the posterior is calculated following the framework of Christensen et al. (2024).¹⁴ The shape and level parameters of the convenience yield function are allowed to attain values within wide bounds to remain maximally agnostic. The monetary and fiscal policy coefficients are likewise unconstrained and can take values across the policy space, so that the data inform the policy schedule. To obtain the unconditional posterior distribution, we use a Markov Chain Monte Carlo procedure with 25,000 draws per chain, employing the tailored randomised block Metropolis-Hastings algorithm of Chib and Ramamurthy (2010) across two chains, achieving an acceptance rate of approximately 25%. Prior distributions are specified as follows. Autoregressive coefficients follow relatively narrow Beta distributions, while standard deviations follow wide inverse-gamma distributions, allowing the data to determine the relative importance of each shock process. The shape and level parameters of the convenience yield function are uniformly distributed with wide bounds, to remain maximally agnostic. The monetary and fiscal policy coefficients are likewise uniform on wide intervals, so that the data inform the policy schedule. Table D.1 in the appendix summarises the prior distributions in detail.

Table 2 presents the estimation results. In both the U.S. and the U.K., the data favour a *fiscally-led policy mix*: marginal tax rates are relatively unreactive to deviations of domestic government debt from steady state, and the Taylor principle is not satisfied once the policy reaction to output is accounted for. Estimated steady-state convenience yields are low, but relatively elastic. The shock process parameters are broadly standard, with persistence of all shock processes except for the convenience yield process being fairly large.

Historical decomposition. We now decompose the historical variation in the key nominal variables of interest: Hegemon (U.S.) inflation, Foreign (U.K.) inflation, and the exchange rate, captured through the terms of trade. Figure 12 presents the contribution of individual shocks to movements in nominal U.S. and U.K. variables since 2015.

The grey bars represent unmodeled factors, and the coloured bars represent the contributions of each shock component as indicated by the legend. The post-Covid fiscal expansion in the U.S.

¹⁴Our estimation is qualitatively fully robust to using simple Maximum Likelihood, a Generalized Method of Moments approach, as well as to including no or alternative historical convenience yield data series in the estimation. In all these approaches, we also construct a likelihood function following Christensen et al. (2024). Appendix E details the results of all our alternative estimation exercises, which deliver qualitatively similar insights.

Policy and convenience yield parameters			Shock mean-reversion and diffusion parameters		
Parameter	Estimate	Std. Error	Parameter	Estimate	Std. Error
ϕ_π	0.4344	1.2585	ϕ_a	0.5049	0.2124
ϕ_π^*	1.0523	1.4908	ϕ_a^*	2.9943	1.4203
ϕ_x	-0.4408	0.0657	ϕ_θ	2.9188	1.6435
ϕ_x^*	-0.6851	0.1557	ϕ_θ^*	0.0668	0.3196
ψ_b	-0.1143	0.5752	ϕ_d	0.0388	0.2599
ψ_b^*	-0.5035	0.6841	ϕ_d^*	0.0201	0.1467
$\bar{\psi}$	0.0023	0.0080	ϕ_κ	0.4865	0.4069
$\bar{\psi}^*$	0.0018	0.0035	σ_a	9.8378	1.8060
			σ_a^*	9.8681	7.5411
			σ_θ	1.8676	0.7111
			σ_θ^*	2.1530	0.3003
			σ_d	2.0780	2.4041
			σ_d^*	2.5379	3.9362
			σ_κ	7.4607	5.8803

Table 2: Estimates for the exact continuous-time Bayesian specification using U.S. and U.K. data since 1998.

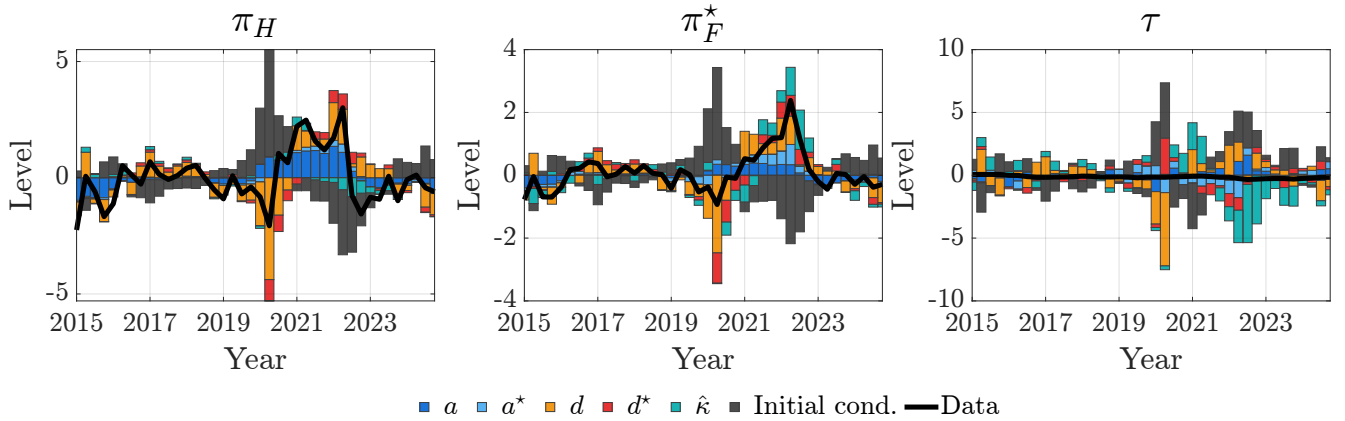


Figure 12: Historical shock decomposition implied by the model.

(orange bars) emerges as the predominant driver of U.S. inflation, accounting for approximately 40% of total excess cyclical inflationary pressure over 2021–2024. Over the same period, the post-Covid fiscal expansion in the U.K. (red bars) accounted for roughly 25% of total inflationary pressure in the U.K., with its weight rising in later quarters. The U.S. fiscal expansion contributed primarily to a depreciation of the U.S. terms of trade, while the U.K. fiscal expansion contributed to a modest appreciation.

The novel mechanism highlighted in this paper concerns the spillover of U.S. deficits (orange bars) to U.K. inflation (centre panel). Through the channels discussed in the impulse response analysis, by which a positive relative wealth effect driven by higher demand for Foreign goods, combined with a relative deterioration of the U.K. fiscal position, the U.S. fiscal expansion transmitted into U.K. price dynamics, contributing cumulatively around two percentage points to the U.K. price level since 2021.

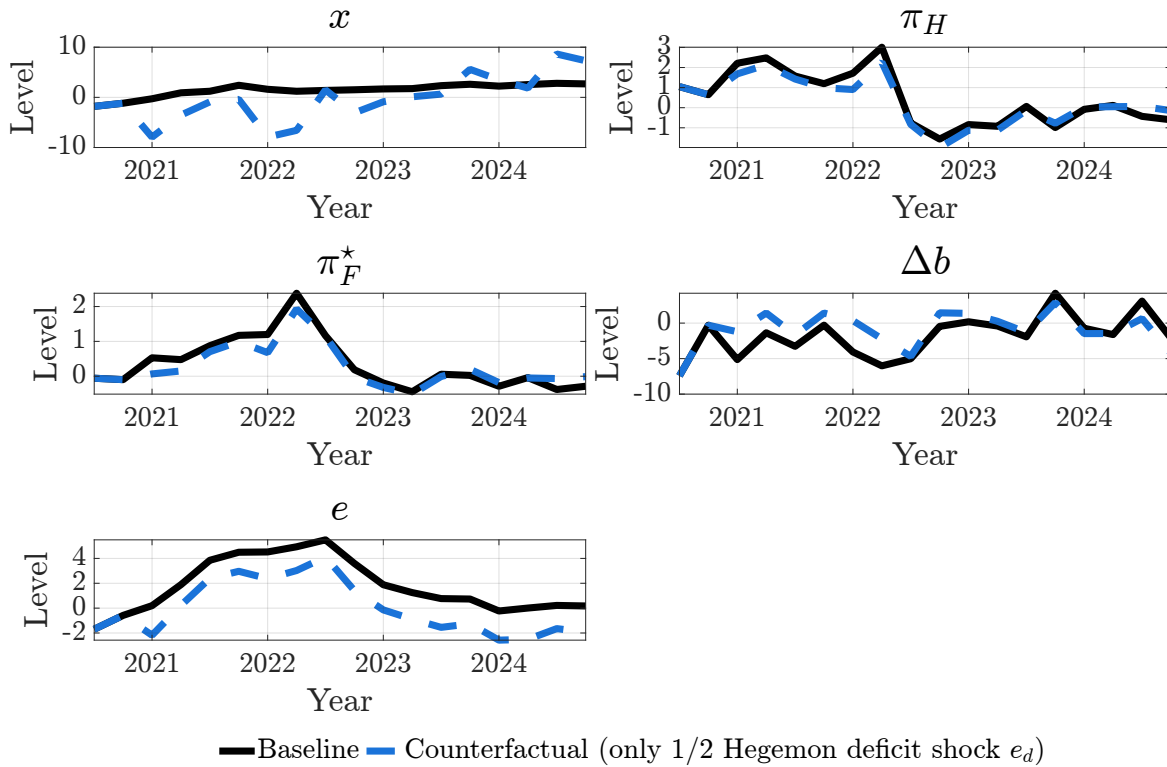


Figure 13: Counterfactual: halving the U.S. fiscal deficit since 2021.

Complementing this decomposition, Figure 13 depicts the hypothetical path of output gaps, inflation rates, Hegemon bond issuance, and the nominal exchange rate had U.S. deficits been halved since 2021. This exercise confirms the intuition from the historical decomposition and illustrates how a more restrained fiscal expansion would have meant a more material and long-lasting recession in the wake of the Covid pandemic, and how U.S. fiscal policy actively contributed to the Dollar depreciation observed in that same period.

5 Conclusion

This paper studies the international transmission of fiscal expansions, with a focus on the bilateral relationship between the U.K. and the U.S. We recover model-consistent fiscal deficit innovations from the market value of U.K. and U.S. sovereign debt over 1975–2024 and use them to study how deficits in a financial Hegemon transmit to exchange rates and inflation abroad. Local projections show that U.S. deficit shocks appreciate the U.K. real exchange rate on impact and raise U.K. consumer prices persistently, with a transmitted magnitude roughly half that of own-country U.K. deficits.

We rationalise these patterns in a continuous-time two-country New Keynesian model in which Hegemon bonds carry a convenience yield. A deficit shock at the Hegemon depreciates its currency, revalues Foreign holdings of Hegemon debt, and raises the effective discount rate on Foreign sovereign debt, tightening the Foreign intertemporal budget constraint even when Foreign surpluses are unchanged. We label this mechanism *fiscal contagion* and show that it operates under a broad range of monetary–fiscal policy mixes. The estimated model attributes roughly two percentage points of cumulative U.K. inflationary pressure since 2021 to spillovers from U.S. fiscal policy. In a counterfactual in which U.S. deficits had been halved, the model would have delivered a materially smaller domestic and Foreign price-level response.

Relative to the existing literature on fiscally driven price-level determination, which has largely focused on closed-economy systems, we provide both an empirical strategy and a quantitative framework for studying the cross-border component. Relative to previous open-economy work on exchange-rate determination and convenience yields, we make the deficit-spillover channel explicit and locate it in the joint dynamics of sovereign bond valuations, discount-rate spillovers, and the monetary–fiscal policy mix.

For policymakers in countries closely integrated with the U.S. sovereign debt market, fiscal sustainability is therefore not a purely domestic question: deficits at the Hegemon transmit to domestic prices and exchange rates through channels that operate even when domestic surpluses are unchanged. Central banks in such economies face a sharper version of the standard output–inflation stabilisation trade-off because restrictive monetary policy alone cannot fully insulate the domestic economy from a Hegemon fiscal expansion when discount-rate spillovers are large. The model suggests that coordination between fiscal and monetary authorities outside the U.S. materially mitigates the inflationary spillover, particularly when Foreign monetary policy commits to defending the attractiveness of Foreign sovereign bonds. We leave for future work whether such coordination is feasible in practice and how it interacts with the broader question of debt-management strategy in an environment of rising global deficits.

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A Additional Local Projection Results

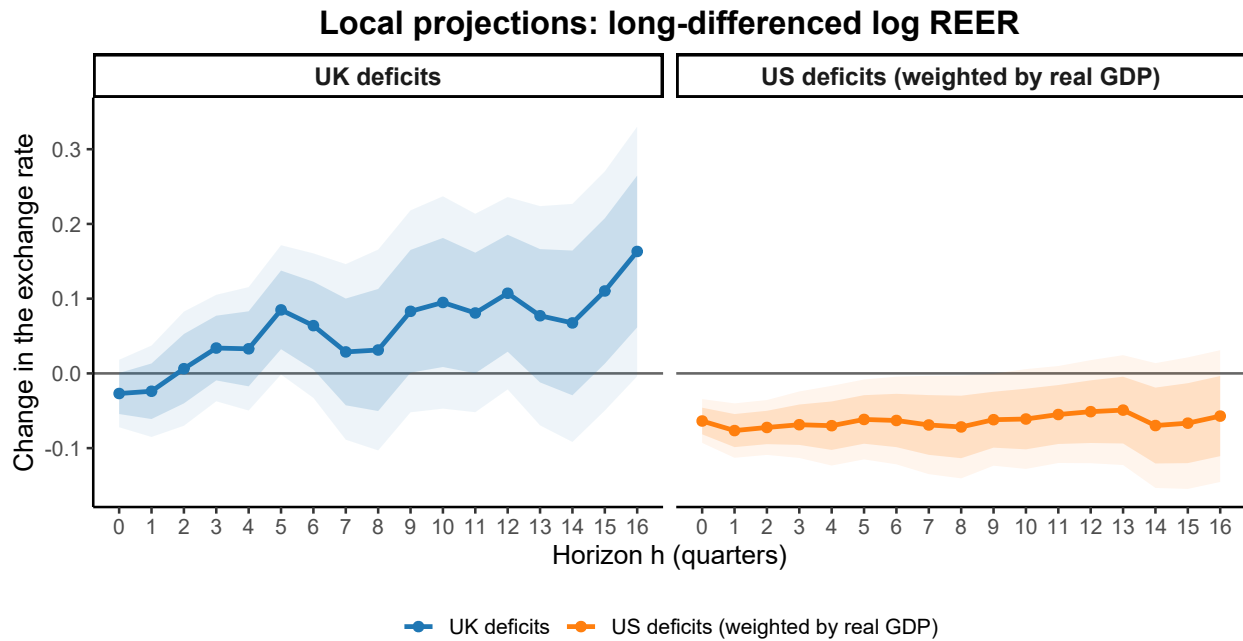


Figure A.1: The effect of deficit surprises in the U.K. and the U.S. on the U.K. REER with quarterly data.

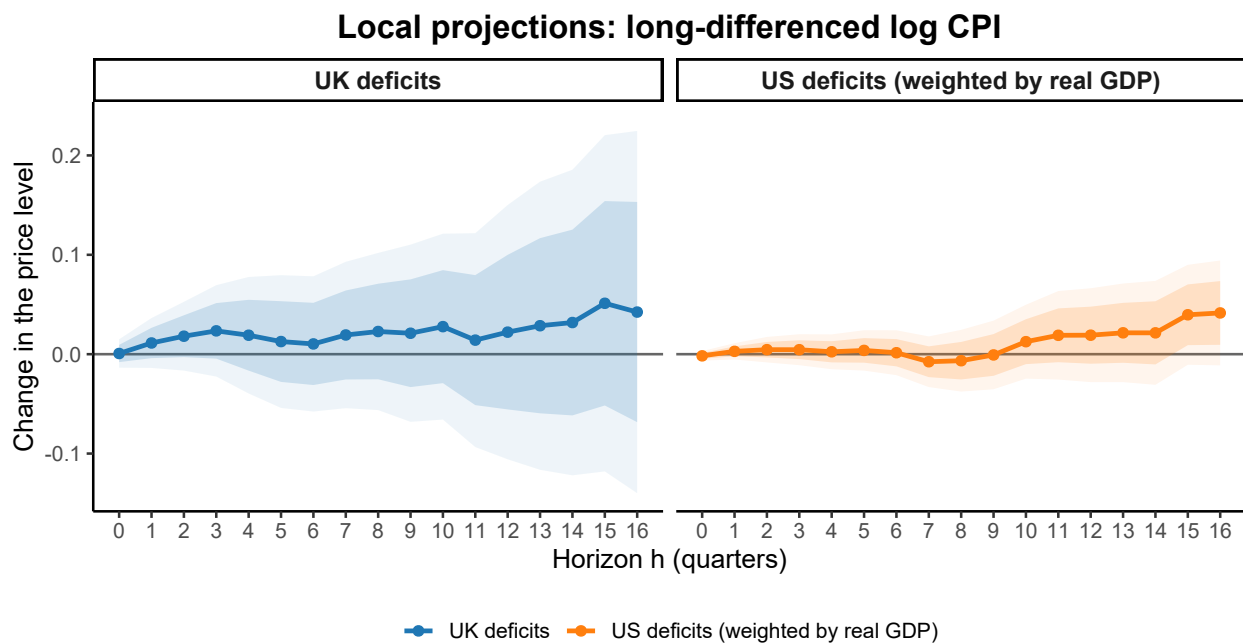


Figure A.2: The effect of deficit surprises in the U.K. and the U.S. on the price level in the U.K, as measured by the CPI, in quarterly data.

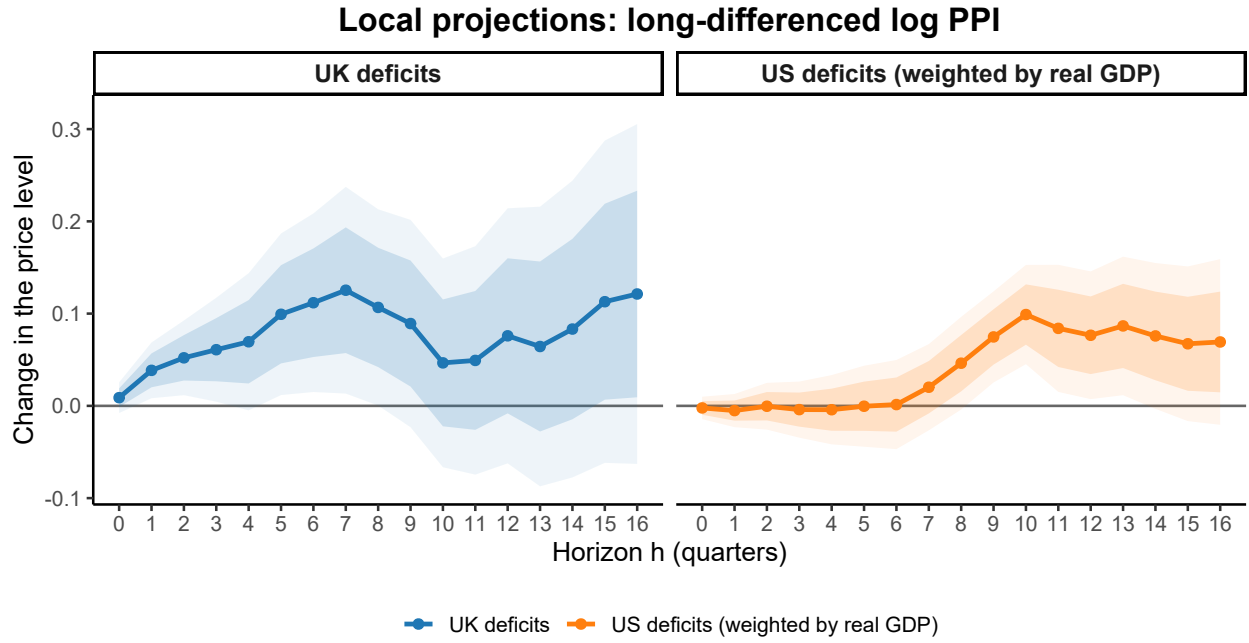


Figure A.3: The effect of deficit surprises in the U.K. and the U.S. on the price level in the U.K, as measured by the PPI, in quarterly data.

B Linking deficits to exchange rates

Complementing section 2.2, we provide additional results akin to the 'reverse regression' approach from Engel and West (2005) and Jiang (2021).

$$\frac{s_{t+k}^i}{B_{t+k}^i} = \alpha^{(k)} + \beta^{(k)} \Delta \log e_t^{(-i)/i} + \delta_1^{(k)} \frac{s_t^i}{B_t^i} + \delta_2^{(k)} \Delta \frac{s_t^i}{B_t^i} + \gamma_1^{(k)} \frac{s_t^{-i}}{B_t^{-i}} + \gamma_2^{(k)} \Delta \frac{s_t^{-i}}{B_t^{-i}} + \varepsilon_{t+k}^i, \quad (\text{B.1})$$

that is, we test the link between the deficit/debt-ratio at time $t + k$ and the change in the real exchange rate as well as the level and the first-difference of *both* domestic and foreign deficit/debt-ratio levels. The results of this exercise are presented in table B.1 and figure B.1a for the U.K. and in table B.2 and figure B.1b for the U.S. In a nutshell, current appreciations are related to smaller local government deficits, but in the case of the U.K., the phenomenon is reversing at very long horizons.

<i>Dependent variable: deficit-to-debt ratio at t+k</i>								
	k=1	k=2	k=3	k=4	k=5	k=6	k=7	k=8
Δ REER	0.129 (0.104)	0.149 (0.092)	0.122 (0.080)	0.296*** (0.086)	0.067 (0.051)	-0.081 (0.054)	-0.086 (0.065)	-0.126** (0.060)
$\frac{s}{B}$ -UK	0.187 (0.256)	-0.491** (0.201)	-0.858*** (0.224)	-1.109*** (0.200)	-0.905** (0.410)	-0.287 (0.260)	-0.142 (0.289)	-0.219 (0.478)
$\Delta \frac{s}{B}$ -UK	0.184 (0.202)	0.090 (0.276)	0.729** (0.348)	0.729* (0.405)	0.689 (0.584)	1.389** (0.626)	0.604 (0.461)	0.706 (0.608)
$\frac{s}{B}$ -US	0.481** (0.194)	0.814*** (0.229)	0.827*** (0.256)	0.897*** (0.188)	0.684 (0.421)	-0.005 (0.281)	-0.125 (0.291)	-0.192 (0.409)
$\Delta \frac{s}{B}$ -US	-0.073 (0.246)	-0.018 (0.220)	-0.647* (0.386)	-0.960 (0.588)	-0.752 (0.673)	-0.824 (0.667)	-0.038 (0.458)	-0.149 (0.560)
Constant	-0.010 (0.010)	-0.024 (0.015)	-0.048*** (0.016)	-0.058*** (0.010)	-0.067*** (0.016)	-0.095*** (0.012)	-0.097*** (0.013)	-0.109*** (0.017)
Obs.	29	28	27	26	25	24	23	22
R ²	0.659	0.436	0.365	0.548	0.287	0.354	0.251	0.387
Adj. R ²	0.585	0.308	0.214	0.435	0.099	0.175	0.030	0.195

Table B.1: UK: [Engel and West \(2005\)](#)-style regressions for the deficit-to-debt ratio. Newey-West standard errors in parentheses. *p<0.1, **p<0.05, ***p<0.01.

<i>Dependent variable: deficit-to-debt ratio at t+k</i>								
	k=1	k=2	k=3	k=4	k=5	k=6	k=7	k=8
Δ REER	-0.051 (0.090)	0.106 (0.131)	0.110 (0.103)	0.077 (0.076)	0.061 (0.093)	0.029 (0.094)	0.025 (0.079)	-0.026 (0.078)
$\frac{s}{B}$ -UK	-0.183*** (0.070)	-0.407*** (0.109)	-0.581*** (0.135)	-0.544*** (0.138)	-0.500*** (0.175)	-0.482** (0.190)	-0.455** (0.181)	-0.412* (0.225)
$\Delta \frac{s}{B}$ -UK	0.284*** (0.096)	0.405* (0.215)	0.413** (0.191)	0.283 (0.220)	0.126 (0.217)	0.131 (0.250)	-0.020 (0.361)	-0.169 (0.432)
$\frac{s}{B}$ -US	0.726*** (0.108)	0.453*** (0.171)	0.345** (0.170)	0.263** (0.128)	0.306** (0.121)	0.350** (0.155)	0.354* (0.193)	0.282 (0.197)
$\Delta \frac{s}{B}$ -US	0.139 (0.123)	0.057 (0.183)	-0.130 (0.197)	-0.322* (0.169)	-0.267* (0.160)	-0.301 (0.219)	0.001 (0.270)	0.140 (0.261)
Constant	-0.038*** (0.012)	-0.079*** (0.018)	-0.100*** (0.016)	-0.103*** (0.015)	-0.095*** (0.016)	-0.088*** (0.016)	-0.087*** (0.019)	-0.093*** (0.026)
Obs.	52	51	50	49	48	47	46	45
R ²	0.624	0.374	0.329	0.293	0.254	0.244	0.230	0.204
Adj. R ²	0.583	0.305	0.253	0.211	0.165	0.152	0.133	0.102

Table B.2: US: [Engel and West \(2005\)](#)-style regressions for the deficit-to-debt ratio. Newey-West standard errors in parentheses. *p<0.1, **p<0.05, ***p<0.01.

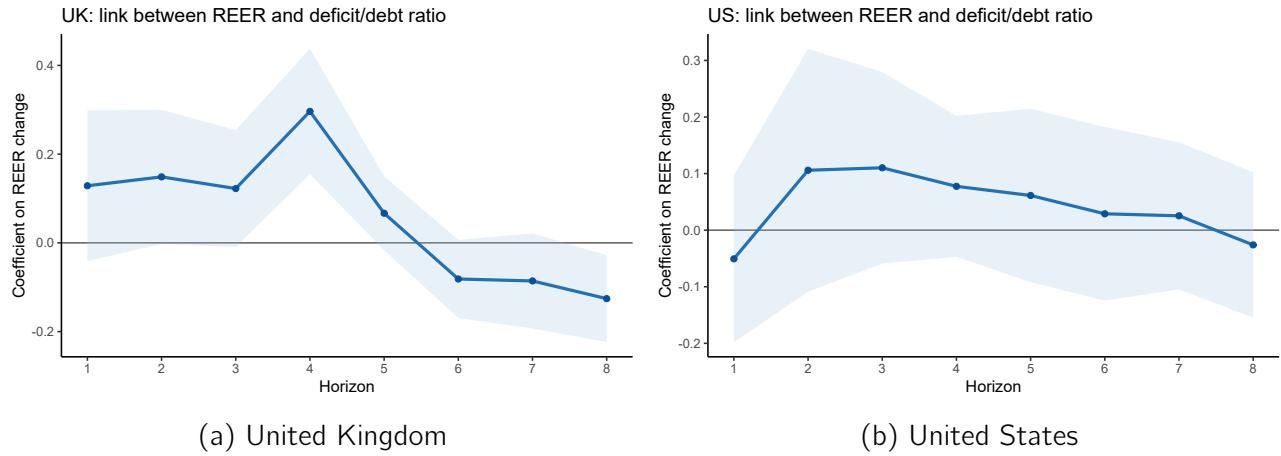


Figure B.1: Engel and West (2005)-style reverse regression

C Omitted derivations from section 3

This appendix records the algebra behind the equations used in Section 3. By convention, hats on positive level variables denote log deviations from the stationary steady state unless explicitly defined otherwise, whereas hats on inflation and interest rates denote level deviations. The holding wedge also uses the level deviation $\hat{\kappa}_t \equiv \kappa_t - \bar{\kappa}$ because κ_t is already a rate. Throughout we impose $\rho = \rho^*$ and hence, by (C.58), $\bar{\delta}_H = \bar{\delta}_F = \rho$.

C.1 Coefficient dictionary for yet unused coefficients

For reference, this subsection collects the algebraic coefficients used in the derivations and the state mappings. Define

$$\mathfrak{g} \equiv \gamma + \mathcal{S}. \quad (\text{C.1})$$

The gap coefficients used in the derivations are

$$\begin{aligned} K_\tau^H &\equiv \mathcal{S} + (1 - \gamma)\Gamma_\xi, & K_\tau^F &\equiv \mathcal{S} - \gamma\Gamma_\xi, \\ \mathcal{C}_\tau &\equiv \gamma + \mathcal{S} - \gamma\Gamma_\xi, & \mathcal{H}_\tau &\equiv 1 + 2\mathcal{S} + h\Gamma_\xi, \\ \mathcal{W}_H &\equiv \mathcal{C}_\tau + \Gamma_\xi + \gamma(\phi_\pi - 1), & \mathcal{W}_F &\equiv \mathcal{C}_\tau + \gamma(\phi_\pi^* - 1), \\ \Lambda_0 &\equiv 1 - \gamma(\phi_\pi + \phi_\pi^*). \end{aligned} \quad (\text{C.2})$$

Finally, the natural-rate derivation uses

$$\begin{aligned} \varrho_H &\equiv (1 - \gamma)\frac{\phi}{1 + \phi}, & \varrho_F &\equiv (1 - \gamma)\frac{\phi^*}{1 + \phi^*}, & \beta_H &\equiv \gamma\frac{\phi}{1 + \phi}, & \beta_F &\equiv \gamma\frac{\phi^*}{1 + \phi^*}, \\ \Upsilon_H &\equiv -\gamma - \mathcal{S}(1 - \lambda_H), & \Upsilon_F &\equiv \gamma + \mathcal{S}(1 - \lambda_F). \end{aligned} \quad (\text{C.3})$$

C.2 Price-index identities and first-order relative-price relations

For the following, we define the price-index and exchange rate composites

$$\begin{aligned} G_t &\equiv (1 - \gamma) + \gamma \text{tot}_t^{1-\eta}, & Z_t &\equiv (1 - \gamma) + \gamma Q_t^{\eta-1} e^{-\Omega_t}, \\ G_t^* &\equiv (1 - \gamma) + \gamma \text{tot}_t^{\eta-1}, & Z_t^* &\equiv (1 - \gamma) + \gamma Q_t^{1-\eta} e^{\Omega_t}. \end{aligned} \quad (\text{C.4})$$

Factoring P_{Ht} out of the Hegemon CPI (9) and P_{Ft}^* out of its Foreign counterpart gives

$$P_t = P_{Ht} G_t^{1/(1-\eta)}, \quad P_t^* = P_{Ft}^* (G_t^*)^{1/(1-\eta)}, \quad (\text{C.5})$$

The law of one price (6) implies $\text{tot}_t^* \equiv P_{Ht}^*/P_{Ft}^* = P_{Ht}/P_{Ft} = \text{tot}_t^{-1}$, so that the stated expression for G_t^* follows. Using $P_{Ft} = E_t P_{Ft}^*$ therefore yields

$$Q_t \equiv \text{tot}_t \left(\frac{G_t^*}{G_t} \right)^{1/(1-\eta)}. \quad (\text{C.6})$$

Taking logs of the exact identity (C.6) gives

$$q_t \equiv \ell_t + \frac{\log G_t^* - \log G_t}{1 - \eta}.$$

Log-linearising this relationship around the symmetric steady state $\bar{\text{tot}} = 1$ gives

$$q_t \simeq (1 - 2\gamma) \ell_t, \quad q_t^n \simeq (1 - 2\gamma) \ell_t^n, \quad \tilde{q}_t \simeq (1 - 2\gamma) \tau_t. \quad (\text{C.7})$$

Similarly, a first-order expansion of (C.5) around the symmetric steady state gives

$$\log P_t - \log P_{Ht} \simeq \gamma \ell_t, \quad \log P_t^* - \log P_{Ft}^* \simeq -\gamma \ell_t, \quad (\text{C.8})$$

Taking time derivatives gives the first-order CPI–PPI relationships in (34). Finally, $\ell_t = \log P_{Ft} - \log P_{Ht}$ implies $\dot{\ell}_t = \pi_{Ft} - \pi_{Ht}$, and $P_{Ft} = E_t P_{Ft}^*$ implies $\pi_{Ft} = \dot{\hat{e}}_t + \pi_{Ft}^*$, so that $\dot{\hat{e}}_t = \dot{\ell}_t - \hat{\pi}_{Ft}^* + \hat{\pi}_{Ht}$.

C.3 Full problem of the Hegemon household

This appendix collects the full setup of the domestic household problem, including the intratemporal allocation block and the derivation of the optimality conditions stated in Section 3.1.

The consumption aggregate is the CES bundle

$$C_t = \left[(1 - \gamma)^{1/\eta} C_{Ht}^{(\eta-1)/\eta} + \gamma^{1/\eta} C_{Ft}^{(\eta-1)/\eta} \right]^{\eta/(\eta-1)}, \quad (\text{C.9})$$

where $1 - \gamma$ is the domestic expenditure share (home bias) and $\eta > 0$ is the intratemporal elasticity

of substitution. Given total nominal expenditure, cost minimisation yields the demand functions

$$C_{Ht} = (1 - \gamma) \left(\frac{P_{Ht}}{P_t} \right)^{-\eta} C_t, \quad C_{Ft} = \gamma \left(\frac{P_{Ft}}{P_t} \right)^{-\eta} C_t. \quad (\text{C.10})$$

With $u(C, N, b_H; \theta) = \log C - \frac{N^{1+\phi}}{1+\phi} + \nu(b_H, C; \theta)$ and ν as in (10), the consumption scaling gives $\nu_C = -\nu_t/C_t$, so that

$$u_C = \frac{1}{C_t} + \nu_C = \frac{1 - \nu_t}{C_t}, \quad u_N = -N_t^\phi, \quad u_{b_H} = \nu_b = \frac{\Psi_t}{C_t}. \quad (\text{C.11})$$

Differentiating (10) with respect to b_{Ht} ,

$$\nu_b = \frac{\bar{\Psi}\theta_t \bar{b}_H}{(1 - \zeta_H)C_t} (1 - \zeta_H) \left(\frac{b_{Ht}}{\bar{b}_H} \right)^{-\zeta_H} \frac{1}{\bar{b}_H} = \frac{\bar{\Psi}\theta_t}{C_t} \left(\frac{b_{Ht}}{\bar{b}_H} \right)^{-\zeta_H} = \frac{\Psi_t}{C_t}. \quad (\text{C.12})$$

Define the Hegemon nominal discount factor

$$\Lambda_t \equiv e^{-\rho t} \frac{u_C}{P_t} = e^{-\rho t} \frac{1 - \nu_t}{C_t P_t}. \quad (\text{C.13})$$

An extra unit of *real* Hegemon bonds delivers, per unit time, the pecuniary return i_t plus the non-pecuniary flow ν_b measured in units of the marginal utility of consumption, namely $\nu_b/u_C = \Psi_t/(1 - \nu_t) = \Phi_t$. The bond first-order condition is therefore

$$-\frac{\dot{\Lambda}_t}{\Lambda_t} = i_t + \Phi_t, \quad \text{with} \quad \lim_{T \rightarrow \infty} \Lambda_T B_{HT} = 0. \quad (\text{C.14})$$

The second condition rules out explosive bond positions. Define the preference-drift composite

$$\chi_t \equiv \frac{\dot{\nu}_t}{1 - \nu_t} = -\frac{d}{dt} \log(1 - \nu_t), \quad (\text{C.15})$$

and analogously $\chi_t^* \equiv \dot{\nu}_t^*/(1 - \nu_t^*)$. From (C.13), $-\dot{\Lambda}_t/\Lambda_t = \rho - \frac{d}{dt} \log(1 - \nu_t) + \dot{C}_t/C_t + \pi_t$. Thus,

$$i_t + \Phi_t = \rho + \chi_t + \frac{\dot{C}_t}{C_t} + \pi_t \quad \Longleftrightarrow \quad \frac{\dot{C}_t}{C_t} = i_t - \pi_t - \rho + \Phi_t - \chi_t, \quad (\text{C.16})$$

which is equivalent to (13). The labour first-order condition $-u_N = u_C W_t/P_t$ is

$$C_t N_t^\phi = (1 - \nu_t) \frac{W_t}{P_t}. \quad (\text{C.17})$$

There is no Euler equation for Foreign bonds. By the segmentation assumption of Section 3.1, $B_{Ft} \equiv 0$ and the Hegemon household's portfolio consists of Hegemon debt alone. It is exactly this missing condition that permits Φ_t and Φ_t^* to move independently outside the steady state.

C.4 Full problem of the Foreign household

The Foreign household maximises

$$\int_0^\infty e^{-\rho^* t} \left\{ \ln C_t^* - \frac{N_t^{*1+\phi^*}}{1+\phi^*} + \nu^*(v_{Ht}^*, C_t^*; \theta_t^*) \right\} dt \quad (C.18)$$

subject to the budget constraint (15). It enjoys a convenience flow on its real quantity of Hegemon debt holdings, given by

$$\nu^*(v_{Ht}^*, C_t^*; \theta_t^*) = \frac{\bar{\Psi}^* \theta_t^* \bar{v}_H^*}{(1 - \zeta_F) C_t^*} \left[\left(\frac{v_{Ht}^*}{\bar{v}_H^*} \right)^{1-\zeta_F} - 1 \right], \quad \zeta_F > 0, \quad (C.19)$$

The Foreign consumption aggregate is symmetric to (C.9) with weights reversed, giving

$$C_{Ft}^* = (1 - \gamma) \left(\frac{P_{Ft}^*}{P_t^*} \right)^{-\eta} C_t^*, \quad C_{Ht}^* = \gamma \left(\frac{P_{Ht}^*}{P_t^*} \right)^{-\eta} C_t^*, \quad (C.20)$$

and the Foreign CPI in (C.5). Let $\Lambda_t^* \equiv e^{-\rho^* t} (1 - \nu_t^*) / (C_t^* P_t^*)$ denote the Foreign nominal (Foreign-currency) discount factor, and recall $v_{Ht}^* = B_{Ht}^* / (E_t P_t^*)$ from (14). Differentiating (C.19) gives $\nu_v^* = \Psi_t^* / C_t^*$ and hence $\nu_v^* / u_{C^*} = \Psi_t^* / (1 - \nu_t^*) = \Phi_t^*$. The Foreign household holds two assets. Its own government's debt bears no convenience flow and no holding wedge, so that

$$-\frac{\dot{\Lambda}_t^*}{\Lambda_t^*} = i_t^* \iff \frac{\dot{C}_t^*}{C_t^*} = i_t^* - \pi_t^* - \rho^* - \chi_t^*, \quad (C.21)$$

Hegemon debt delivers the Foreign-currency pecuniary return $i_t - \dot{E}_t / E_t$, net of the intermediation wedge χ_t , plus the convenience flow Φ_t^* , so that

$$-\frac{\dot{\Lambda}_t^*}{\Lambda_t^*} = i_t - \frac{\dot{E}_t}{E_t} - \chi_t + \Phi_t^* \iff \frac{\dot{C}_t^*}{C_t^*} = \left(i_t - \frac{\dot{E}_t}{E_t} - \chi_t \right) - \pi_t^* - \rho^* + \Phi_t^* - \chi_t^*, \quad (C.22)$$

Differencing (C.21) and (C.22) gives $i_t^* = i_t - \dot{E}_t / E_t - \chi_t + \Phi_t^*$, which is the modified UIP condition (16). Because the convenience argument is v_{Ht}^* , already expressed in Foreign consumption units, no exchange rate normalisation constant appears anywhere in the Foreign block. The deterministic transversality conditions are $\lim_{T \rightarrow \infty} \Lambda_T^* B_{FT}^* = 0$ and $\lim_{T \rightarrow \infty} \Lambda_T^* (B_{HT}^* / E_T) = 0$.

C.5 Details on the portfolio derivations

Differencing the Hegemon and Foreign Euler equations on their respective own-currency bonds and substituting the UIP condition (16) yields the international risk-sharing condition (Itskhoki and Mukhin, 2021),

$$\frac{\dot{C}_t}{C_t} - \frac{\dot{C}_t^*}{C_t^*} - \frac{\dot{Q}_t}{Q_t} = (\rho^* - \rho) + (\Phi_t - \Phi_t^* + \chi_t) + (\chi_t^* - \chi_t). \quad (C.23)$$

For this derivation, define $\Xi_t \equiv \log[(1 - \nu_t)/(1 - \nu_t^*)]$. Then, $\dot{\Xi}_t = \chi_t^* - \chi_t$, while (18) implies $\varpi_t = \Omega_t - \Xi_t$. Under $\rho = \rho^*$, subtracting $\dot{\Xi}_t$ from $\dot{\Omega}_t$ in (C.23) therefore gives the law (19).

Write $\hat{b}_{Ht} \equiv \log(b_{Ht}/\bar{b}_H)$. Then, $(b_{Ht}/\bar{b}_H)^{1-\zeta_H} = \exp[(1 - \zeta_H)\hat{b}_{Ht}] \approx 1 + (1 - \zeta_H)\hat{b}_{Ht}$, so (10) gives

$$\nu_t \approx \frac{\bar{\Psi}\theta_t\bar{b}_H}{(1 - \zeta_H)C_t}(1 - \zeta_H)\hat{b}_{Ht} = \frac{\bar{\Psi}\theta_t\bar{b}_H}{C_t}\hat{b}_{Ht}. \quad (\text{C.24})$$

Because $\hat{b}_{Ht}$ is already first order, variation in the prefactor enters only through products such as $\hat{\theta}_t\hat{b}_{Ht}$, which are second order. Evaluating the prefactor at the steady state and using $\bar{\nu} = 0$ gives

$$\nu_t \simeq s_H\hat{b}_{Ht}, \quad \nu_t^* \simeq s_F\hat{v}_{Ht}^*, \quad s_H \equiv \frac{\bar{\Psi}\bar{b}_H}{\bar{C}}, \quad s_F \equiv \frac{\bar{\Psi}^*\bar{v}_H^*}{\bar{C}^*}. \quad (\text{C.25})$$

Differentiating, $\chi_t \approx \dot{\nu}_t = s_H\dot{\hat{b}}_{Ht}$ and $\chi_t^* \approx s_F\dot{\hat{v}}_{Ht}^*$, and $\Xi_t \approx \nu_t^* - \nu_t$. From (11), $\hat{\Psi}_t = \hat{\theta}_t - \zeta_H\hat{b}_{Ht}$, and since $\Phi_t = \Psi_t/(1 - \nu_t)$,

$$\hat{\Phi}_t = \hat{\Psi}_t - \log(1 - \nu_t) = \hat{\theta}_t - \zeta_H\hat{b}_{Ht} + s_H\hat{b}_{Ht} = \hat{\theta}_t - \mathcal{A}_H\hat{b}_{Ht}, \quad (\text{C.26})$$

and symmetrically $\hat{\Phi}_t^* = \hat{\theta}_t^* - \mathcal{A}_F\hat{v}_{Ht}^*$, which is (22).

As for bond market clearing, totally differentiating (20) at $\bar{b} = \bar{b}_H + \bar{v}_H^*$ and $\bar{Q} = 1$ gives

$$\hat{b}_t = \omega_H\hat{b}_{Ht} + \omega_F[\hat{v}_{Ht}^* + (1 - 2\gamma)(\ell_t^n + \tau_t)]. \quad (\text{C.27})$$

The reference allocation is conditioned on the same $\hat{b}_t$ and the same $\hat{v}_{Ht}^*$, so it satisfies $\hat{b}_t = \omega_H\hat{b}_{Ht}^n + \omega_F[\hat{v}_{Ht}^* + (1 - 2\gamma)\ell_t^n]$. Subtracting,

$$0 = \omega_H(\hat{b}_{Ht} - \hat{b}_{Ht}^n) + \omega_F(1 - 2\gamma)\tau_t \implies \hat{b}_{Ht} - \hat{b}_{Ht}^n = -\frac{\omega_F(1 - 2\gamma)}{\omega_H}\tau_t, \quad (\text{C.28})$$

Because the Foreign position is a state shared by the two allocations, only the Hegemon position adjusts. Finally, applying (C.26) and (C.25),

$$\begin{aligned} \Phi_t - \Phi_t^n &= \bar{\Phi}(\hat{\Phi}_t - \hat{\Phi}_t^n) = -\bar{\Phi}\mathcal{A}_H(\hat{b}_{Ht} - \hat{b}_{Ht}^n) = \bar{\Phi}\mathcal{A}_H\frac{h\omega_F}{\omega_H}\tau_t = \Gamma_\Phi\tau_t, \\ \Phi_t^* - \Phi_t^{*n} &= -\bar{\Phi}^*\mathcal{A}_F(\hat{v}_{Ht}^* - \hat{v}_{Ht}^{*n}) = 0, \\ \nu_t - \nu_t^n &= s_H(\hat{b}_{Ht} - \hat{b}_{Ht}^n) = -\Gamma_\xi\tau_t, \quad \nu_t^* - \nu_t^{*n} = 0, \end{aligned} \quad (\text{C.29})$$

and therefore $\xi_t \equiv \Xi_t - \Xi_t^n = (\nu_t^* - \nu_t^{*n}) - (\nu_t - \nu_t^n) = \Gamma_\xi\tau_t$. Since $\varpi_t^n = \varpi_t$ by construction of the reference allocation, $\tilde{\Omega}_t = \xi_t = \Gamma_\xi\tau_t$, which completes (25). Differentiating (C.28) also gives the drift relation used in the IS derivation,

$$\chi_t - \chi_t^n = s_H(\hat{b}_{Ht} - \hat{b}_{Ht}^n) = -\Gamma_\xi\tau_t, \quad \chi_t^* - \chi_t^{*n} = 0. \quad (\text{C.30})$$

C.6 Full setup of production

In each country, a unit continuum of differentiated intermediate goods is aggregated by competitive final good producers via a Dixit–Stiglitz technology, $Y_{Ht} = [\int_0^1 Y_{Ht}(j)^{(\varepsilon-1)/\varepsilon} dj]^{\varepsilon/(\varepsilon-1)}$ with $\varepsilon > 1$, and symmetrically abroad. Cost minimisation yields

$$Y_{Ht}(j) = \left[\frac{p_{Ht}(j)}{P_{Ht}} \right]^{-\varepsilon} Y_{Ht}, \quad P_{Ht} \equiv \left[\int_0^1 p_{Ht}(j)^{1-\varepsilon} dj \right]^{1/(1-\varepsilon)}, \quad (\text{C.31})$$

with a symmetric Foreign sector. Each Hegemon intermediate producer j operates $Y_{Ht}(j) = A_t N_t(j)$. Under flexible prices, cost minimisation and the isoelastic demand give the standard constant markup pricing rule $p_{Ht}(j) = \frac{\varepsilon}{\varepsilon-1} \frac{W_t}{A_t}$, i.e. $mc_t^n = 1$. This condition characterises the reference allocation only. Under sticky prices $mc_t \neq 1$ and the marginal cost wedge is carried explicitly in (C.51). Per-period nominal profits, net of adjustment costs, are

$$\Pi_t = P_{Ht} Y_{Ht} - W_t N_t - P_{Ht} \frac{\vartheta_H}{2} \pi_{Ht}^2 Y_{Ht}, \quad \Pi_t^* = P_{Ft}^* Y_{Ft}^* - W_t^* N_t^* - P_{Ft}^* \frac{\vartheta_F}{2} (\pi_{Ft}^*)^2 Y_{Ft}^*, \quad (\text{C.32})$$

so that household labour and profit income together equal

$$W_t N_t + \Pi_t = P_{Ht} Y_{Ht} \left(1 - \frac{\vartheta_H}{2} \pi_{Ht}^2 \right), \quad W_t^* N_t^* + \Pi_t^* = P_{Ft}^* Y_{Ft}^* \left(1 - \frac{\vartheta_F}{2} (\pi_{Ft}^*)^2 \right). \quad (\text{C.33})$$

The adjustment costs use resources, so goods market clearing is

$$\left(1 - \frac{\vartheta_H}{2} \pi_{Ht}^2 \right) Y_{Ht} = C_{Ht} + C_{Ht}^*, \quad \left(1 - \frac{\vartheta_F}{2} (\pi_{Ft}^*)^2 \right) Y_{Ft}^* = C_{Ft} + C_{Ft}^*. \quad (\text{C.34})$$

Productive labour and profit income satisfy (C.33). In addition, the cross-border fee paid by Foreign holders has the equal-and-opposite Hegemon financial-sector receipt $\Pi_t^I = \varkappa_t B_{Ht}^*$. Because this receipt is a financial redistribution rather than production income, it is kept separate from Π_t and does not enter either goods-resource constraint. Including the counterpart receipt therefore closes the holding-wedge accounting without changing the government budgets, bond-market-clearing conditions, or resource constraints.

Hegemon firms are owned by Hegemon households, whose nominal discount factor is Λ_t in (C.13). Equity bears no convenience flow, so by (C.14) any nominal claim without a convenience service must be discounted at $-\dot{\Lambda}_t/\Lambda_t = i_t + \Phi_t$. Foreign firms are owned by Foreign households and, by (C.21), discount at i_t^* .

C.7 Rotemberg inflation dynamics

Lemma 1 (Rotemberg inflation dynamics). *Producer price inflation in the two countries satisfies*

$$\left[i_t + \Phi_t - \pi_{Ht} - \frac{\dot{Y}_{Ht}}{Y_{Ht}} \right] \pi_{Ht} = \kappa_H (mc_t - 1) + \dot{\pi}_{Ht}, \quad (C.35)$$

$$\left[i_t^* - \pi_{Ft}^* - \frac{\dot{Y}_{Ft}^*}{Y_{Ft}^*} \right] \pi_{Ft}^* = \kappa_F (mc_t^* - 1) + \dot{\pi}_{Ft}^*, \quad (C.36)$$

where $\kappa_H \equiv (\varepsilon - 1)/\vartheta_H$, $\kappa_F \equiv (\varepsilon - 1)/\vartheta_F$, and $mc_t \equiv \frac{\varepsilon}{\varepsilon - 1} \frac{W_t}{P_{Ht} A_t}$ is real marginal cost.

Proof. The state variable is $p_{Ht}(j)$, the control variable is $\dot{p}_{Ht}(j)$, and the co-state variable is μ_t . The Hegemon firm solves

$$V_0(p_{H0}) = \max_{\{p_{Ht}(j)\}_{t \geq 0}} \int_0^\infty \exp\left\{-\int_0^t [i_s + \Phi_s] ds\right\} [\Pi_t(p_{Ht}(j)) - \Theta_t(j)] dt, \quad (C.37)$$

with $\Theta_t(j) = \frac{\vartheta_H}{2} (\dot{p}_{Ht}(j)/p_{Ht}(j))^2 P_{Ht} Y_{Ht}$. We can form the current value Hamiltonian as

$$\mathcal{H} = \left[p_{Ht}(j) \left(\frac{p_{Ht}(j)}{P_{Ht}} \right)^{-\varepsilon} Y_{Ht} - \frac{W_t}{A_t} \left(\frac{p_{Ht}(j)}{P_{Ht}} \right)^{-\varepsilon} Y_{Ht} \right] - \frac{\vartheta_H}{2} \left(\frac{\dot{p}_{Ht}(j)}{p_{Ht}(j)} \right)^2 P_{Ht} Y_{Ht} + \mu_t \dot{p}_{Ht}(j). \quad (C.38)$$

The presence of convenience yields only affects the *discount rate* in the co-state equation, and the instantaneous profit and adjustment cost terms are unchanged. The necessary first-order conditions for firm optimality, together with the transversality condition $\lim_{T \rightarrow \infty} \exp\{-\int_0^T (i_s + \Phi_s) ds\} \mu_T p_{HT}(j) = 0$, are

$$\vartheta_H \frac{\dot{p}_{Ht}(j)}{p_{Ht}(j)} \frac{P_{Ht}}{p_{Ht}(j)} Y_{Ht} = \mu_t, \quad (C.39)$$

$$\dot{\mu}_t = (i_t + \Phi_t) \mu_t - \mathcal{H}_{p_H}, \quad (C.40)$$

where

$$\mathcal{H}_{p_H} = (1 - \varepsilon) \left(\frac{p_{Ht}(j)}{P_{Ht}} \right)^{-\varepsilon} Y_{Ht} + \varepsilon \frac{W_t}{P_{Ht} A_t} \left(\frac{p_{Ht}(j)}{P_{Ht}} \right)^{-\varepsilon} Y_{Ht} + \vartheta_H \left(\frac{\dot{p}_{Ht}(j)}{p_{Ht}(j)} \right)^2 \frac{P_{Ht}}{p_{Ht}(j)} Y_{Ht}. \quad (C.41)$$

We restrict our attention to symmetric domestic firm pricing. Imposing $p_{Ht}(j) = P_{Ht}$ and writing $\pi_{Ht} = \dot{P}_{Ht}/P_{Ht}$, condition (C.39) becomes $\mu_t = \vartheta_H \pi_{Ht} Y_{Ht}$ and condition (C.40) becomes

$$\dot{\mu}_t = (i_t + \Phi_t) \mu_t - \left[(1 - \varepsilon) Y_{Ht} + \varepsilon \frac{W_t}{P_{Ht} A_t} Y_{Ht} + \vartheta_H \pi_{Ht}^2 Y_{Ht} \right]. \quad (C.42)$$

Differentiating $\mu_t = \vartheta_H \pi_{Ht} Y_{Ht}$ gives $\dot{\mu}_t = \vartheta_H (\dot{\pi}_{Ht} Y_{Ht} + \pi_{Ht} \dot{Y}_{Ht})$. Substituting into (C.42) and dividing by $\vartheta_H Y_{Ht}$,

$$\dot{\pi}_{Ht} + \pi_{Ht} \frac{\dot{Y}_{Ht}}{Y_{Ht}} = (i_t + \Phi_t) \pi_{Ht} + \frac{\varepsilon - 1}{\vartheta_H} - \frac{\varepsilon}{\vartheta_H} \frac{W_t}{P_{Ht} A_t} - \pi_{Ht}^2. \quad (C.43)$$

Since $\frac{\varepsilon-1}{\vartheta_H} - \frac{\varepsilon}{\vartheta_H} \frac{W_t}{P_{Ht}A_t} = -\kappa_H(mc_t - 1)$ with $\kappa_H = (\varepsilon - 1)/\vartheta_H$ and $mc_t = \frac{\varepsilon}{\varepsilon-1} \frac{W_t}{P_{Ht}A_t}$, rearranging the final expression gives the claimed result in the statement of the lemma for Hegemon producers. The result for Foreign producers follows by a symmetric argument, with discount rate i_t^* in place of $i_t + \Phi_t$, giving (C.36). The proof applies to deterministic paths that are differentiable away from the impact date. $\square$

C.8 Formal derivation of output

Derivation of (C.51)-(C.52). **Step 1. Hegemon demand and goods market clearing.** Using (6) and $Q_t = E_t P_t^*/P_t$,

$$\frac{P_{Ht}^*}{P_t^*} = \frac{P_{Ht}/E_t}{P_t Q_t/E_t} = \frac{P_{Ht}}{P_t} \frac{1}{Q_t} \implies \left(\frac{P_{Ht}^*}{P_t^*}\right)^{-\eta} = \left(\frac{P_{Ht}}{P_t}\right)^{-\eta} Q_t^\eta. \quad (C.44)$$

Substituting (C.10) and (C.20) into the Hegemon resource constraint in (C.34),

$$\left(1 - \frac{\vartheta_H}{2} \pi_{Ht}^2\right) Y_{Ht} = \left(\frac{P_{Ht}}{P_t}\right)^{-\eta} [(1 - \gamma) C_t + \gamma Q_t^\eta C_t^*]. \quad (C.45)$$

The level risk-sharing relation $\log C_t - \log C_t^* - q_t = \Omega_t$ from (18) gives $C_t^* = C_t Q_t^{-1} e^{-\Omega_t}$, so that the bracket equals $C_t Z_t$ with Z_t as in (C.4). Using $P_{Ht}/P_t = G_t^{1/(\eta-1)}$ from (C.5),

$$\left(1 - \frac{\vartheta_H}{2} \pi_{Ht}^2\right) Y_{Ht} = G_t^{\eta/(1-\eta)} C_t Z_t \iff C_t = Y_{Ht} \left(1 - \frac{\vartheta_H}{2} \pi_{Ht}^2\right) \frac{G_t^{\eta/(\eta-1)}}{Z_t}. \quad (C.46)$$

Step 2. Labour supply with a marginal cost wedge. From the definition of mc_t , $W_t/P_t = \frac{\varepsilon-1}{\varepsilon} mc_t A_t (P_{Ht}/P_t)$. Combining with (C.17) and $N_t = Y_{Ht}/A_t$,

$$C_t \left(\frac{Y_{Ht}}{A_t}\right)^\phi = (1 - \nu_t) \frac{\varepsilon - 1}{\varepsilon} mc_t A_t G_t^{1/(\eta-1)}. \quad (C.47)$$

Substituting (C.46) into (C.47) and using $G_t^{1/(\eta-1) - \eta/(\eta-1)} = G_t^{-1}$,

$$Y_{Ht}^{1+\phi} = A_t^{1+\phi} \frac{\varepsilon - 1}{\varepsilon} mc_t (1 - \nu_t) G_t^{-1} Z_t \left(1 - \frac{\vartheta_H}{2} \pi_{Ht}^2\right)^{-1}, \quad (C.48)$$

and taking the $(1 + \phi)$ -th root gives (C.51).

Step 3. Foreign output. The Foreign block is symmetric. We have $(P_{Ft}^*/P_t^*)^{-\eta} = (P_{Ft}/P_t)^{-\eta} Q_t^\eta$, and the Foreign resource constraint is

$$\left(1 - \frac{\vartheta_F}{2} (\pi_{Ft}^*)^2\right) Y_{Ft}^* = \left(\frac{P_{Ft}}{P_t}\right)^{-\eta} [\gamma C_t + (1 - \gamma) Q_t^\eta C_t^*]. \quad (C.49)$$

Inverting the risk-sharing relation gives $C_t = C_t^* Q_t e^{\Omega_t}$, so that

$$\gamma C_t + (1 - \gamma) Q_t^\eta C_t^* = C_t^* [\gamma Q_t e^{\Omega_t} + (1 - \gamma) Q_t^\eta] = C_t^* Q_t^\eta [(1 - \gamma) + \gamma Q_t^{1-\eta} e^{\Omega_t}] = C_t^* Q_t^\eta Z_t^*. \quad (C.50)$$

Note that the substitution must be made into the bracket in its pre-substitution form, since otherwise spurious powers $Q_t^{2-\eta}$ and $e^{2\Omega_t}$ appear. Combining (C.50) with $(P_{Ft}/P_t)^{-\eta}Q_t^\eta = (P_{Ft}^*/P_t^*)^{-\eta} = (G_t^*)^{\eta/(1-\eta)}$ gives $(1 - \frac{\vartheta_E}{2}(\pi_{Ft}^*)^2)Y_{Ft}^* = (G_t^*)^{\eta/(1-\eta)}C_t^*Z_t^*$, and repeating the algebra of Steps 1 and 2 with starred variables gives

$$Y_{Ht} = A_t \left[\frac{\varepsilon-1}{\varepsilon} m c_t (1 - \nu_t) \right]^{\lambda_H} G_t^{-\lambda_H} Z_t^{\lambda_H} \left(1 - \frac{\vartheta_H}{2} \pi_{Ht}^2 \right)^{-\lambda_H}, \quad (C.51)$$

$$Y_{Ft}^* = A_t^* \left[\frac{\varepsilon-1}{\varepsilon} m c_t^* (1 - \nu_t^*) \right]^{\lambda_F} (G_t^*)^{-\lambda_F} (Z_t^*)^{\lambda_F} \left(1 - \frac{\vartheta_E}{2} (\pi_{Ft}^*)^2 \right)^{-\lambda_F}. \quad (C.52)$$

C.9 Stationary steady state

Set $\bar{A} = \bar{A}^* = 1$, $\bar{\theta} = \bar{\theta}^* = 1$, $\bar{\pi} = \bar{\pi}^* = 0$, $\bar{E} = 1$ and $\bar{\dot{E}}/\bar{E} = 0$. The convenience yield specifications (10) and (C.19) are normalised so that

$$\bar{\nu} = \bar{\nu}^* = 0, \quad \bar{\Phi} = \frac{\bar{\Psi}}{1 - \bar{\nu}} = \bar{\Psi}, \quad \bar{\Phi}^* = \frac{\bar{\Psi}^*}{1 - \bar{\nu}^*} = \bar{\Psi}^*. \quad (C.53)$$

Evaluating the three Euler equations (13), (C.21) and (C.22) at zero consumption growth gives

$$\bar{r} = \rho - \bar{\Phi}, \quad \bar{r}^* = \rho^*, \quad \bar{r} = \rho^* + \bar{\kappa} - \bar{\Phi}^*, \quad (C.54)$$

and the first and third of these deliver (17), namely $\bar{\kappa} = \bar{\Phi}^* - \bar{\Phi}$ under $\rho = \rho^*$.

At $\bar{\text{tot}} = 1$ we have $\bar{G} = \bar{G}^* = 1$, and with $\bar{\Omega} = 0$ also $\bar{Z} = \bar{Z}^* = 1$. Setting $\pi_{Ht} = 0$ in (C.35) gives $\bar{m}\bar{c} = \bar{m}\bar{c}^* = 1$, so (C.51)-(C.52) reduce to

$$\bar{Y}_H = \left(\frac{\varepsilon-1}{\varepsilon} \right)^{\lambda_H}, \quad \bar{Y}_F^* = \left(\frac{\varepsilon-1}{\varepsilon} \right)^{\lambda_F}, \quad (C.55)$$

while goods market clearing and $\bar{\Omega} = 0$ require $\bar{Y}_H = \bar{C} = \bar{C}^* = \bar{Y}_F^* \equiv \bar{Y}$.

The steady-state fiscal positions satisfy $\bar{s} = \bar{r}\bar{b}$ and $\bar{s}^* = \bar{r}^*\bar{b}^*$ with

$$\bar{r} = \bar{r} - \bar{\pi} = \rho - \bar{\Phi}, \quad \bar{r}^* = \bar{r}^* - \bar{\pi}^* = \rho^*, \quad \bar{r}_v = \bar{r} - \bar{\kappa} - \bar{\dot{E}}/\bar{E} - \bar{\pi}^* = \rho^* - \bar{\Phi}^*. \quad (C.56)$$

The steady-state discount coefficients in the two Rotemberg problems are

$$\bar{\delta}_H \equiv \bar{r} + \bar{\Phi} - \bar{Y}_H/\bar{Y}_H - \bar{\pi}_H, \quad \bar{\delta}_F \equiv \bar{r}^* - \bar{Y}_F^*/\bar{Y}_F^* - \bar{\pi}_F^*, \quad (C.57)$$

and stationarity, zero trend inflation, (C.56) and $\rho = \rho^*$ imply

$$\bar{\delta}_H = \rho, \quad \bar{\delta}_F = \rho^*, \quad \text{hence} \quad \bar{\delta}_H = \bar{\delta}_F = \rho = \rho^*. \quad (C.58)$$

The Hegemon discounts its surpluses at $\rho - \bar{\Phi} < \rho$, so that a given surplus stream supports a larger debt stock. This convenience rent is the first-order role of the *level* $\bar{\Phi}$ in the model. The

Foreign investor's real return on Treasuries net of the wedge, $\bar{r}_v = \rho^* - \bar{\Phi}^*$, is exactly the rate at which its own convenience yield compensates the shortfall relative to its rate of time preference. By contrast, the firm discount rates in (C.57) are $\bar{\delta}_H = \bar{r} + \bar{\Phi} = \rho$ and $\bar{\delta}_F = \bar{r}^* = \rho^*$, so that the convenience yield in $\bar{\delta}_H$ is exactly offset by the lower nominal rate it induces and has no effect on the Phillips curve.

C.10 Intermediary equilibrium objects

Log-linearised output. Define $y_{Ht} \equiv \log(Y_{Ht}/\bar{Y}_H)$ and $y_{Ft}^* \equiv \log(Y_{Ft}^*/\bar{Y}_F^*)$. At the symmetric steady state, $\bar{G} = \bar{G}^* = \bar{Z} = \bar{Z}^* = 1$ and

$$\log G_t \simeq \gamma(1-\eta)\ell_t, \quad \log G_t^* \simeq \gamma(\eta-1)\ell_t, \quad \log Z_t \simeq \gamma[(\eta-1)q_t - \Omega_t], \quad \log Z_t^* \simeq \gamma[(1-\eta)q_t + \Omega_t]. \quad (\text{C.59})$$

The adjustment cost factors contribute $\lambda_H \vartheta_H \pi_{Ht}^2/2 = O(2)$ and drop, and $\log(1 - \nu_t) = -\nu_t + O(2)$. Taking log deviations of (C.51)–(C.52), using the first-order relation $q_t \simeq (1 - 2\gamma)\ell_t$ and $\gamma(\eta - 1)(1 + (1 - 2\gamma)) = 2\gamma(1 - \gamma)(\eta - 1) = \mathcal{S}$, gives

$$y_{Ht} = a_t + \lambda_H [\widehat{mc}_t - \nu_t + \mathcal{S}\ell_t - \gamma\Omega_t], \quad (\text{C.60})$$

$$y_{Ft}^* = a_t^* + \lambda_F [\widehat{mc}_t^* - \nu_t^* - \mathcal{S}\ell_t + \gamma\Omega_t]. \quad (\text{C.61})$$

Marginal cost gaps. The reference allocation sets $\widehat{mc}_t^n = \widehat{mc}_t^{*n} = 0$ and is evaluated at the same $a_t, a_t^*, \varpi_t, \hat{b}_t, \hat{v}_{Ht}^*$. Subtracting it from (C.60) and using (C.29),

$$x_t = \lambda_H [\widehat{mc}_t + \Gamma_\xi \tau_t + \mathcal{S}\tau_t - \gamma\Gamma_\xi \tau_t] = \lambda_H [\widehat{mc}_t + (\mathcal{S} + (1 - \gamma)\Gamma_\xi)\tau_t], \quad (\text{C.62})$$

and symmetrically $x_t^* = \lambda_F [\widehat{mc}_t^* - (\mathcal{S} - \gamma\Gamma_\xi)\tau_t]$. Inverting gives

$$\widehat{mc}_t = (1 + \phi)x_t - K_\tau^H \tau_t, \quad \widehat{mc}_t^* = (1 + \phi^*)x_t^* + K_\tau^F \tau_t, \quad (\text{C.63})$$

with K_τ^H and K_τ^F as in (C.2).

Consumption gaps. Define $c_t \equiv \log(C_t/\bar{C})$ and $c_t^* \equiv \log(C_t^*/\bar{C}^*)$. Taking log deviations of (C.46) and using (C.59),

$$c_t = y_{Ht} - \frac{\eta}{1-\eta}\gamma(1-\eta)\ell_t - \gamma[(\eta-1)(1-2\gamma)\ell_t - \Omega_t] = y_{Ht} - \mathfrak{g}\ell_t + \gamma\Omega_t, \quad (\text{C.64})$$

with $\mathfrak{g} \equiv \gamma[\eta + (\eta - 1)(1 - 2\gamma)]$. Expanding,

$$\mathfrak{g} = \gamma\eta + \gamma(\eta - 1)(1 - 2\gamma) = \gamma + 2\gamma(\eta - 1) - 2\gamma^2(\eta - 1) = \gamma + \mathcal{S}, \quad (\text{C.65})$$

and symmetrically $c_t^* = y_{Ft}^* + \mathbf{g}\ell_t - \gamma\Omega_t$. Subtracting the reference allocation and using $\tilde{\Omega}_t = \Gamma_\xi \tau_t$ gives

$$\tilde{c}_t = x_t - \mathcal{C}_\tau \tau_t, \quad \tilde{c}_t^* = x_t^* + \mathcal{C}_\tau \tau_t, \quad \mathcal{C}_\tau \equiv \gamma + \mathcal{S} - \gamma\Gamma_\xi. \quad (\text{C.66})$$

Allocation restriction. By definition $\tilde{\Omega}_t = \tilde{c}_t - \tilde{c}_t^* - \tilde{q}_t$ with $\tilde{q}_t \simeq (1 - 2\gamma)\tau_t$ to first order. Substituting (C.66),

$$\Gamma_\xi \tau_t = x_t - x_t^* - (2\mathcal{C}_\tau + (1 - 2\gamma))\tau_t, \quad 2\mathcal{C}_\tau + (1 - 2\gamma) = 2\gamma + 2\mathcal{S} - 2\gamma\Gamma_\xi + 1 - 2\gamma = 1 + 2\mathcal{S} - 2\gamma\Gamma_\xi, \quad (\text{C.67})$$

so that $x_t - x_t^* = [1 + 2\mathcal{S} + (1 - 2\gamma)\Gamma_\xi]\tau_t = \mathcal{H}_\tau \tau_t$, which is (33).

Natural terms of trade. Setting $\widehat{mc}_t = \widehat{mc}_t^* = 0$ in (C.60)-(C.61), the production side gives

$$y_{Ht}^n - y_{Ft}^{*n} = a_t - a_t^* - \lambda_H \nu_t^n + \lambda_F \nu_t^{*n} + \mathcal{S}(\lambda_H + \lambda_F)\ell_t^n - \gamma(\lambda_H + \lambda_F)\Omega_t^n. \quad (\text{C.68})$$

Independently, the reference counterpart of (C.64) and its Foreign analogue, together with $\Omega_t^n = \log C_t^n - \log C_t^{*n} - q_t^n$ and the first-order relation $q_t^n \simeq (1 - 2\gamma)\ell_t^n$, give

$$\Omega_t^n = (y_{Ht}^n - y_{Ft}^{*n}) - 2\mathbf{g}\ell_t^n + 2\gamma\Omega_t^n - (1 - 2\gamma)\ell_t^n \implies y_{Ht}^n - y_{Ft}^{*n} = (1 + 2\mathcal{S})\ell_t^n + (1 - 2\gamma)\Omega_t^n, \quad (\text{C.69})$$

where we used $2\mathbf{g} + (1 - 2\gamma) = 1 + 2\mathcal{S}$. Equating (C.68) and (C.69),

$$(1 + 2\mathcal{S} - \mathcal{S}(\lambda_H + \lambda_F))\ell_t^n = a_t - a_t^* - \lambda_H \nu_t^n + \lambda_F \nu_t^{*n} - \Delta \Omega_t^n, \quad \Delta = (1 - 2\gamma) + \gamma(\lambda_H + \lambda_F). \quad (\text{C.70})$$

Substituting $\Omega_t^n = \Xi_t^n + \varpi_t = \nu_t^{*n} - \nu_t^n + \varpi_t$ and collecting terms gives

$$(1 + 2\mathcal{S} - \mathcal{S}(\lambda_H + \lambda_F))\ell_t^n = a_t - a_t^* + (\Delta - \lambda_H)\nu_t^n + (\lambda_F - \Delta)\nu_t^{*n} - \Delta \varpi_t. \quad (\text{C.71})$$

Finally, $\nu_t^n = s_H \hat{b}_{Ht}^n$, $\nu_t^{*n} = s_F \hat{v}_{Ht}^*$ and the reference version of (C.27),

$$\hat{b}_{Ht}^n = \frac{1}{\omega_H} [\hat{b}_t - \omega_F \hat{v}_{Ht}^* - \omega_F (1 - 2\gamma) \ell_t^n], \quad (\text{C.72})$$

give, after collecting the ℓ_t^n terms and using $s_H \omega_F (1 - 2\gamma) / \omega_H = \Gamma_\xi$, equation (32), with Λ_n , L_b and L_v as defined before.

Natural rates. The Hegemon natural rate is defined as the real rate that (13) implies along the reference path,

$$r_t^n \equiv \rho + \frac{\dot{C}_t^n}{C_t^n} - \Phi_t^n + \chi_t^n, \quad \bar{r}^n = \rho - \bar{\Phi} = \bar{r}, \quad \hat{r}_t^n = \dot{c}_t^n - \bar{\Phi} \hat{\Phi}_t^n + \chi_t^n. \quad (\text{C.73})$$

Substituting the reference version of (C.60) into (C.64),

$$c_t^n = a_t - \lambda_H \nu_t^n + \Upsilon_H \ell_t^n + \gamma(1 - \lambda_H) \Omega_t^n, \quad \Upsilon_H \equiv \lambda_H \mathcal{S} - \mathfrak{g} = -\gamma - \mathcal{S}(1 - \lambda_H), \quad (\text{C.74})$$

where the second equality uses (C.65). Taking drifts, using $\chi_t^n = \dot{\nu}_t^n$ and $\dot{\Omega}_t^n = \dot{\Xi}_t^n + \dot{\omega}_t = (\chi_t^{*n} - \chi_t^n) + \dot{\omega}_t$,

$$\dot{c}_t^n = \dot{a}_t - \lambda_H \chi_t^n + \Upsilon_H \dot{\ell}_t^n + \gamma(1 - \lambda_H) [\chi_t^{*n} - \chi_t^n + \dot{\omega}_t]. \quad (\text{C.75})$$

Inserting (C.75) into (C.73) and collecting the χ_t^n terms,

$$\hat{r}_t^n = \dot{a}_t - \bar{\Phi} \hat{\Phi}_t^n + \underbrace{(1 - \lambda_H)(1 - \gamma)}_{\varrho_H} \chi_t^n + \underbrace{\gamma(1 - \lambda_H)}_{\beta_H} \chi_t^{*n} + \beta_H \dot{\omega}_t + \Upsilon_H \dot{\ell}_t^n, \quad (\text{C.76})$$

where $1 - \lambda_H = \phi/(1 + \phi)$. The Foreign natural rate is defined from (C.21) as $r_t^{n*} \equiv \rho^* + \dot{c}_t^{*n} + \chi_t^{*n}$, with $\bar{r}^{n*} = \rho^* = \bar{r}^*$. The reference Foreign consumption level is

$$c_t^{*n} = a_t^* - \lambda_F \nu_t^{*n} + \Upsilon_F \ell_t^n - \gamma(1 - \lambda_F) \Omega_t^n, \quad \Upsilon_F \equiv \mathfrak{g} - \lambda_F \mathcal{S} = \gamma + \mathcal{S}(1 - \lambda_F), \quad (\text{C.77})$$

and the same steps give

$$\hat{r}_t^{n*} = \dot{a}_t^* + \beta_F \chi_t^n + \varrho_F \chi_t^{*n} - \beta_F \dot{\omega}_t + \Upsilon_F \dot{\ell}_t^n. \quad (\text{C.78})$$

The conditional-reference objects entering (C.76)–(C.78) are $\chi_t^n = s_H \hat{b}_{Ht}^n$, $\chi_t^{*n} = s_F \hat{v}_{Ht}^*$, $\hat{\Phi}_t^n = \hat{\theta}_t - \mathcal{A}_H \hat{b}_{Ht}^n$, and ℓ_t^n from (32). Their underlying predetermined-state paths, including $\dot{\omega}_t$, are inherited inputs rather than outcomes of separate flexible-price dynamic laws. The natural rate contains no sticky-price relative price.

C.11 IS curves

Subtracting the reference relation in (C.73) from the actual Euler equation (13),

$$\tilde{c}_t = (i_t - \pi_t) - r_t^n + (\Phi_t - \Phi_t^n) - (\chi_t - \chi_t^n) = (i_t - \pi_t) - r_t^n + \Gamma_\Phi \tau_t + \Gamma_\xi \dot{\tau}_t, \quad (\text{C.79})$$

using (C.29) and (C.30). The monetary rule (31) gives $(i_t - \pi_t) - \bar{r} = (\phi_\pi - 1)\hat{\pi}_t + \phi_x x_t$ exactly, since the rule has a constant intercept, and $r_t^n - \bar{r} = \hat{r}_t^n$. Using $\tilde{c}_t = x_t - \mathcal{C}_\tau \tau_t$ from (C.66) and $\hat{\pi}_t = \hat{\pi}_{Ht} + \gamma(\dot{\ell}_t^n + \dot{\tau}_t)$ from (34),

$$\dot{x}_t - \mathcal{C}_\tau \dot{\tau}_t = (\phi_\pi - 1)\hat{\pi}_{Ht} + \gamma(\phi_\pi - 1)(\dot{\ell}_t^n + \dot{\tau}_t) + \phi_x x_t - \hat{r}_t^n + \Gamma_\Phi \tau_t + \Gamma_\xi \dot{\tau}_t, \quad (\text{C.80})$$

which rearranges to (37) with $\mathcal{W}_H = \mathcal{C}_\tau + \Gamma_\xi + \gamma(\phi_\pi - 1)$ as previously defined.

Repeating the argument for Foreign, and using $\chi_t^* - \chi_t^{*n} = 0$ and $\Phi_t^* - \Phi_t^{*n} = 0$ from (C.29),

gives

$$\dot{x}_t^* = (\phi_\pi^* - 1)\hat{\pi}_{Ft}^* + \phi_x^* x_t^* - \hat{r}_t^{n*} - \mathcal{W}_F \dot{r}_t - \gamma(\phi_\pi^* - 1)\dot{\ell}_t^n, \quad \mathcal{W}_F \equiv \mathcal{C}_\tau + \gamma(\phi_\pi^* - 1). \quad (\text{C.81})$$

The asymmetry $\mathcal{W}_H - \mathcal{W}_F = \Gamma_\xi + \gamma(\phi_\pi - \phi_\pi^*)$ has two sources, namely differing monetary responses and the fact that only the Hegemon portfolio position responds to the terms of trade gap.

Using the consumption-gap relations (C.66), the portfolio closures (25), the allocation restriction (33), the natural-rate definitions (C.76)–(C.78), the terms-of-trade equation (40), and the relative-wealth law (19), the Foreign IS curve (C.81) is implied by the Hegemon IS curve (37) and is therefore not a separate condition of the equilibrium system in Section 3.7.

C.12 Phillips curves

Write (C.35) as $\pi_{Ht} = \delta_t \pi_{Ht} - \kappa_H (mc_t - 1)$ with $\delta_t \equiv i_t + \Phi_t - \pi_{Ht} - \dot{Y}_{Ht}/Y_{Ht}$ and $\bar{\delta}_H = \rho$. Setting $\pi_{Ht} = \bar{\pi}_H + \hat{\pi}_{Ht}$ and $\delta_t = \bar{\delta}_H + \hat{\delta}_t$,

$$\hat{\pi}_{Ht} = \bar{\delta}_H \bar{\pi}_H + \bar{\delta}_H \hat{\pi}_{Ht} + \bar{\pi}_H \hat{\delta}_t + \hat{\delta}_t \hat{\pi}_{Ht} - \kappa_H \widehat{mc}_t. \quad (\text{C.82})$$

The product $\hat{\delta}_t \hat{\pi}_{Ht}$ is second order and is dropped at first order, and the zero-inflation target $\bar{\pi}_H = 0$ removes the remaining two terms, so that $\hat{\pi}_{Ht} = \bar{\delta}_H \hat{\pi}_{Ht} - \kappa_H \widehat{mc}_t$. Since $\overline{mc} = 1$, the level deviation and the log deviation of marginal cost coincide to first order, and since the reference allocation satisfies $mc_t^n = 1$ identically, the deviation is the gap. Substituting (C.63) gives (38), and the same steps with $\delta_t^* = i_t^* - \pi_{Ft}^* - \dot{Y}_{Ft}^*/Y_{Ft}^*$ and $\bar{\delta}_F = \rho^*$ give (39).

C.13 Terms of trade dynamics

Write the modified UIP condition (16) in deviations. Using $\bar{r} - r^* = \bar{x} - \bar{\Phi}^*$ from (C.54), $\bar{E}/\bar{E} = 0$ and $\Phi_t^* = \bar{\Phi}^*(1 + \hat{\Phi}_t^*)$,

$$\hat{r}_t - \hat{r}_t^* - \hat{x}_t + \bar{\Phi}^* \hat{\Phi}_t^* = \hat{e}_t, \quad \hat{r}_t \equiv i_t - \bar{r}. \quad (\text{C.83})$$

Substituting the monetary rules $\hat{r}_t = \phi_\pi \hat{\pi}_t + \phi_x x_t$ and $\hat{r}_t^* = \phi_\pi^* \hat{\pi}_t^* + \phi_x^* x_t^*$, the CPI–PPI identities $\hat{\pi}_t = \hat{\pi}_{Ht} + \gamma \dot{\ell}_t$, $\hat{\pi}_t^* = \hat{\pi}_{Ft}^* - \gamma \dot{\ell}_t$ and $\hat{e}_t = \dot{\ell}_t - \hat{\pi}_{Ft}^* + \hat{\pi}_{Ht}$ from (34),

$$\phi_\pi (\hat{\pi}_{Ht} + \gamma \dot{\ell}_t) - \phi_\pi^* (\hat{\pi}_{Ft}^* - \gamma \dot{\ell}_t) + \phi_x x_t - \phi_x^* x_t^* + \bar{\Phi}^* \hat{\Phi}_t^* - \hat{x}_t = \dot{\ell}_t + \hat{\pi}_{Ht} - \hat{\pi}_{Ft}^*, \quad (\text{C.84})$$

and collecting the $\dot{\ell}_t$ terms gives (40), with $\Lambda_0 = 1 - \gamma(\phi_\pi + \phi_\pi^*)$ and $\hat{\Phi}_t^* = \hat{\theta}_t^* - \mathcal{A}_F \hat{v}_{Ht}^*$.

C.14 Debt dynamics

Real government debt $b_t = B_t/P_t$ obeys $\dot{b}_t = (i_t - \pi_t)b_t - s_t$, from the nominal flow constraint $\dot{B}_t = i_t B_t - T_t$ and $s_t = T_t/P_t$. Write $b_t = \bar{b} e^{\hat{b}_t}$ and $s_t = \bar{s} + \bar{b} \hat{s}_t$ with $\bar{s} = \bar{r} \bar{b}$, so that

$$\hat{b}_t = (i_t - \pi_t) - (\bar{r} + \hat{s}_t) e^{-\hat{b}_t} \simeq \bar{r} \hat{b}_t + [(i_t - \pi_t) - \bar{r}] - \hat{s}_t, \quad (\text{C.85})$$

where the first-order form drops the second-order product $\hat{s}_t \hat{b}_t$. The monetary rule (31) gives $(i_t - \pi_t) - \bar{r} = (\phi_\pi - 1)\hat{\pi}_t + \phi_x \hat{x}_t$ exactly, since the rule has a constant intercept, and the CPI-PPI relation (34) gives $\hat{\pi}_t = \hat{\pi}_{Ht} + \gamma \hat{\ell}_t$. Substituting these together with the fiscal rule $\hat{s}_t = \psi_b \hat{b}_t - d_t$ from (30) delivers (41). The Foreign law (42) follows from the same steps with $\bar{r}^*$, $\hat{\pi}_t^* = \hat{\pi}_{Ft}^* - \gamma \hat{\ell}_t$ and $\hat{s}_t^* = \psi_b^* \hat{b}_t^* - d_t^*$.

C.15 The international budget constraint

Let $b_{Ft}^* = B_{Ft}^*/P_t^*$ and $w_t^* = b_{Ft}^* + v_{Ht}^*$ denote Foreign household real financial wealth in Foreign consumption units. Dividing (15) by P_t^* and using the quotient rule,

$$\dot{w}_t^* = (i_t^* - \pi_t^*)b_{Ft}^* + \left(i_t - \frac{\dot{E}_t}{E_t} - \kappa_t - \pi_t^*\right)v_{Ht}^* + \frac{W_t^* N_t^* + \Pi_t^* - T_t^*}{P_t^*} - C_t^* - \bar{\mathfrak{Z}}. \quad (\text{C.86})$$

Foreign bond market clearing is $b_t^* = b_{Ft}^*$, since Hegemon households hold no Foreign claims. Subtracting the Foreign government's constraint from (C.86), the terms in $(i_t^* - \pi_t^*)b_t^*$ and in $s_t^* = T_t^*/P_t^*$ cancel, leaving

$$\dot{v}_{Ht}^* = \left(i_t - \frac{\dot{E}_t}{E_t} - \kappa_t - \pi_t^*\right)v_{Ht}^* + \frac{W_t^* N_t^* + \Pi_t^*}{P_t^*} - C_t^* - \bar{\mathfrak{Z}}, \quad (\text{C.87})$$

Using (C.33), this becomes the nonlinear balance of payments equation

$$\dot{v}_{Ht}^* = \left(i_t - \frac{\dot{E}_t}{E_t} - \kappa_t - \pi_t^*\right)v_{Ht}^* + \frac{P_{Ft}^*}{P_t^*} \left(1 - \frac{\vartheta_F}{2} (\pi_{Ft}^*)^2\right) Y_{Ft}^* - C_t^* - \bar{\mathfrak{Z}}. \quad (\text{C.88})$$

The term $-\kappa_t v_{Ht}^*$ in the Foreign external return has the Hegemon counterpart

$$\frac{\Pi_t^I}{P_t} = \kappa_t \frac{B_{Ht}^*}{P_t} = \kappa_t Q_t v_{Ht}^*.$$

Thus, the wedge is a cross-country financial transfer and not a leakage from global income. Multiplying the Foreign payment $\kappa_t v_{Ht}^*$ by the real exchange rate Q_t gives exactly the Hegemon receipt. At the steady state, $\bar{\Pi}^I = \bar{\kappa} \bar{B}_H^*$ and, under $\bar{Q} = 1$, $\bar{\Pi}^I/\bar{P} = \bar{\kappa} \bar{v}_H^*$. If $\bar{\kappa} = 0$, both the payment and receipt are zero in steady state, although shocks to κ_t can still move returns and international wealth.

Given the two goods market clearing conditions, the two government budget constraints and the two bond market clearing conditions, (C.87) is the one household budget constraint that Walras' law does not deliver. It is therefore an independent equilibrium condition, and it is what determines the level of relative consumption.

Steady state and the transfer normalisation. At the steady state $\bar{P}_F^*/\bar{P}^* = 1$ and $\bar{Y}_F^* = \bar{C}^* = \bar{Y}$, the external-return term in (C.87) is $\bar{r}_v \bar{v}_H^*$ and is exactly offset by $-\bar{\mathfrak{T}}$. To permit a non-zero steady-state net foreign asset position while retaining a symmetric real allocation, the budget constraints (8) and (15) carry this constant lump-sum transfer $\bar{\mathfrak{T}} = \bar{r}_v \bar{v}_H^*$ from the Foreign to the Hegemon household, with $\bar{r}_v = \rho^* - \bar{\Phi}^*$ from (C.56). The transfer also closes the Hegemon side: consolidating (8) with the Hegemon government budget and bond-market clearing shows that the Hegemon household pays net interest $(\bar{\Gamma} - \bar{\mathfrak{K}}) \bar{v}_H^* = \bar{r}_v \bar{v}_H^*$ to Foreign holders in steady state, which the transfer receipt exactly offsets, so that $\bar{C} = \bar{Y}$ on both sides. This is the standard device for closing a two-country model with asymmetric steady-state portfolios, and being constant it drops out of every first-order condition.

Linearisation. Write $\dot{v}_{Ht}^* = \bar{v}_H^* \hat{v}_{Ht}^*$ to first order. The three terms on the right of (C.88) expand to

$$\begin{aligned} \left(\dot{i}_t - \frac{\dot{E}_t}{E_t} - \mathfrak{x}_t - \pi_t^* \right) v_{Ht}^* &= \bar{r}_v \bar{v}_H^* + \bar{v}_H^* (\hat{i}_t - \dot{e}_t - \hat{\mathfrak{x}}_t - \hat{\pi}_t^*) + \bar{r}_v \bar{v}_H^* \hat{v}_{Ht}^*, \\ \frac{P_{Ft}^*}{P_t^*} Y_{Ft}^* - C_t^* &= \bar{Y} [\gamma \ell_t + y_{Ft}^* - c_t^*], \quad -\bar{\mathfrak{T}} = -\bar{r}_v \bar{v}_H^*, \end{aligned} \quad (\text{C.89})$$

where the second line uses the first-order relation $\log P_{Ft}^* - \log P_t^* \simeq \gamma \ell_t$ from (C.8) and the fact that the adjustment cost factor is second order. From (C.64) applied to Foreign, $y_{Ft}^* - c_t^* = -\mathfrak{g} \ell_t + \gamma \Omega_t$, so that the linearised trade balance is

$$\frac{TB_t^*}{\bar{Y}} = \gamma \ell_t + y_{Ft}^* - c_t^* = (\gamma - \mathfrak{g}) \ell_t + \gamma \Omega_t = -\mathcal{S} \ell_t + \gamma \Omega_t, \quad (\text{C.90})$$

using $\mathfrak{g} = \gamma + \mathcal{S}$. The sign pattern is the expected one. For $\eta > 1$, a rise in the relative price of Foreign goods worsens the Foreign trade balance, and a rise in Hegemon relative to Foreign consumption improves it. The real return term expands, using the monetary rule and (34), as

$$\hat{i}_t - \dot{e}_t - \hat{\pi}_t^* = \phi_\pi (\hat{\pi}_{Ht} + \gamma \dot{\ell}_t) + \phi_x x_t - (\dot{\ell}_t + \hat{\pi}_{Ht} - \hat{\pi}_{Ft}^*) - (\hat{\pi}_{Ft}^* - \gamma \dot{\ell}_t) = (\phi_\pi - 1) \hat{\pi}_{Ht} + \phi_x x_t + [\gamma \phi_\pi - (1 - \gamma)] \dot{\ell}_t. \quad (\text{C.91})$$

Dividing (C.89) by $\bar{v}_H^*$ and retaining the $-\hat{\mathfrak{x}}_t$ term delivers (43). Finally, from (C.25) and (18),

$$\Omega_t = \Xi_t + \varpi_t = (\nu_t^* - \nu_t) + \varpi_t = s_F \hat{v}_{Ht}^* - s_H \hat{b}_{Ht} + \varpi_t, \quad (\text{C.92})$$

An equivalent form in terms of the reference position follows from (C.28), namely $\Omega_t = s_F \hat{v}_{Ht}^* - s_H \hat{b}_{Ht}^n + \varpi_t + \Gamma_\xi \tau_t$.

The relative-wealth law. Substituting (C.26) into (19) and using $\bar{\Phi} - \bar{\Phi}^* + \bar{\pi} = 0$ gives the reduced relative-wealth law (44), with $\hat{b}_{Ht}$ recovered from (C.27). Note the stabilising force acting on the net foreign asset position. A larger $\hat{v}_{Ht}^*$ lowers $\hat{\Phi}_t^*$ by $\mathcal{A}_F$, which through (40) raises the return Foreign households require on Hegemon debt.

C.16 Impact conditions and solution concept

At $t = 0$, the predetermined nominal positions satisfy $B_{0+} = B_{0-}$, $B_{0+}^* = B_{0-}^*$ and $B_{H,0+}^* = B_{H,0-}^*$, while the accumulated relative-wealth state satisfies $\varpi_{0+} = \varpi_{0-}$. Admissible Rotemberg price-setting paths satisfy $P_{H,0+} = P_{H,0-}$ and $P_{F,0+}^* = P_{F,0-}^*$. The nominal exchange rate and forward-looking variables may jump, and; the induced impact valuations are $b_{0+} = B_{0-}/P_{0+}$, $b_{0+}^* = B_{0-}^*/P_{0+}^*$ and $v_{H,0+}^* = B_{H,0-}^*/(E_{0+}P_{0+}^*)$, subject also to bond market clearing and the mappings stated in Section 3.7.

D Prior distributions for the estimation in section 4

E Alternative estimation approaches

Here, we present some complementary estimation results, utilizing other estimation methods. In addition, we here at times use convenience yield data to show the robustness of our historical decomposition to the inclusion of such data from different sources. One way in which we obtain the convenience yield data series is as follows: we first measure the U.S. convenience yield as the difference between AAA corporate bonds and the yield offered on 10-year treasuries, and we subsequently measure the U.S. convenience yield as the sum of the aforementioned spread and CIP deviations between the U.S. and the U.K. as estimated by Du et al. (2018). Alternatively, we also use the treasury richness series of Fleckenstein and Longstaff (2024) for the baseline U.S. convenience yield.

Parameter	Prior imposed on	Distribution	Mean	Std. Dev.
Structural parameters				
ϕ_π	ϕ_π	Normal	0.98	1.0000
ϕ_π^*	ϕ_π^*	Normal	0.98	1.0000
ϕ_x	ϕ_x	Normal	0.00	0.2887
ϕ_x^*	ϕ_x^*	Normal	0.00	0.2887
ψ_b	ψ_b	Normal	-0.05	0.5887
ψ_b^*	ψ_b^*	Normal	-0.05	0.5887
$\bar{\Psi}$	$\bar{\Psi}$	Inverse Gamma	0.002	0.010
$\bar{\Psi}^*$	$\bar{\Psi}^*$	Inverse Gamma	0.001	0.010
Continuous-time mean-reversion parameters				
ϕ_a	$\rho_a = \exp(-h\phi_a)$	Beta	0.85	0.08
ϕ_a^*	$\rho_a^* = \exp(-h\phi_a^*)$	Beta	0.85	0.08
ϕ_θ	$\rho_\theta = \exp(-h\phi_\theta)$	Beta	0.50	0.18
ϕ_θ^*	$\rho_\theta^* = \exp(-h\phi_\theta^*)$	Beta	0.50	0.18
ϕ_d	$\rho_d = \exp(-h\phi_d)$	Beta	0.50	0.18
ϕ_d^*	$\rho_d^* = \exp(-h\phi_d^*)$	Beta	0.50	0.18
ϕ_κ	$\rho_\kappa = \exp(-h\phi_\kappa)$	Beta	0.50	0.18
Continuous-time shock diffusions				
σ_a	$\sigma_{\eta,a} = c(\phi_a)\sigma_a$	Inverse Gamma	0.50	0.30
σ_a^*	$\sigma_{\eta,a}^* = c(\phi_a^*)\sigma_a^*$	Inverse Gamma	0.50	0.30
σ_θ	$\sigma_{\eta,\theta} = c(\phi_\theta)\sigma_\theta$	Inverse Gamma	0.50	0.30
σ_θ^*	$\sigma_{\eta,\theta}^* = c(\phi_\theta^*)\sigma_\theta^*$	Inverse Gamma	0.50	0.30
σ_d	$\sigma_{\eta,d} = c(\phi_d)\sigma_d$	Inverse Gamma	0.50	0.30
σ_d^*	$\sigma_{\eta,d}^* = c(\phi_d^*)\sigma_d^*$	Inverse Gamma	0.50	0.30
σ_κ	$\sigma_{\eta,\kappa} = c(\phi_\kappa)\sigma_\kappa$	Inverse Gamma	0.50	0.30
Measurement-error standard deviations				
stderr_x	stderr_x	Inverse Gamma	0.30	0.10
stderr_{x^*}	stderr_{x^*}	Inverse Gamma	0.15	0.05
stderr_{gb}	stderr_{gb}	Inverse Gamma	0.45	0.15
stderr_{gb^*}	stderr_{gb^*}	Inverse Gamma	0.40	0.13
$\text{stderr}_{gb,H}$	$\text{stderr}_{gb,H}$	Inverse Gamma	0.70	0.20
stderr_θ	stderr_θ	Inverse Gamma	$0.15s_\theta$	$0.05s_\theta$
stderr_{θ^*}	stderr_{θ^*}	Inverse Gamma	$0.15s_{\theta^*}$	$0.05s_{\theta^*}$

Notes: For the continuous-time shock processes, h denotes the sampling interval and $c(\phi) = \sqrt{[1 - \exp(-2\phi h)]/(2\phi)}$. The reported means and standard deviations apply to the objects in the second column. Consequently, they are not the moments of ϕ_j or of the unconditional marginal prior for σ_j . The prior for ϕ_j is induced by the Beta prior on $\rho_j = \exp(-h\phi_j)$ and includes the corresponding Jacobian. The prior for σ_j is conditional on ϕ_j , is induced by the inverse-gamma prior on the quarterly innovation standard deviation $\sigma_{\eta,j} = c(\phi_j)\sigma_j$, and also includes the corresponding Jacobian. Here, s_θ and s_{θ^*} denote the sample standard deviations of the respective observables.

Table D.1: Prior distributions for the continuous-time estimation in Section 4.3.

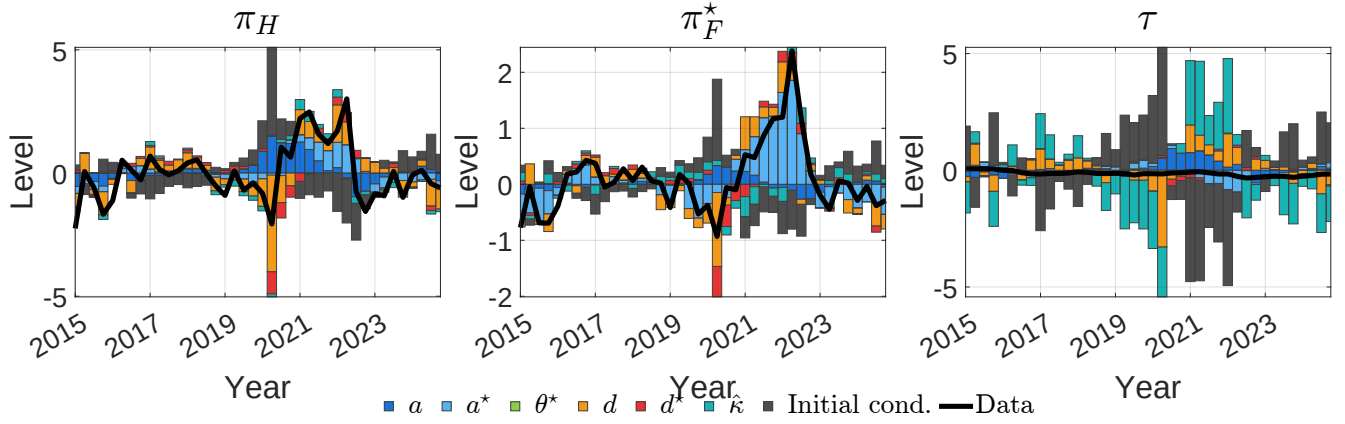


Figure E.1: Historical shock decomposition implied by the model when using ML, without using convenience yield estimates to discipline estimation.

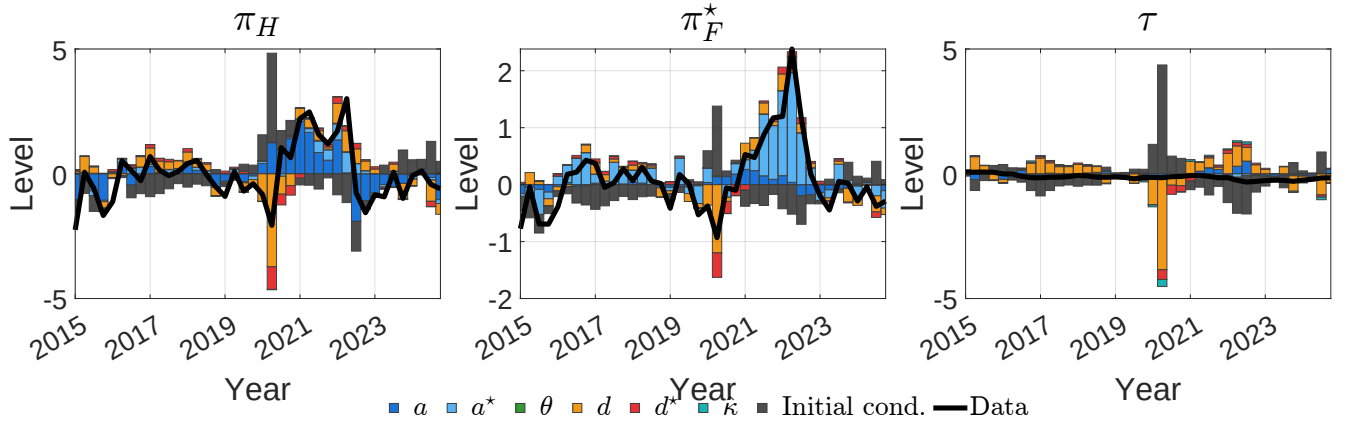


Figure E.2: Historical shock decomposition implied by the model when using ML, using convenience yield estimates to discipline estimation.

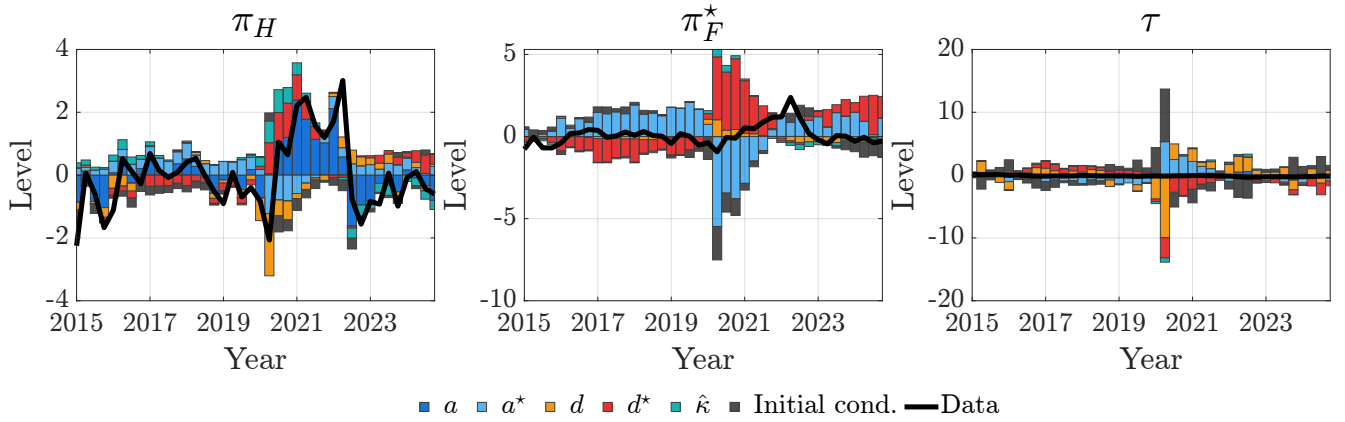


Figure E.3: Historical shock decomposition implied by the model when using ML, when using an alternative convenience yield specification following [Fleckenstein and Longstaff \(2024\)](#).

E.1 ML without convenience yields

E.2 ML with convenience yields

E.3 ML with alternative convenience yield definition

E.4 GMM

We here present the supplementary results from the alternative estimation specification, where we use a Generalized Method of Moments estimator based on exact ED-SSR moments. The estimator matches the contemporaneous covariance matrix and the first-order autocovariances of the observables.

Policy and convenience yield parameters				Shock mean-reversion and diffusion parameters			
Parameter	Estimate	S.e.	Efficient s.e.	Parameter	Estimate	S.e.	Efficient s.e.
ϕ_π	1.0216	0.1124	0.0354	ϕ_a	0.0936	0.2976	0.0891
ϕ_π^*	1.1260	0.2035	0.0897	ϕ_a^*	0.0507	0.0067	0.0044
ϕ_χ	-0.6780	0.0693	0.0244	ϕ_θ	0.0222	0.4778	0.1419
ϕ_χ^*	-1.0000	0.1608	0.0634	ϕ_θ^*	0.0200	9.4030	2.1790
ψ_b	-0.0557	0.0971	0.0456	ϕ_d	0.0200	0.0128	0.0079
ψ_b^*	-0.9857	0.3219	0.1097	ϕ_d^*	0.0386	0.1337	0.0746
$\bar{\Psi}$	0.0044	0.0030	0.0019	ϕ_κ	1.3364	0.1445	0.0885
$\bar{\Psi}^*$	0.0033	0.0018	0.0010	σ_a	9.7962	6.4520	1.4110
				σ_a^*	10.0000	4.0060	1.8040
				σ_θ	5.4823	0.0992	0.0194
				σ_θ^*	10.0000	0.0917	0.0134
				σ_d	0.8519	1.0970	0.4554
				σ_d^*	4.9452	1.9250	0.6961
				σ_κ	10.0000	1.8920	0.5928

Table E.2: GMM estimates using U.S. and U.K. data since 1998.

The estimation uses 107 observations, 27 moments, and 22 parameters, leaving five over-identifying restrictions. The weighting matrix is constructed using a Newey–West HAC estimator with a Bartlett kernel and bandwidth four. The value of the GMM criterion is $Q = 22869$. The Hansen test yields a p -value of 0.0000, and therefore strongly rejects the over-identifying restrictions. The moment Jacobian has full column rank, with $\text{rank}(G) = 22$.

E.5 Additional IRFs

We here present additional impulse responses across policy regimes in the baseline case with low convenience yields.

E.6 Comparison across all specifications

Moment	Sample	Model	Standardized residual
$\text{Var}(x_t)$	4.23133	122.052	135.231
$\text{Cov}(x_t, \pi_{H,t})$	0.390887	11.4491	38.712
$\text{Cov}(x_t, \pi_{F,t}^*)$	0.250965	2.53306	12.756
$\text{Cov}(x_t, \tau_t)$	-0.046349	24.9988	463.803
$\text{Cov}(x_t, gb_t)$	-2.90146	-45.6978	-31.490
$\text{Cov}(x_t, gb_t^*)$	-3.19430	7.73783	8.072
$\text{Var}(\pi_{H,t})$	1.15963	9.27155	21.656
$\text{Cov}(\pi_{H,t}, \pi_{F,t}^*)$	0.518981	2.07891	8.154
$\text{Cov}(\pi_{H,t}, \tau_t)$	0.029041	1.44468	69.342
$\text{Cov}(\pi_{H,t}, gb_t)$	-1.72983	-3.74258	-2.544
$\text{Cov}(\pi_{H,t}, gb_t^*)$	-0.486936	-0.632728	-0.286
$\text{Var}(\pi_{F,t}^*)$	0.358416	1.74821	11.576
$\text{Cov}(\pi_{F,t}^*, \tau_t)$	0.005394	-0.076798	-4.678
$\text{Cov}(\pi_{F,t}^*, gb_t)$	-0.812466	-1.12073	-0.808
$\text{Cov}(\pi_{F,t}^*, gb_t^*)$	-0.232080	-0.186223	0.177
$\text{Var}(\tau_t)$	0.023974	7.57942	1469.277
$\text{Cov}(\tau_t, gb_t)$	0.043430	-6.23885	-114.621
$\text{Cov}(\tau_t, gb_t^*)$	0.082804	3.10254	77.105
$\text{Var}(gb_t)$	10.4501	43.1583	8.203
$\text{Cov}(gb_t, gb_t^*)$	3.91033	-13.7942	-8.602
$\text{Var}(gb_t^*)$	7.50175	22.8641	6.952
$\text{Cov}(x_t, x_{t-1})$	3.52586	38.8542	52.666
$\text{Cov}(\pi_{H,t}, \pi_{H,t-1})$	0.493574	8.84012	49.786
$\text{Cov}(\pi_{F,t}^*, \pi_{F,t-1}^*)$	0.198167	1.67343	19.345
$\text{Cov}(\tau_t, \tau_{t-1})$	0.022845	0.768324	146.949
$\text{Cov}(gb_t, gb_{t-1})$	1.74359	15.2953	11.477
$\text{Cov}(gb_t^*, gb_{t-1}^*)$	2.43605	-0.545275	-1.826

Table E.3: Fit of the contemporaneous covariance and first-order autocovariance moments used in the GMM estimation. The standardized residual is the difference between the model-implied and sample moment divided by the estimated standard deviation of the corresponding moment.

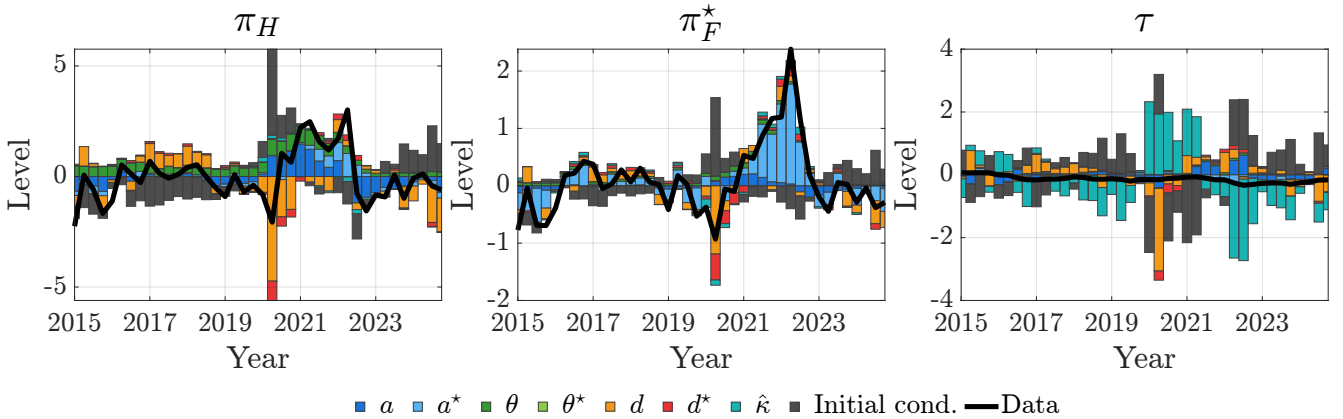


Figure E.4: Historical shock decomposition implied by the model when using GMM, main convenience yield specification.

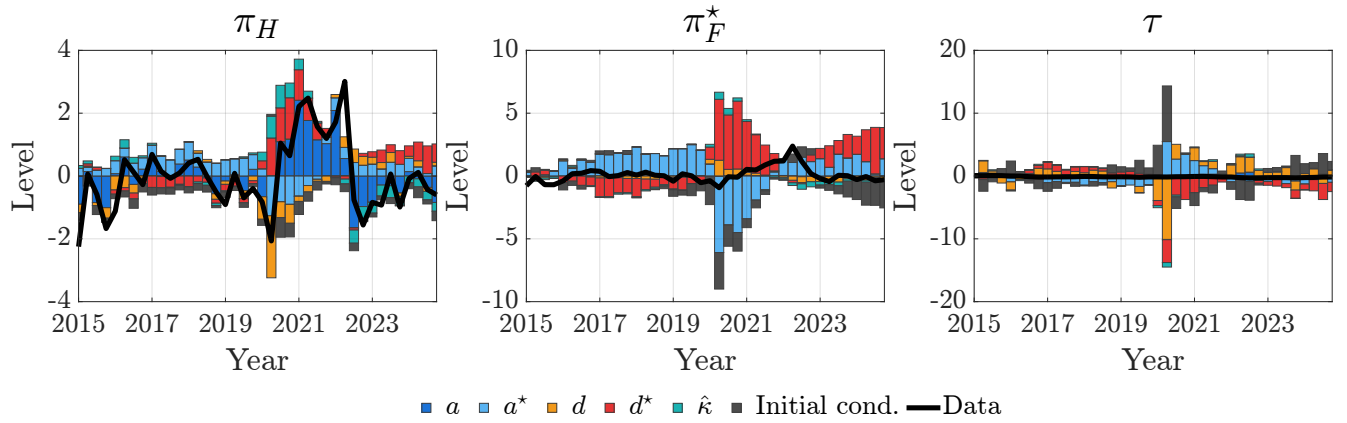


Figure E.5: Historical shock decomposition implied by the model when using GMM, alternative convenience yield specification.

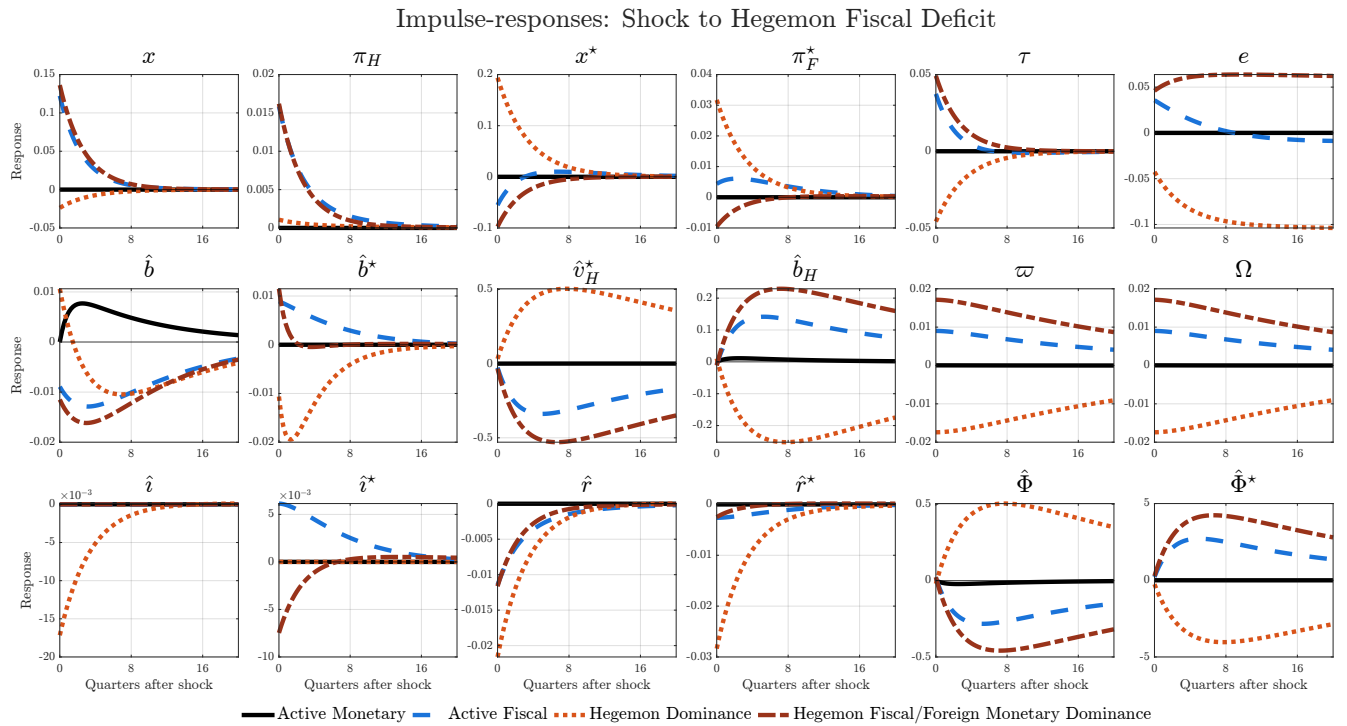


Figure E.6: Additional impulse response functions following an expansionary Hegemon fiscal transfer shock - low convenience yield calibration.

Impulse-responses: Shock to Hegemon Fiscal Deficit

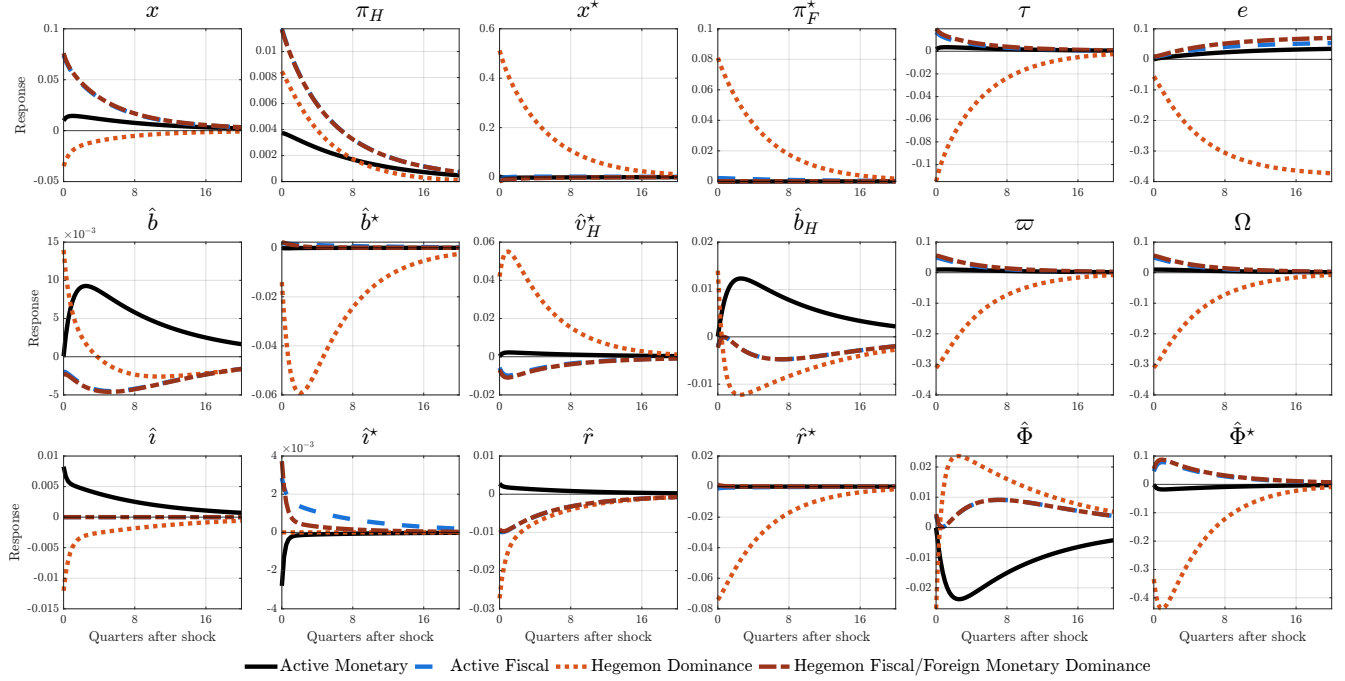


Figure E.7: Additional impulse response functions following an expansionary Hegemon fiscal transfer shock - high convenience yield calibration.

Cross-version contribution of Hegemon deficit shock e_d

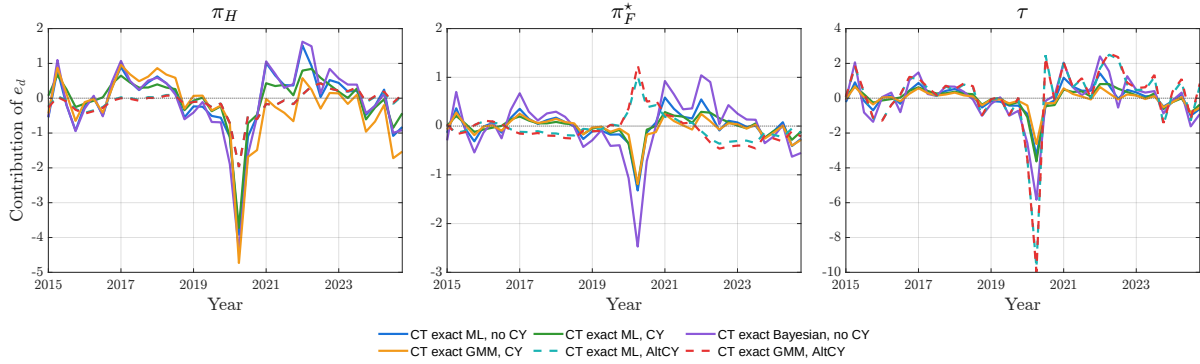


Figure E.8: Comparison of all six estimation approaches - ML without CY data, ML with main CY data, Bayesian without CY data, GMM with main CY data, ML with alternative CY data, GMM with alternative CY data.